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Coupling of hydrodynamic and electric interactions in adsorption of colloidal particles

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Abstract

The role of the coupling between hydrodynamic and electric interactions in adsorption of colloid particles is reviewed. First the general formulation of the problem of fluid motion connected with the macroscopic suspension flow as well as microscopic particle motion is presented together with the general description of the colloid particle adsorption. Then two limiting cases are considered: motion of uncharged particles and electric interactions of stationary particles. The theoretical consideration are supported by illustrative experimental results. Next the electrokinetic effects in particle adsorption process are considered and the theoretical background for the description of these effects is given. The effect of the electroviscous drag on the particle aggregation and adsorption kinetics is discussed. Also the electrokinetic lift force, which is an example of the non-linear electrokinetic effect, is considered and the possible implications on particle adsorption are examined. In the following part of the review the influence of electro-hydrodynamic coupling on the particle transport to the interface is considered. First the 'inverse salt effect' for the initial adsorption flux when the interface is not covered by adsorbed particles is examined. Then, the electro-hydrodynamic scattering effect which occurs when the interface is partially covered by particles is considered and its experimental evidence concerning the adsorption kinetics and the local structure of adsorbate is considered. Also a novel application of the scattering effect to measure interparticle forces the 'colloid particle collider' is presented

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and some experimental results obtained using this technique are shown. All results reported in this review suggest strong coupling of the hydrodynamic and electric forces and their pronounced influence on colloid particle adsorption. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Colloid particle; Adsorption; Hydrodynamic; Electric double layer; Interactions

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1. Introduction

Interactions of macromolecules, colloid and bio-particles (proteins, viruses, bacteria, etc.) with solid–liquid interface leading to adsorption are of the practical

importance for many modern technological processes. A quantitative analysis of particle adsorption phenomena also has significance for polymer and colloid science, biophysics and medicine. It enables better understanding and controlling protein separation processes, enzyme immobilization, biofouling of transplants, adhesion of bacteria and viruses, etc.

The behavior of many colloidal systems is successfully explained by the well established DLVO theory [1,2]. This theory describes the stability of the colloidal system, i.e. its tendency to minimize the surface area of the system by clustering single particles to form larger aggregates, in terms of the interplay of attractive van der Waals interactions and repulsive forces of the electrical double layers (EDL) surrounding particles in the polar media. The application of the DLVO theory to various colloidal systems has been extensively discussed in many monographs [3,4] where various aspects concerning the origin of the electric charge on particle surfaces, the influence of electrolyte valency and concentration have been examined. However, the DLVO theory describes a colloidal system in the equilibrium. In particular, the energy of interactions between particles is calculated assuming the quasistatic approach of particles from infinity to a given distance [2]. In reality particles are dispersed in a fluid medium and are subjected to Brownian motion. Additionally in many practical applications particles of the colloidal system are also subjected to the external macroscopic flow. Hence, the hydrodynamic effects should also be taken into account in the description of behavior of any colloidal dispersion. The effect of hydrodynamics on colloid stability was first recognized by Derjaguin and Muller [5] and Spielman [6]. They explained that the rate of fast, barrierless aggregation was always below one predicted by Smoluchowski [7] as a result of the increased hydrodynamic drag exerted on a particle while approaching the second one. The increased viscous resistance to the particle motion in the neighborhood of other surfaces is often referred to as the hydrodynamic force (interaction).

The colloidal particle adsorption process can be regarded as the limiting case of the heterocoagulation and thus it can also be described in terms of the DLVO theory. However, in the case of the adsorption of colloidal particle at macroscopic surfaces the role of the hydrodynamic interactions is even more pronounced than for homocoagulation of like particles [8,9].

The theories of the hydrodynamic interactions and the electric double layer interactions have been developed independently. However, there is ample experimental evidence that in colloidal systems these interactions are coupled and should be considered simultaneously.

The aim of this review is to show that in many aspects of the colloidal particle adsorption there exists the coupling between the hydrodynamic and electric interactions. It is shown that in the case of charged particles immersed in a polar medium, the effect of the electric field should be considered already at the stage of formulation of the equation describing motion of the fluid around particles. That leads to the electrokinetics effects like electroviscous (or electrokinetic) drag or electrokinetic lift which may influence adsorption of colloid particles at interfaces. The coupling between hydrodynamics and electric interactions directly determines

the rate of colloidal particle adsorption. At early stages of adsorption, when the surface is not covered by particles, strong fluid flow enhances the effect of attractive EDL forces between oppositely charged particles and interface increasing the adsorption rate. In contrast, when the interface is already partially covered by deposited particles, the repulsive EDL forces between adsorbed and adsorbing particles enhance the effect of 'hydrodynamic' shadowing and thus considerably decrease the adsorption rate.

The paper is organized as follows. First the general considerations concerning the description of the macroscopic and microscopic fluid flows occurring in the colloidal particle adsorption process are presented. Then, the quantitative methods of description of the adsorption process are presented. The theoretical considerations are accompanied by some experimental examples of the model colloid adsorption in the well described hydrodynamic conditions.

Next two limiting cases are discussed, the hydrodynamic interactions of uncharged particles and interfaces and the electric interactions between charged particles and interfaces in a stagnant fluid. Some selected experimental results supporting the theoretical predictions are also presented.

Then the role of electrokinetic effects on colloidal particle adsorption is discussed. The experimental data concerning the influence of the electrokinetic drag on particle aggregation and adhesion are compared with theoretical predictions. The electrokinetic lift which can influence the colloid particle adsorption process in non-aqueous media is also discussed.

In the next chapter the consequences of the direct coupling of hydrodynamic forces and electric forces on the adsorption of colloidal particles as predicted by the transport equations are studied. The 'inverse salt effect' for the initial adsorption rates and the 'electro-hydrodynamic shadowing' for the later stages of adsorption are discussed in some details.

2. General description

2.1. Motion of fluid

Any motion of the Newtonian fluid can be described in terms of the Navier–Stokes equation:

$$\rho\left(\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{v}\cdot\nabla\boldsymbol{v}\right) = -\nabla p + \mu\nabla^2\boldsymbol{v} + \boldsymbol{F} \tag{1}$$

where $\boldsymbol{v}$ is the fluid flow velocity, ρ is the fluid density, p is the pressure, μ is the dynamic viscosity, $\boldsymbol{F}$ is the body volume force exerted on the fluid and t is the time. For the incompressible liquids the Navier–Stokes equation is complemented by the continuity equation:

$$\nabla\cdot\boldsymbol{v} = 0 \tag{2}$$

To solve Eqs. (1) and (2) for the fluid velocity $\boldsymbol{v}$, the initial velocity field and boundary condition must be specified. The Navier–Stokes equation is a highly non-linear equation and complete analytical solutions exists only for simple geometries. In most cases it is necessary to apply sophisticated numerical algorithms. In many processes the steady state is established very fast and it is sufficient to consider the steady fluid motion. Then, the time derivative $\partial \boldsymbol{v}/\partial t$ can be neglected.

In general the presence of the dispersed phase changes the properties of the liquid. If the concentration of the dispersed colloidal particles is high the fluid usually exhibits viscoelastic behavior [10]. For the moderate concentrated suspension it is sufficient to adjust the viscosity of the fluid for the presence of colloidal particles [9].

If the volume fraction of the particles is $< 5\%$, which corresponds to a particle concentration of the order of 10^{12} cm^{-3} in the case of typical colloids, the influence of particles on the flow around the macroscopic interfaces can be neglected. Therefore, the hydrodynamic problem of the flow connected with the motion of colloidal suspension can be decomposed into problem of the macroscopic flow of the uniform liquid towards the interfaces where particles are adsorbing (often referred to as collectors) and the microscopic flows between moving particles and the interfaces. The latter are the origin of the hydrodynamic interactions (cf. Fig. 1).

The review of the macroscopic flow fields being the solution of the Navier–Stokes equation for the collectors of various geometries relevant for the practical applications of adsorption of colloidal particles was presented in the monographs by van de Ven [9], Elimelech [11] in the review articles by Adamczyk et al. [8,12].

Among numerous collector geometries we discuss in some details the 'stagnation-point flow' collector which appeared to be particularly useful for theoretical and experimental studies of colloidal particles adsorption. The first experimental cell, referred to as an 'impinging jet', was constructed by Dąbroś and van de Ven [13]. Since then it has found the numerous applications in the research on the

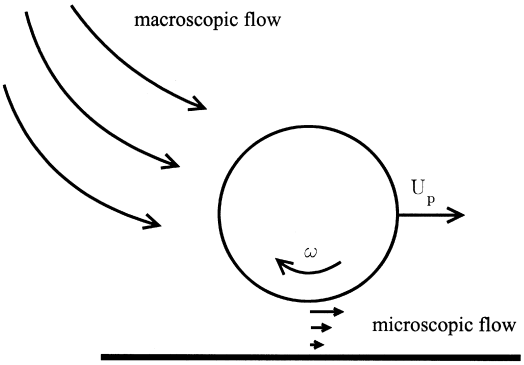


Fig. 1. Schematic presentation of macroscopic and microscopic flows involved in the colloidal particle adsorption process.

adsorption of colloidal systems [12–22]. The principle of the cell is illustrated in Fig. 2a. The fluid velocity components in the close vicinity of the stagnation point can be described by [13]:

$$v_r = \alpha_r(Re, h/r_c)\frac{V_m}{r_c^2}rz$$

$$v_z = -\alpha_r(Re, h/r_c)\frac{V_m}{r_c^2}z^2 \tag{3}$$

where r is the radial distance from the symmetry axis, z is the distance from the adsorbing interface, α_r, is a flow parameter dependent on the cell geometry and the flow intensity expressed by the Reynolds number $Re = Q/\pi r_c \nu$, where Q is the volume flow, r_c is the capillary radius, ν is the kinematic viscosity of the fluid and $V_m = Q/\pi r_c^2$, is the mean linear velocity. This dependence is illustrated in Fig. 2b where the values of α_r, as a function of the Reynolds number for various values of the ratio of the distance between the capillary outlet and the adsorption interface to the capillary radius (h/r_c) is presented. The results shown in Fig. 2b were obtained by the numerical solution of the Navier–Stokes equation. The details of calculations are given elsewhere [23]. Far from the stagnation point the fluid velocity field deviate from the one described by Eq. (3).

Recently Adamczyk et al. [23,24] developed the new experimental cell in which

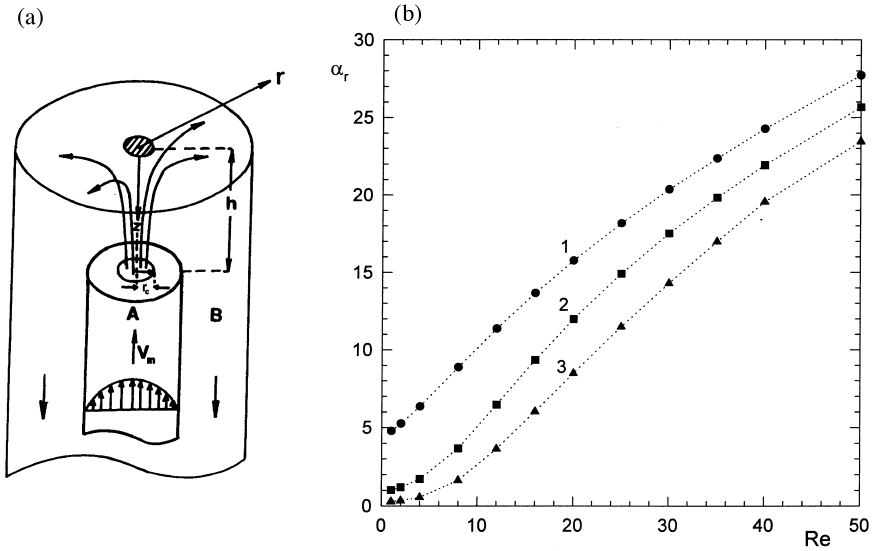


Fig. 2. The stagnation point flow 'impinging jet' experimental cell, part a — schematic view, part b — the dependence of the flow parameter α_r on the Reynolds number of the flow calculated for various cell geometry. Spline curves (1) for $h/r_c = 1$; (2) for $h/r_c = 2$; and (3) for $h/r_c = 3$ fitted to the points obtained from the numerical solution of the Navier–Stokes equation.

the suspension jet has plane-parallel geometry. This type of flow geometry enables microscopic observations of the adsorption process over areas exceeding by order of magnitude those for the impinging jet cells with radial geometry. The flow field in this new cell, referred to as the slot impinging jet, near the stagnation line can be described by:

$$v_x = 2\alpha_s(Re,h/d_c)\frac{V_m}{d^2}xz$$

$$v_z = -\alpha_s(Re,h/d_c)\frac{V_m}{d^2}z^2 \tag{4}$$

where x is the distance from the symmetry plane, α_s, is the flow parameter which, as in the case of cylindrical geometry, is a function of the cell geometry and the Reynolds number of the flow here expressed as $Re = Q/2l\nu$, where l is the depth of the channel assumed much larger than its width $2d$ and $V_m = Q/2dl$. The principle of this cell is illustrated in Fig. 3a and the dependence of the flow parameter α_s on the Reynolds number and the ratio of the distance between the channel outlet and the adsorption surface to the channel half width (h/d) is shown in Fig. 3b. Here again the results were obtained by the numerical solution of the Navier–Stokes equation in the appropriate geometry [23].

The experimental results obtained using these experimental cells will be discussed extensively in due course.

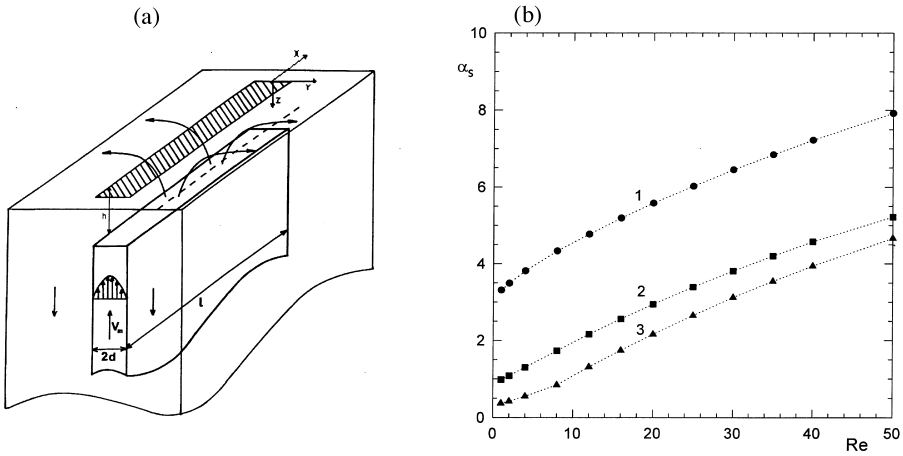


Fig. 3. The stagnation line flow 'slot impinging jet' experimental cell, part a — schematic view, part b — the dependence of the flow parameter α_s on the Reynolds number of the flow calculated for various cell geometry. Spline curves (1) for $h/d_c = 1$; (2) for $h/d_c = 2$; and (3) for $h/d_c = 3$ fitted to the points obtained from the numerical solution of the Navier–Stokes equation.

Another flow field important from the theoretical and practical point of view is a simple shear flow for which fluid velocity components are defined as:

$$v_x = Gz$$

$$v_z = 0 \tag{5}$$

where G is the shear rate. Any flow near the interface can be decomposed in local Cartesian coordinates into a stagnation point flow and a simple shear flow.

The solution of Eq. (1) for a microscopic flow is greatly simplified by the fact that the size of particles in a colloidal system is of the order of micrometer or less. Thus, the resulting Reynolds number associated with the microscopic flows, $Re = aU_p/\nu$ seldom exceeds 10^{-2}, a is the particle radius, U_p is the typical velocity of the particle motion. Therefore, the Navier–Stokes equation can be linearized and as a result the Stokes or creeping flow equation is obtained:

$$\mu\nabla^2 \boldsymbol{v} = \nabla p - \boldsymbol{F} \tag{6}$$

The detailed form of the Stokes equation and the methods of the solution for the hydrodynamic problems important from the point of view of colloidal adsorption will be discussed in the proceeding sections.

In most cases, in particular in polar media, colloidal particles as well as surfaces of macroscopic collectors bear an electric charge. The origin of this charge depends on the nature of the interface. Defects of the crystalline structure at the interface, adsorption of ions from the solution and/or dissociation of ionogenic sites at the interface may be the origin of the interfacial charge. The review on this topic can be found in the monographs of Lyklema [3] and Hunter [4]. Because the colloidal dispersion as a whole is electrically neutral, the interfacial charge must be neutralized by the excess of ions of opposite charge in the suspension. These ions concentrated in the neighborhood of the charged interface form the layer of locally charged fluid which is usually referred to as the electric double layer (EDL). Thus, in the region of the electric double layer the volume force term in Eq. (6) can be expressed as:

$$\boldsymbol{F} = \rho_e \boldsymbol{E} = -\rho_e \nabla\psi \tag{7}$$

where ρ_e is the charge density, $\boldsymbol{E}$ is the electric field within the double layer and ψ is the electric potential. Therefore, the creeping flow equation can be transformed to:

$$\mu\nabla^2 \boldsymbol{v} = \nabla p + \rho_e \nabla\psi \tag{8}$$

and is complemented by the continuity equation, Eq. (2). The electric potential is related to the charge density by the Poisson equation:

$$\nabla^2\psi = -\frac{\rho_e}{\varepsilon} \tag{9}$$

where ε is the dielectric permittivity of the medium.

On the other hand the charge density is related to the distribution of positive and negative ions by:

$$\rho = \sum_i n_i z_i e \tag{10}$$

where n_i is the number concentration and z_i the valency for i-type ion, e is the unit charge. The distribution of ions in the double layer can be found from the convective diffusion equation. For binary electrolyte:

$$\frac{\partial n_i}{\partial t} = \nabla \cdot \left(D_i \nabla n_i + n_i \frac{z_i e}{kT} D_i \nabla\psi - n_i \boldsymbol{v} \right) \tag{11}$$

where, D_i the diffusion coefficient for ions i, k is the Boltzmann constant and T the temperature in Kelvin. For the steady-state problems the time derivative at the left hand side of Eq. (11) can be neglected.

The system of Eqs. (2) and (8)–(11) together with appropriate boundary conditions provides the general framework for every hydrodynamic, electric and electrokinetic phenomena involving colloidal particles. The limiting case $\rho_e = 0$ describes the microscopic flows, i.e. the hydrodynamic interactions of uncharged particles, while the other limiting case $\boldsymbol{v} = 0$ describes the quasistatic electric double layer interactions between colloidal particles or between a colloidal particle and the surface of macroscopic collector. This second limit, i.e. $\boldsymbol{v} = 0$ is a subject of the DLVO theory. In general the hydrodynamic and electric interactions are coupled. The degree of this coupling can be evaluated from the criterion:

$$Sm = \frac{(\kappa l_{ch})^2 kTE_{ch}}{\mu\, v_{ch}\, e} \tag{12}$$

where $\kappa = \sqrt{\frac{2Ie^2}{\varepsilon kT}}$ is the Debye–Hückel parameter, $1/\kappa$ having the sense of the double layer thickness, E_{ch} is the magnitude of the electric field, l_{ch} is the characteristic range of the velocity field and v_{ch} is the average velocity. The ionic strength of the suspension I is expressed in terms of the ionic concentration as $I = \frac{1}{2}\sum_i z_i^2 n_{i\infty}$ where $n_{i\infty}$ are the ionic concentrations in the bulk.

If $Sm \to \infty$ the electric interaction prevails, when $Sm \to 0$ the problem reduces to the hydrodynamic interactions between uncharged bodies. If $Sm \approx 1$ the coupling between the hydrodynamic and electric phenomena is at its maximum.

Knowing the distribution of the fluid velocity, electric potential and ionic concentration around the particle one can find the force and torque due to

hydrodynamic and electrostatic interactions acting on the particle according to:

$$\boldsymbol{F}_p = \int_{S_p} \underline{\underline{\boldsymbol{\sigma}}} \cdot d\boldsymbol{S}_p$$

$$\boldsymbol{T}_p = \int_{S_p} \boldsymbol{r}_p \times \underline{\underline{\boldsymbol{\sigma}}} \cdot \mathrm{d}\boldsymbol{S}_p \tag{13}$$

where $\boldsymbol{r}_p$ is the position vector, $\underline{\underline{\boldsymbol{\sigma}}}$ is the stress tensor and the integration is performed over the particle surface.

The total force and torque acting on a particle close to the interface can be obtained by taking into account in Eq. (13) also other interactions present in the system like van der Waals, steric interactions due to adsorbed polymer layers, etc.

The stress tensor $\underline{\underline{\boldsymbol{\sigma}}}$ can be found as a sum of the viscous stress and Maxwell stress tensors:

$$\underline{\underline{\boldsymbol{\sigma}}} = -p\underline{\underline{\boldsymbol{I}}} - \frac{\varepsilon}{2}E^2\underline{\underline{\boldsymbol{I}}} + \varepsilon \boldsymbol{EE} + \nu(\nabla \boldsymbol{v} + \nabla \boldsymbol{v}^+) \tag{14}$$

where $\underline{\underline{\boldsymbol{I}}}$ is the unit tensor.

The information concerning the total force and torque acting on a particle during its motion close to the interface is crucial for the description of the adsorption process. This is discussed in Section 2.2.

2.2. Motion of particles

Two different concepts of the description of the motion of colloidal particles during the adsorption process can be distinguished: Lagrangian methods in which single particle motion is followed along its trajectory and Eulerian methods in which the concentration or the probability of finding particle at a given point in space is considered.

In the Lagrangian description of the particle motion the attention is focused on a single particle trajectory which can be described in terms of the second Newton law of mechanics. As it was shown above for colloidal particles suspended in fluid, when the Reynolds number based on particle dimension fulfills the condition $\mathrm{Re} \ll 1$, the inertia effects can be neglected and the particle motion is described by the creeping flow equation — Eq. (6). In the case of weak coupling between the hydrodynamic and electric effects the particle velocity is a linear function of the total force and torque acting on the particle:

$$\frac{\mathrm{d}\boldsymbol{r}_p}{dt} = \boldsymbol{U}_p = \underline{\underline{\boldsymbol{M}}}_t \cdot \boldsymbol{F}_p + \underline{\underline{\boldsymbol{M}}}_{rt} \cdot \boldsymbol{T}_p \tag{15}$$

where $\underline{\underline{\boldsymbol{M}}}_t$ and $\underline{\underline{\boldsymbol{M}}}_{tr}$ are the translational and rotational mobility matrices dependent on the position of the particle relatively to interface and other particles in the system. They can be found from the solution of the solution of the system of Eqs.

(2) and (8)–(11). The inverse of the translational mobility matrix, e.g. $\underline{\underline{\boldsymbol{R}}}_t = \underline{\underline{\boldsymbol{M}}}_t^{-1}$ is often referred to as the resistant matrix and is a measure of the viscous resistance exerted by the fluid on the moving body. Far from the interfaces the mobility and the diffusion of a particle is isotropic, i.e. the mobility and diffusion matrices assume diagonal form:

$$\underline{\underline{\boldsymbol{M}}}_t = \frac{1}{6\pi\mu a}\underline{\underline{\boldsymbol{I}}} \tag{16}$$

The total force and torque acting on a particle consist of external (e.g. gravity) and hydrodynamic forces due to the macroscopic fluid motion. Thus, in the simplest case of the spherical uncharged particle moving in the unbound fluid Eq. (15) reduces to the well known Stokes formula:

$$\boldsymbol{U}_p = f_{St}\boldsymbol{F}_p = \frac{1}{6\pi\mu a}\boldsymbol{F}_p \tag{17}$$

where a is the particle radius and thus $R_{st} = 6\pi\mu a$ is the Stokes resistance coefficient.

In the vicinity of interfaces the hydrodynamic forces and torques are modified and the mobility and diffusion matrices assume complicated and particle position dependent forms. Some examples of these hydrodynamic interactions are discussed in the subsequent sections. At the small distance to the interface, besides hydrodynamic and external forces, also specific surface interactions (e.g. electric, van der Waals, steric interactions) determine the rate of the particle motion.

Since the mobility matrix and/or resistance matrix result from the solution of the system of Eqs. (2) and (8)–(11) they depend not only on the relative positions of the particle and interface but also on their electric properties as well as on the hydrodynamic and electric properties of the dispersing medium. So in general:

$$\underline{\underline{\boldsymbol{M}}}_t = \underline{\underline{\boldsymbol{M}}}_t(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_n,\sigma_1,\ldots,\sigma_n,\mu,\kappa) \tag{18}$$

where $\boldsymbol{r}_1,\ldots,\boldsymbol{r}_n$ are positions of all objects in the system and $\sigma_1,\ldots,\sigma_n$, their surface charges. The same concerns the rotational mobility matrix. This coupling between hydrodynamic and electric effects leads to the whole class of phenomena known as the electrokinetic. The electrophoresis, elecroosmosis, and streaming potential (including sedimentation potential) belong to this class of phenomena and they are extensively studied [4]. However, the theoretical and experimental effort concerning the influence of the electrokinetic phenomena on adsorption of colloids is still far from being complete.

Integrating Eq. (15) one obtains the particle trajectory. However, the description of motion of particles in terms of this equation is limited, as it disregards the Brownian motion of particles. Therefore, it can be applied either for relatively large particles ($a > 3$ μm in water) or for smaller particles in more viscous media. There is no conceptual difficulty in incorporating the thermal motion into the

theory. The equation of motion then becomes the stochastic Langevin type equation and is the base of the Brownian Dynamics (BD) method for simulating colloidal particles adsorption [25,26]. The stochastic trajectory equation can be written in the form:

$$r_{pi}(t + \Delta t) = r_{pi}(t) + U_{pi}\Delta t + \left(\nabla \cdot \underline{\underline{\boldsymbol{D}}}\right)_i \Delta t + R_i \tag{19}$$

where U_{pi} are the components of the particle velocity vector given by Eq. (15), r_{pi} are the components of the particle position vector and R_i is a random displacement. The distribution of the random displacements is usually assumed to be normal with the mean and variance given by [26]:

$$\begin{aligned} &\langle R_i \rangle = 0 \\ &\langle R_i R_j \rangle = 2\underline{\underline{D}}_{ij}\Delta t \end{aligned} \tag{20}$$

where $\underline{\underline{D}}_{ij}$ is the element of the diffusion matrix connected with the translational mobility matrix by the Stokes–Einstein relation:

$$\underline{\underline{\boldsymbol{D}}} = kT\underline{\underline{\boldsymbol{M}}}_{tr} \tag{21}$$

The problem of predicting adsorption rates reduces to calculating the probability of capture of a particle by the interface, given a certain initial position of the particle. To obtain the statistically meaningful results the procedure of the trajectory calculation has to be repeated for several thousand particles whose initial positions are chosen at random at a certain distance to the interface.

In contrast to Lagrangian methods in which the trajectories of individual particles are followed, the Eulerian methods are concerned with obtaining the distribution of particles in space and, for non-spherical particles, also the orientational distribution. In its most general form these distributions are described by the set of coupled Fokker–Planck equations [27,28]. If the distribution of particles in the suspension is known, their flux towards the interface can be calculated and thus, the adsorption rate can be determined.

If we consider a suspension of spherical non-interacting particles suspended in a fluid, the mass balance equation (derived from the more general Fokker–Plank equation) assumes the form:

$$\begin{aligned} &\frac{\partial n_p}{\partial t} + \nabla \cdot \boldsymbol{j}_p = Q_p \\ &\boldsymbol{j}_p = -\underline{\underline{\boldsymbol{D}}} \cdot \nabla n_p + \boldsymbol{U}_p n_p = -\underline{\underline{\boldsymbol{D}}} \cdot \nabla n_p + \left(\underline{\underline{\boldsymbol{M}}}_t \cdot \boldsymbol{F}_p + \underline{\underline{\boldsymbol{M}}}_{tr} \cdot \boldsymbol{T}_p\right) n_p \end{aligned} \tag{22}$$

where n_p is the concentration of particles, $\boldsymbol{j}_p$ is the particle flux vector and Q_p is a source term. Eq. (17) is usually known as the convective–diffusion equation. For a given geometry of the collector it can be solved numerically using standard numerical techniques and under some simplifying assumption also the analytical

solutions are possible. The detailed discussion of the analytical approximate solutions of Eq. (22) can be found in Adamczyk et al. [8], van de Ven [9] and Elimelech et al. [11].

Basing on the form of Eqs. (15) and (22) describing the motion of particles in the vicinity of adsorbing interfaces, one can infer that the direct coupling between hydrodynamic interactions and electric forces inevitably occurs due to the terms $\underline{\underline{\boldsymbol{M}}}_t \cdot \boldsymbol{F}_p$ and $\underline{\underline{\boldsymbol{M}}}_{tr} \cdot \boldsymbol{T}_p$ as for charged particles the total force and torque contains the EDL interactions components. Thus one cannot consider the EDL force separately from the hydrodynamic flows.

2.3. Adsorption kinetics

Adsorption of colloidal particles occurs in two stages. In the first stage the interface is initially clean and the adsorption rate is determined by the mechanism of transport of particles to the interface and by interactions between them. As it was pointed out above, the particle flux to the interface can be determined from the solution of the convective–diffusion equation [Eq. (22)]. The adsorption kinetics, i.e. the change of the number of particle adsorbed at interface in time is then governed by:

$$\frac{\mathrm{d}n_s}{dt} = j_\perp \tag{23}$$

where n_s is particle surface concentration, i.e. the number of adsorbed particles per unit area, and $j_\perp$ is the component of the particle flux normal to the surface. At the early stages of adsorption $n_s \approx 0$, therefore we can always neglect desorption and treat adsorption as irreversible. It is convenient to introduce the dimensionless variables and to express Eq. (23) in the form:

$$\frac{\mathrm{d}\Theta}{d\tau} = 1 \tag{24}$$

where $\Theta = \dfrac{\pi a^2 N_s}{S}$ is the dimensionless surface coverage, N_s is the number of particles at a surface S, $\tau = \dfrac{t\pi a^2 j_0}{0.547}$ is the dimensionless time, j_0 is the normal component of initial particle flux to uncovered interface, being a solution of Eq. (22) at the interface and numerical factor in the denominator corresponds to the maximum coverage for the random irreversible adsorption [12].

As the adsorption process continues the lateral interactions between adsorbed and adsorbing particles start to play a role. In the simplest case of hard spheres there are only the excluded volume interactions and their consequence is that part of the interface, already taken by adsorbed particles, is inaccessible for the incoming ones. For reversible or partially reversible systems the situation is

additionally complicated by the particle desorption. For this adsorption stage the kinetics can be described by the equation [12]:

$$\frac{d\Theta}{d\tau} = B(\Theta) - \bar{K}_d\Theta \tag{25}$$

where $B(\Theta)$ is sometimes called the available surface function or the blocking parameter and is a measure of the probability of the adsorption for covered surface and $\bar{K}_d$ is the dimensionless desorption constant. For the uncovered surface, i.e. for $\Theta \to 0$, $B(\Theta) = 1$ and for the irreversible adsorption $\bar{K}_d = 0$.

The probability of particle adsorption, when the interface is partially covered, is determined by the lateral interactions between adsorbed and adsorbing particles. These interactions are very difficult to account for as they are extremely dependent on the local distribution of adsorbed particles.

In general the Eulerian methods are more suitable to describe the initial stages of the adsorption process when the coverage of the interface by adsorbed particles is small. Dąbroś and van de Ven [29] made an attempt to extend this method to account for the moderate coverage of the adsorbed particles in the convective–diffusion equation by using the concept of the mean field force. This force originating from the presence of adsorbed particles is then considered in Eq. (22) and determines the flux of adsorbing particles. The modification of the macroscopic fluid flow by the layer of adsorbed particles is treated in the similar manner.

For high coverages of the interface by adsorbed particles the Sequential Brownian Dynamics simulation method (SBD) [12,25,26] based on the Lagrangian principle of calculating particle trajectories is the only method of determining the adsorption rate. The SBD rely on following the simulated trajectory of virtual particle until it leaves the simulation area or it is deposited at the interface. At the latter case the position of the deposited particle is stored and the interaction between it and particles simulated later has to be considered. However, this method requires the detailed knowledge of the functional form of the many-body interaction force which is not yet available. The approximations are made usually assuming the superposition of two-body interactions or the interaction with the nearest neighbor only.

For some geometries of the flow other computer simulation methods of the adsorption process like the Random Sequential Adsorption (RSA) [30–32] or the Diffusional Random Sequential Adsorption (DRSA) [33,34] are applied. In the RSA method the trajectory of a particle is not followed but the probability of the adsorption at the given spot at the interface is evaluated taking into account the interactions with the neighboring particles. If the interactions favor the adsorption the particle remains adsorbed at the spot, otherwise it is removed from the system. The second method (DRSA) allows for the diffusion of the particle around the initially chosen spot after adsorption fails at the first attempt. In both methods the effect of flow, i.e. the hydrodynamic interactions are originally not taken into account.

In the absence of external flow the available surface function $B(\Theta)$ can be described in terms of the lateral interactions between particles as:

$$B(\Theta) = 1 - \frac{1}{\Delta S}\int_{\Delta S}(1 - \exp(-\phi_v/kT))\mathrm{d}S \tag{26}$$

where $\phi(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_n) > 0$ is the interaction energy of adsorbing (virtual) particles with all previously adsorbed particles at the positions r_i and the integration is performed over the area of adsorbing interface. For the non-spherical particles the interaction energy additionally depends on their mutual orientation; $\phi_v \to \infty$ when the particles overlap.

In general $B(\Theta)$ can be expressed in terms of virial expansion, i.e.:

$$B(\Theta) = 1 - C_1\Theta + C_2\Theta^2 + C_3\Theta^3 + \ldots \tag{27}$$

For the RSA adsorption of the non-interacting hard spheres the coefficients C_i in the virial expansion have a simple geometrical interpretation and can be determined analytically with the simple result [31]:

$$\begin{aligned} C_1 &= 4 \\ C_2 &= \frac{6\sqrt{3}}{\pi} \\ C_3 &= 1.407 \end{aligned} \tag{28}$$

The first coefficient C_1 has a meaning of the area blocked by a single adsorbed particle (scaled by particle cross sectional area πa^2) thus, this area is four times larger than particle geometrical cross section. For interacting particles the calculation of virial coefficient in Eq. (27) involves the evaluation of multiple integrals and it can only be performed numerically. Extensive discussion of this topic for adsorption of spherical and non-spherical particles can be found in Adamczyk et al. [12,35].

Adamczyk et al. generalized the RSA method to account for the electrostatic [17] and hydrodynamic interactions [18,36] between the adsorbing particle and one already present at the interface. They used the concept of the effective surface blocked by a single adsorbed particle. The size of this surface strongly depends on the flow intensity and the range of the electric double layer interaction and thus has to be found from the solution of Eqs. (2) and (8)–(11). Therefore, the solution of the creeping flow equation, Poisson equation and convective–diffusion equation for the concentration of ions around the interacting particles provides the necessary information on the hydrodynamic interactions and coupled with them electric interactions. The evaluation of these interactions is essential for the description of adsorption of colloidal particles using any of the methods described above.

In the next two sections two limiting cases will be discussed. First the hydrodynamic interaction between the uncharged particles will be discussed ($Sm = 0$) together with various methods of solution of the creeping flow equation. Then the electric interactions between stagnant particles will be considered and various

approximates and exact methods of the evaluation of these equations will be reviewed.

3. Hydrodynamic interactions between uncharged bodies

3.1. Particle–particle interactions

The creeping flow (Stokes) equation for the uncharged particles assumes a form:

$$\mu\nabla^2 \boldsymbol{v} = \nabla p \tag{29}$$

The Stokes equation and the continuity equation, Eq. (2) together with the appropriate boundary conditions on the particle surfaces and macroscopic interfaces form a closed system.

On the surface of the rigid particle or interface the no-slip boundary conditions are usually assumed while for liquid/liquid interface the condition of equal stress at both sides of the interface is used. The solution of the system of Eqs. (2) and (29) is the precondition to determine the hydrodynamic interactions between uncharged colloidal particles. As a first example we discuss the solution of the creeping flow equation for two uncharged particles approaching each other in the stagnant liquid. The hydrodynamic interactions between two spherical particles, although not directly involved in the colloidal particle adsorption process, determine the rate of particle aggregation (coagulation) and thus, one of the most important features of the colloidal system, its stability. Measuring the aggregation rate of particles, the theory of the hydrodynamic interaction can be easily verified.

The problem of the motion of two hard spheres along the line connecting their centers was solved by Stimson and Jeffery [37]. In order to do it they introduced the set of bispherical polar coordinates illustrated in Fig. 4 and they obtained analytic solution in terms of polynomial series. Brenner [38] showed that for particles of equal size a, the Stokes formula connecting the particle velocity and the total force acting on it has to be modified according to:

$$\boldsymbol{U}_p = \underline{\underline{\boldsymbol{M}}}_t(h) \cdot \boldsymbol{F}_p = \frac{1}{6\pi\mu a} F_s(H) \boldsymbol{F}_p \tag{30}$$

where:

$$F_s^{-1}(H) = \frac{4}{3}\sinh(\alpha) \sum_{n=1}^{\infty} \frac{n(n+1)}{(2n-1)(2n+3)} \times \left[\frac{4\cosh^2\left(n+\frac{1}{2}\right)\alpha + (2n+1)^2\sinh^2(\alpha)}{2\sinh(2n+1)\alpha - (2n+1)\sinh(2\alpha)} - 1 \right] \tag{31}$$

with $\cosh(\alpha) = \frac{H}{2} + 1$ and $H = \frac{h}{a}$ where $h = r - 2a$ is the surface-to-surface distance between particles. Therefore, according to the Stokes–Einstein relationship [Eq. (21)] the particle diffusion coefficient also has to be corrected to account for the hydrodynamic interactions and can be expressed as:

$$D(H) = D_\infty F_S(H) \tag{32}$$

where D_∞ is the particle diffusion coefficient in an infinitely dilute suspension. The dependence of the correction function F_s on the interparticle distance is illustrated in Fig. 5. Honig et al. [39] introduced approximate interpolating function which can be conveniently used:

$$F_s(H) \cong \frac{6H^2 + 4H}{6H^2 + 13H + 2} \tag{33}$$

The results of the approximate formula together with the small distance limit of Eq. (33) which can be also obtained from the lubrication theory approximation [40]:

$$F_s(H) = 2H \quad (H \to 0) \tag{34}$$

are compared in Fig. 5 with the results of exact calculations of Brenner. As it can be seen in Fig. 5 the hydrodynamic interaction between approaching particles are long ranged. At the distance of 10 particle radii the diffusion coefficient is still

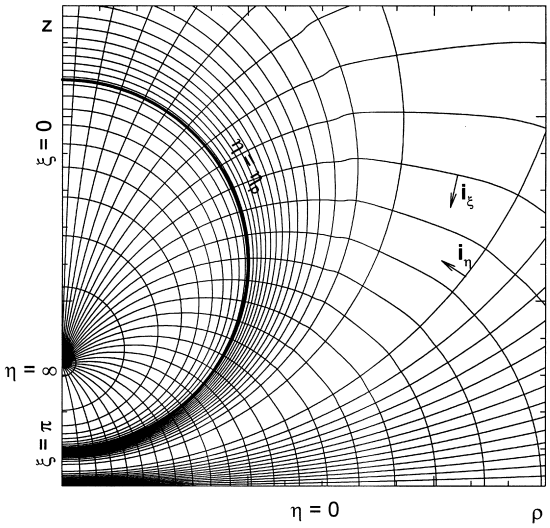


Fig. 4. Bispherical system of coordinates (η,ζ,ϕ) used for the analytical solution of hydrodynamic and/or electrostatic problems in sphere–sphere and sphere–plane geometry. The cylindrical coordinates are expressed in terms of bispherical ones as: $\rho = \sin\xi/(\cosh\eta - \cos\xi)$, $\phi = \phi$, $z = \sinh\eta/(\cosh\eta - \cos\xi)$.

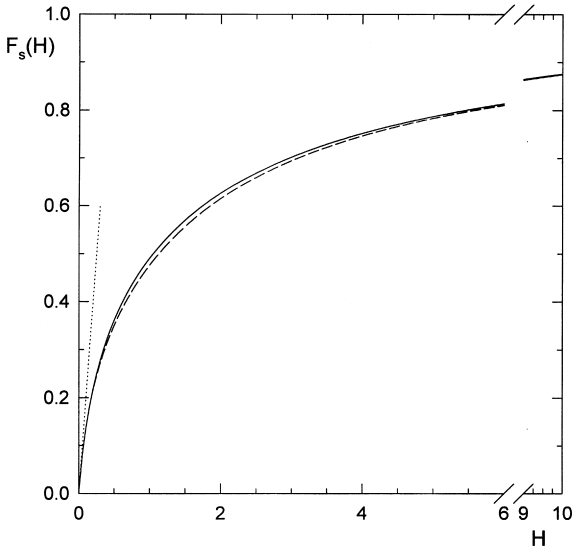


Fig. 5. The dependence of the hydrodynamic correction function for the mobility of the solid sphere approaching the identical one on the distance between them. Solid line Eq. (31), dashed line Eq. (33) and dotted line lubrication approximation, Eq. (34).

$< 90\%$ of its bulk value. While the approximate expression of Honig et al. provides very good approximation to the exact calculations at all distances between particles, the lubrication theory approximation can be applied when the distance does not exceed 10% of particle radius.

Derjaguin and Muller [5], Spielman [6] and Honig et al [39] were the first who considered the influence of the hydrodynamic interactions on the aggregation rate of colloidal particles. The convective–diffusion equation governing the aggregation can be written in a form:

$$\frac{\mathrm{d}}{dH}\left[(2+H)^2 F_s(H)\left(\frac{\mathrm{d}n_p}{dH} + n_p\frac{\mathrm{d}\Phi_s}{dH}\right)\right] = 0 \tag{35}$$

where Φ_s is the interparticle interaction energy (in kT units). Eq. (35) can readily be solved and the steady-state flux J_{p0} of particles towards a central particle, being the measure of the aggregation rate, can be computed according to:

$$J_{p0} = \frac{8\pi n_b D_\infty a}{\displaystyle\int_0^\infty \frac{\exp(\Phi_s(H))}{F_s(H)(2+H)^2}} \tag{36}$$

where n_b is the particle bulk concentration. If we consider only the case of fast aggregation, i.e. there is no repulsive interactions of any kind between particles, the

only contribution to Φ_s comes from the van der Waals forces. They can be evaluated using the formulae derived by Schenkel and Kitchener [41]:

$$\Phi_s(H) = -\frac{A}{12H}\left[\frac{1}{1+1.77p_0}\right] \quad \text{for } 0 < p_0 < 2$$

$$\Phi_s(H) = -\frac{A}{H}\left[\frac{2.45}{60p_0} - \frac{2.17}{180p_0^2} + \frac{0.59}{420p_0^2}\right] \quad \text{for } 0.5 < p_0 \tag{37}$$

where A is the Hamaker constant (in kT units), $p_0 = 2\pi H\frac{a}{\lambda}$ and λ is the wavelength corresponding to the characteristic electronic oscillation of atoms. If in Eq. (36) both hydrodynamic and van der Waals interactions are neglected, it reduces to the Smoluchowski formula for the fast aggregation rate:

$$J_{Sm} = 16\pi\, n_b\, D_\infty\, a \tag{38}$$

Fig. 6 presents comparison of the dependence of the fast aggregation rate (scaled vs. Smoluchowski rate) on the value of the Hamaker constant for two different particle radii. The hydrodynamic interactions are taken into account (solid lines) and omitted (dashed lines) in the calculations. Results presented in Fig. 6 suggest that the hydrodynamic interaction provides significant retardation to the aggrega-

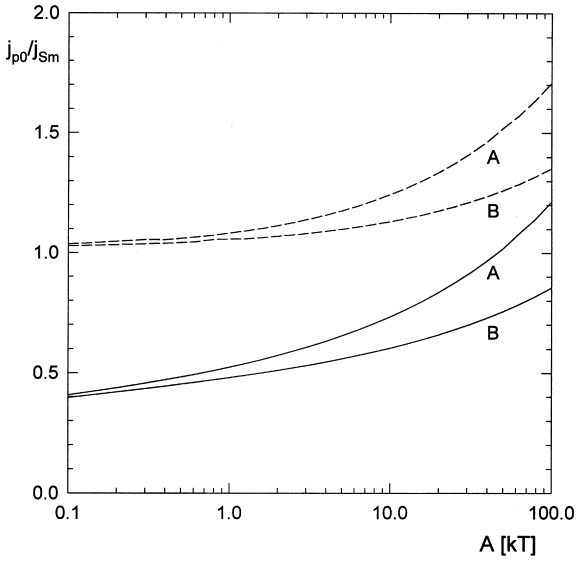


Fig. 6. Dependence of the fast aggregation rated normalized by the Smoluchowski aggregation rate on the value of Hamaker constant calculated, taking into account the hydrodynamic interaction (solid lines) and neglecting them (dashed lines). Curves A for a = 100 nm, curves B = 500 nm.

tion decreasing its rate approximately by half. Indeed, the experimentally observed aggregation rates are always below the Smoluchowski rate (with exception of extremely high Hamaker constants for metallic particles) [42,43].

3.2. Particle–wall interactions

Because of its relevance to practical applications the hydrodynamic problem of the uncharged spherical particle moving close to the solid surface has been a subject of numerous papers. Maude [44] and Bart [45] considered the problem of a sphere approaching rigid plane interface, O'Neill solved the problem of the spherical particle sliding (moving without rotation) along the rigid plane [46,47], Dean and O'Neill gave the solution for the problem of the particle rotating at the rigid wall [48], while Goren and O'Neill solved the general hydrodynamic problem of the particle moving at the rigid wall in a flow which velocity is of the second degree in the local coordinates [49]. The stagnation point flow, and the simple shear flow are the special cases of this general flow field. In all these papers the bispherical polar coordinates were applied and the fluid velocity fields were obtained analytically in terms of orthogonal functions expansions. These expansions are, however, slow converging if the distance between a particle and a rigid wall approaches zero. Therefore, Goldman, Cox and Brenner studied the problem of the spherical particle moving at the rigid wall in the lubrication approximation valid for the distances $< 20\%$ of particle radius [50–52].

The velocity of the spherical particle moving in the stagnant fluid perpendicularly to the rigid wall can be obtained from:

$$U_{p\perp} = \frac{F_1(H)}{6\pi\mu a} F_{p\perp} = \frac{1}{R_{St}\lambda_{\perp}(H)} F_{p\perp} \tag{39}$$

where $F_1(H)$ and $\lambda_{\perp}(H)$ are the correction function for the mobility and the Stokes resistance coefficient respectively, H is the width of the particle–wall gap normalized by particle radius and $F_{p\perp}$ is the component of the force causing particle motion directed perpendicularly to the wall and equal to the viscous drag force. The correction functions can be computed from [44,45]:

$$\lambda_{\perp}(H) = \frac{1}{F_1(H)} = \frac{4}{3}\sinh(\alpha) \sum_{n=1}^{\infty} \frac{n(n+1)}{(2n-1)(2n+3)} \left[\frac{Y_n}{T_n} - 1 \right] \tag{40}$$

where $\alpha = \cosh^{-1}(H)$ and

$$Y_n = 2\sinh(2n+1)\alpha + (2n+1)\sinh 2\alpha$$
$$T_n = 4\sinh^2\left(n + \frac{1}{2}\right)\alpha - (2n+1)^2\sinh^2\alpha \tag{41}$$

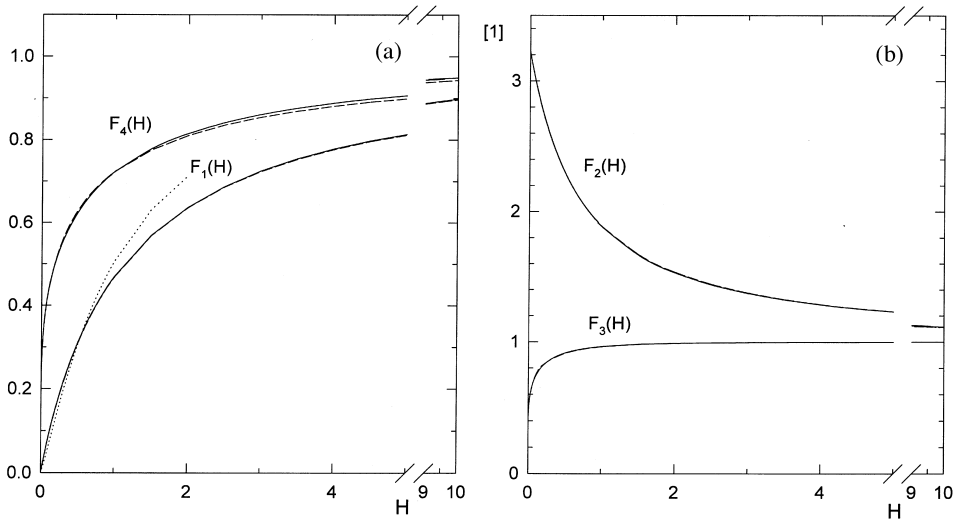


Fig. 7. The hydrodynamic correction functions for the spherical particle in the vicinity of the solid wall. Part a — the correction for a mobility of a freely moving particle perpendicular $F_1(H)$ and parallel $F_4(H)$ to the wall. Part b — the correction for the particle velocity moving towards the wall in the stagnation point flow $F_3(H)$ and along the wall in the simple shear flow $F_2(H)$. Solid lines, the results of exact solution of the creeping flow equation in the bispherical coordinates, dashed curves — the respective interpolation formulae (see text) and dotted line the result of the lubrication approximation.

The convenient approximation to Eq. (40) stems from the lubrication approximation for $H < 1$ [52]:

$$F_1(H) = \frac{H}{\left(1 + 0.971H - \frac{1}{5}\ln H\right)} \tag{42}$$

while the interpolation function:

$$F_1(H) = \frac{19H^2 + 4H}{19H^2 + 26H + 4} \tag{43}$$

gives a good approximation to the exact formula for the entire range of the gap width. The correction functions $F_1(H)$ calculated using Eqs. (40), (42) and (43) are compared in Fig. 7a.

The velocity of spherical particle moving in the stagnant fluid parallel to the rigid wall can be obtained from:

$$U_{p\|} = \frac{F_4(H)}{R_{st}} F_{p\|} = \frac{1}{R_{st}\lambda_{\|}(H)} F_{p\|} \tag{44}$$

where $F_4(H)$ and $\lambda_{\|}(H)$ are the correction function for the mobility and the Stokes resistance coefficient respectively and $F_{p\|}$ is the component of the force causing particle motion directed perpendicularly to the wall. The correction function $F_4(H)$ calculated by Goren and O'Neill from the solution of the creeping flow equation in the bispherical polar coordinates is illustrated in Fig. 7a together with the interpolating function:

$$F_4(H) = \frac{1}{1.062 - 0.516\ln(H)} \quad H < 0.11$$
$$F_4(H) = \left(\frac{H}{2.639 + H}\right)^{1/4} \quad H \geq 0.11 \tag{45}$$

The theoretical expressions describing the mobility of the particle close to a solid wall were verified experimentally by Adamczyk et al. [53] and Małysa and van de Ven [54] using multiple frame photography technique. In Fig. 8 their experimental results are compared with the results of theoretical calculations suggesting a very good agreement between them. Fig. 8a additionally illustrates that the reason for the increased hydrodynamic drag in the presence of macroscopic interfaces is viscous resistance to the fluid motion exerted by the solid interface. When the solid interface is replaced by the totally mobile free interface (e.g. clean air/water interface) the resistance is much smaller. For example for $H \sim 0.05$ the resistance for the motion towards the solid wall is 10 times higher than the resistance for the motion to the free interface.

As it is compared in Figs. 7 and 8 the correction to the mobility for the movement perpendicular to the wall is much bigger than for the motion parallel to the wall. During the motion towards the rigid wall the liquid is squeezed out from the gap between particle and interface and that requires much more energy than dragging the fluid by the particle near the rigid wall.

Since colloidal particle adsorption usually occurs in the presence of macroscopic flow it is important to describe also the influence of the hydrodynamic interactions on trajectory of particle moving in the external flow. When a freely moving particle is subjected to a flow it follows the streamlines and the particle velocity is equal to the local fluid velocity. Due to the presence of the rigid interface the particle lags the fluid flow, i.e. the particle velocity differs from the local fluid velocity at the same point. As it was mentioned before, the fluid flow can be decomposed in the local Cartesian coordinates into the stagnation flow and simple shear flow, therefore, it is enough to discuss motion of particles in these flow fields.

In the stagnation flow if $v_{\perp}$ is the component of the fluid velocity perpendicular to the rigid wall, then the corresponding component of the particle velocity $u_{\perp}$, can be expressed as:

$$u_{\perp} = F_1(H)F_2(H)v_{\perp} \tag{46}$$

The correction function $F_2(H)$ was obtained from the analytical solution of the

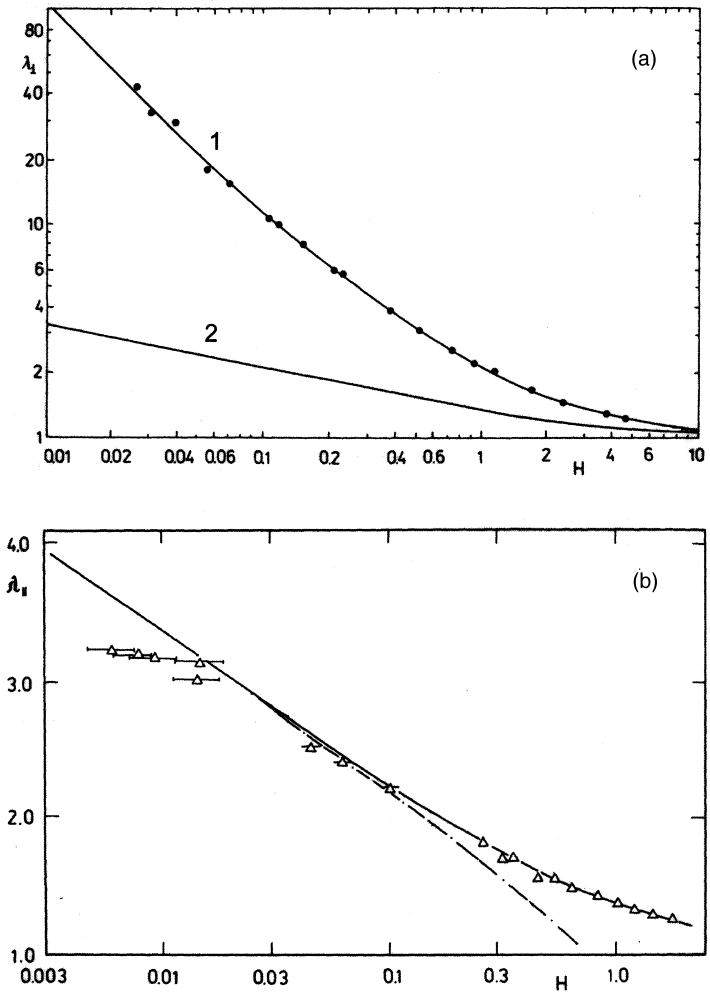


Fig. 8. The comparison between experimentally determined and theoretically calculated, from the Stokes equation, the resistance coefficient for the motion of spherical particle in the vicinity of the interface (solid lines). Part a, curve 1, for the motion towards the rigid wall; curve 2, for the motion towards free interface. Experimental data from Adamczyk et al. [53] shown as solid circles. Part b, motion along the rigid wall; triangles represent the experimental resistance coefficient given in Małysa and van de Ven [54] and dash–dotted line the lubrication approximation.

creeping flow equation by Goren and O'Neill [49] and its dependence on the particle–wall is shown in Fig. 7b. The $F_2(H)$ can be well approximated by:

$$F_2(H) = 1 + \frac{1.79}{(0.828 + H)^{1.167}} \tag{47}$$

The interactions with the wall starts to influence the particle motion at distances $H < 2$ and for example at $H \cong 0.25$ the particle velocity is two times lower than the local fluid velocity. Particles moving parallel to the wall also rotate with the angular velocity being a function of the particle–wall gap. The rotational velocity increases with decreasing gap width.

In the simple shear flow particle-velocity is a function of the shear rate G and a distance to the wall H:

$$u_{\parallel} = F_3(H)v_{\parallel} = F_3(H)Ga(1 + H) \tag{48}$$

where the dependence of the correction function $F_3(H)$ on the particle–wall separation determined by Goren and O'Neill [49] is illustrated in Fig. 7b. It can be well approximated by the interpolation function:

$$\begin{aligned} F_3(H) &= \frac{1}{0.754 - 0.256\ln(H)} \quad H < 0.137 \\ F_3(H) &= 1 - \frac{0.304}{(1 + H)^3} \quad H \geq 0.137 \end{aligned} \tag{49}$$

The relationship between particle distance from the wall and its velocity in the simple shear flow [Eq. (48)] can be used for the experimental determination of the position of the particle relatively to the interface. Having measured particle velocity in the simple shear flow of the controlled intensity with the optical microscope, one can invert Eq. (48) and obtain particle–wall separation. Several authors have used this technique to study motion of colloidal particles at interfaces under the influence of the electric, electrokinetic, van der Waals, gravitational and steric forces [55–58].

To account for the hydrodynamic interactions in the description of the process of adsorption of colloidal particles one has to apply the functional form of the hydrodynamic correction functions described by Eqs. (39)–(43) and (45)–(49) in the equations of the motion of colloidal particles near the interfaces, i.e. Eq. (15) and/or Eq. (22). As the illustration of this procedure the convective–diffusion equation for the concentration of particles in the stagnation point (or stagnation line) flows, presented in the previous section, is discussed. Introducing Eq. (3) or Eq. (4) for the macroscopic flow field and Eqs. (39)–(43) and (45)–(49) to account for the hydrodynamic interactions in the convective–diffusion Eq. (22) one obtains in the steady-state [12,23]:

$$\begin{aligned} &\frac{\partial}{\partial H}\left\{F_1(H)\left[\frac{\partial \bar{n}_p}{\partial H} + \frac{1}{2}PeF_2(H)(H+1)^2\bar{n}_p + \bar{F}_{\text{tot}}(H)\bar{n}_p\right]\right. \\ &\qquad \left. -PeF_3(H)(H+1)\bar{n}_p\right\} = 0 \end{aligned} \tag{50}$$

where $\bar{n}_p$ is the dimensionless particle concentration scaled vs. its bulk value n_b, $\bar{F}_{\text{tot}}(H)$ is the total force expressed in units kT/a, consisting of external and specific surface forces and Pe is the dimensionless Peclet number characterizing the mutual relation of convection and diffusion effects. For the impinging jet or slot geometry given by $Pe = 2\alpha_{r,s}(Re)\dfrac{a^3 V_m}{D_\infty(r,d)^2}$ where the flow parameters α_r, for the impinging jet and α_s for the slot impinging jet geometry are defined in Eqs. (3) and (4), respectively. The component of the particle flux normal to the interface is given by:

$$j_\perp = F_1(H)\left[\frac{\partial \bar{n}_p}{\partial H} + \frac{1}{2}PeF_2(H)(H+1)^2\bar{n}_p + \bar{F}_{tot}(H)\bar{n}_p\right] \tag{51}$$

As the boundary condition for Eq. (50) one usually assumes the perfect sink boundary condition $\bar{n}_p = 0$ at the interface (i.e. for $H = 0$) and $\bar{n}_p = 1$ in the bulk for $H \to \infty$.

One can clearly see that the convective–diffusion equation in the impinging jet (slot) geometry does not contain the coordinate parallel to the interface, thus, the particle transfer to the interface does not depend on the location in the neighborhood of the stagnation point (or line) and therefore, the interface is uniformly accessible for particle adsorption. This simplifies the theoretical analysis and makes the experimental cells based on the impinging jet or slot impinging jet geometry particulary useful for experimental verification of theoretical models of the adsorption [12–23]. However, as calculations of the flow field in the impinging jet (slot) cell show [23], the condition of the uniform accessibility is only fulfilled close to the stagnation point (line), usually for $r/r_c = x/d \leq 0.3$. The area of this region diminishes as Re increases.

Solving Eq. (50) using standard numerical methods one can readily obtain the particle flux being a measure of the initial particle adsorption rate. Fig. 9 illustrates the dependence of the dimensionless particle flux, often referred to as the Sherwood number $Sh = \dfrac{j_\perp a}{D_\infty n_b}$, on the Peclet number with, and without the hydrodynamic interactions taken into account, calculated for various particle sizes ranging from $a = 50$ nm to $a = 1$ μm. In the calculations of the particle flux it was assumed that particles interact with smooth rigid interface only via van der Waals interactions. The dependence of the energy of the van der Waals interactions on particle–wall separation was approximated by [59,60]:
2

$$\Phi_d = -\frac{Ad}{H}\frac{\bar{\lambda}}{\bar{\lambda} + 11.11H} \quad \text{for } Ha < 100 \text{ nm} \tag{52}$$

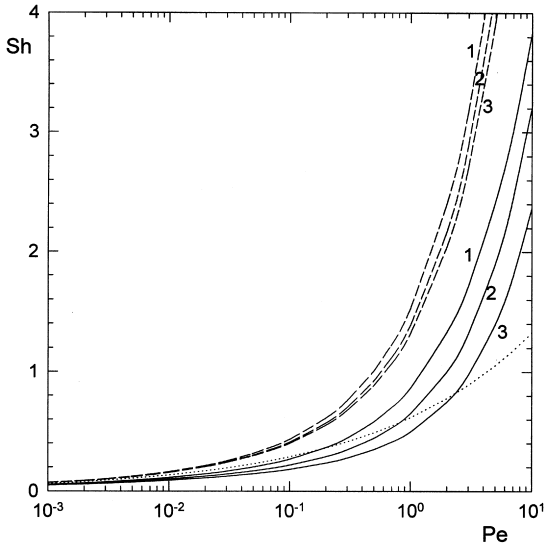


Fig. 9. The normalized particle flux being the solution of the convective–diffusion equation for the stagnation point/line flow [Eq. (50)] shown as a function of the Peclet number, calculated with (solid lines) and without (dashed lines) the hydrodynamic interactions taken into account. Curves: (1) for $a = 1$ μm; (2) for $a = 500$ nm; and (3) for $a = 50$ nm; $Ad = 0.4$. Dotted line represents the Smoluchowski–Levich approximation, Eq. (54).

and

$$\Phi_d = Ad\left[\frac{2.45\bar{\lambda}}{10\pi}\left(\frac{H-1}{H^2} - \frac{H+3}{(H+2)^2}\right) - \frac{2.17\bar{\lambda}^2}{120\pi^2}\left(\frac{H-2}{H^3} - \frac{H+4}{(H+2)^3}\right) + \frac{0.59\bar{\lambda}^3}{840\pi^3}\left(\frac{H-3}{H^4} - \frac{H+5}{(H+2)^4}\right)\right] \quad \text{for } Ha > 50 \text{ nm} \tag{53}$$

where $Ad = A/6\ kT$ and $\bar{\lambda} = \lambda/a$. The value $A = 10^{-20}$ J ($Ad = 0.4$) was assumed for Hamaker constant and $\lambda = 100$ nm.

The results are compared with the analytical result:

$$Sh = 0.616Pe^{1/3} \tag{54}$$

obtained by applying the Smoluchowski–Levich approximation for point-like particles, i.e. assuming that the effect of the hydrodynamic interactions is balanced by van der Waals interactions. As it can be seen from the results presented in Fig. 9 the hydrodynamic interaction leads to a significant retardation of the colloidal particle adsorption. For low values of the Peclet number all curves tend asymptotically to the one predicted by the Smoluchowski–Levich approximation. For high *Pe* numbers the particle flux to the interface increases rapidly mainly due to the interception mechanism of adsorption of colloidal particles of finite size [17,61].

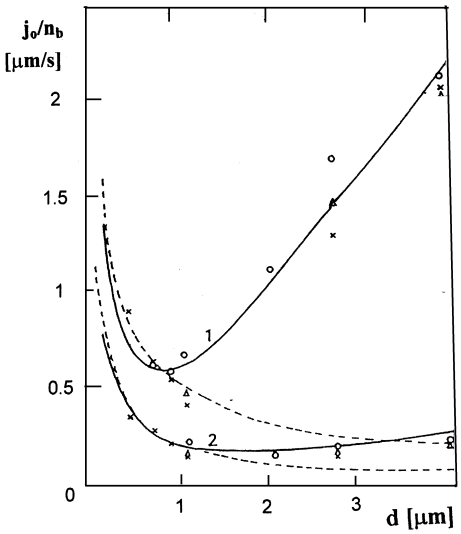


Fig. 10. The normalized initial adsorption flux as a function of particle size. The points represent the experimental results obtained in the impinging jet cell using monodisperse latex suspensions with high salt concentration (10^{-3} M). The solid lines represent the numerical solution of Eq. (50), the dashed lines are the results derived using the Smoluchowski–Levich approximation. (1) $Re = 150$, (2) $Re = 30$.

The convective–diffusion theory of the colloidal particle adsorption which takes into account the hydrodynamic interactions is well confirmed experimentally. Fig. 10 shows the comparison of the dependence of the adsorption rate expressed in terms of the particle flux normalized by the bulk concentration of particles on the particle diameter $d = 2a$ for various flow intensities in the impinging jet experimental cell. The points illustrate the experimental results for the adsorption of monodispersed polystyrene latex particles [14] and solid lines represent the theoretical predictions from the solution of the convective–diffusion equation, Eq. (50). The experiments were performed under high salt concentration (10^{-3} M) when the electric double layer interactions are screened and particles can be treated, to a good approximation, as neutral. The adsorption was totally irreversible since no particle desorption was observed even under the intense flow. As Fig. 10 suggests the agreement of the experimental data with the theoretical predictions is very good. The Smoluchowski–Levich approximation (dashed lines) correctly describes adsorption of small particles $a < 0.5$ μm. For large particles and high Re (intense flows), when the adsorption rate is determined by the hydrodynamic interception effects, the Smoluchowski–Levich approximation completely fails. Under these conditions the experimental results for the deposition of model colloidal particles in the impinging jet cell [12] shown in Fig. 10 can be quantitatively described by the full convective–diffusion theory provided that the hydrodynamic forces are taken into account.

If the motion of colloidal particles is characterized by higher Reynolds numbers

it cannot be described by the creeping flow equation. As it was pointed out in Małysa and Warszyński [62] the behavior of colloidal particles at interfaces under the condition of intense flows is to a much lesser extent determined by the hydrodynamic interactions.

3.3. Interactions of two particles at the interface

At the initial stage of the deposition process, when the surface is free of attached particles, the adsorption rate is determined only by a particle–interface interaction. During the course of the adsorption process more and more particles can attach to the interface. When the coverage of the interface by deposited particles increases, one has to take into account also the interactions of moving particles with the attached ones as well as the influence of the deposited particles on the flow field which deviates from the original one for empty surface. On the other hand the flowing particles can exert the influence on the adsorbed ones leading to their desorption. Therefore, the proper description of the hydrodynamic interaction of moving colloidal particles with ones already deposited at the interface is of importance for the quantitative analysis of the adsorption process.

Due to the complicated geometry the solution of the creeping flow equation for the problem of many particles at interface is prohibitively complex and has been obtained only for the case of two particles by applying elaborate numerical techniques. In general the solution of the creeping flow (Stokes) equation can be expressed in the integral form as:

$$\boldsymbol{v}(\boldsymbol{x}) = \boldsymbol{v}_0(\boldsymbol{x}) + \sum_{l=1}^{2} \int_{S_l} \underline{\underline{\boldsymbol{G}}}(\boldsymbol{x},\boldsymbol{y}_l) \cdot \boldsymbol{f}(\boldsymbol{y}_l) \mathrm{d}S_l \tag{55}$$

where $\boldsymbol{v}_0(\boldsymbol{x})$ denotes the fluid flow field at a given point in the absence of the particles fulfilling the no-slip boundary condition ($\boldsymbol{v}_0 = 0$) at the interface, $\boldsymbol{y}$ is a vector pointing at the surface element $\mathrm{d}S_l$, $\underline{\underline{\boldsymbol{G}}}(\boldsymbol{x},\boldsymbol{y})$ is the Oseen tensor accounting for the presence of the rigid wall [63], $\boldsymbol{f}(y_l)$ is the force density at a point $\boldsymbol{y}_l$ at the $l(l = 1, 2)$ particle surface and S_l is the surface of particle l.

The total force $\boldsymbol{F}_l$ and torque $\boldsymbol{T}_l$ exerted by particle l on the fluid are given by the equations:

$$\boldsymbol{F}_l = \int_{S_l} \boldsymbol{f}(\boldsymbol{y}_l) \mathrm{d}S_l \tag{56}$$

and

$$\boldsymbol{T}_l = \int_{S_l} (\boldsymbol{y} - \boldsymbol{R}_l) \times \boldsymbol{f}(\boldsymbol{y}_l) \mathrm{d}S_l \tag{57}$$

where $\boldsymbol{R}_l$ specifies the position of the center of particle l.

In order to obtain a closed system of equations one has to take into account boundary conditions at the particle surface:

$$\boldsymbol{v}(\boldsymbol{y}_l) = \boldsymbol{u}_l + \boldsymbol{\Omega}_l \times (\boldsymbol{y}_l - \boldsymbol{R}_l) \tag{58}$$

where $\boldsymbol{u}_l$ and $\boldsymbol{\Omega}_l$ are the translational and rotational velocities of the particle l, respectively. For the particle firmly adsorbed at the interface both velocities are equal to zero.

Eqs. (55)–(58) form a closed system, provided that the external force and torque or the velocities of the particles as well as the external flow field $\boldsymbol{v}_0(x)$ are specified.

Dąbroś and van de Ven [64,65] developed two convenient methods of the numerical solution of the system of Eqs. (55)–(58) for the problem of two spherical particles at a rigid interface. In the first, the boundary element method (BEM), the surface of each particle is divided into a finite number of triangular elements. The surface integrals can be replaced by the sums of the integrals over the elements [64]. In the second method the particle surfaces are divided into a set of subunits and the surface integration in Eqs. (55)–(57) can be replaced by summation over these subunits. A more detailed discussion of the subunit method with the application to the problem of the collision of two colloidal particles at the rigid interface can be found in Dąbroś and van de Ven [66]. The multisubunit method together with analytical formulae stemming from the lubrication approximation for close particle–particle and particle–wall separations [66] were extensively applied to obtain theoretical particle trajectories presented in this paper.

The suitability of the creeping flow theory to the surface collisions of two uncharged spherical particles was confirmed by experiment. Fig. 11 presents experimental results for the trajectory $Z = f(Y)$ of a sphere sedimenting along the wall past a second sphere attached to the wall [67]. It can be seen that during its motion along the interface the moving particle is repelled by the sitting one due to the hydrodynamic interactions. When the initial particle–wall gap width is greater then the particle diameter ($Z_i > 2$) the trajectory of the moving particle is symmetrical, i.e. there is only the hydrodynamic interaction between the moving and sitting particle. When the mobile particle is initially closer to the wall the trajectory becomes non-symmetrical due to the presence of particle–wall hydrodynamic interactions and particle–particle hard sphere repulsion. That result agrees well with the calculations performed according to method of Dąbroś and van de Ven represented by solid lines in Fig. 11. The calculations were done by applying the Lagrangian method of determining the trajectory of the moving particle. Only hydrodynamic and hard sphere interactions were taken into account in the computations. Further analysis of this result is reported in Fig. 12 where the trajectories of colloidal particle colliding in the simple shear flow with the adsorbed particle are presented. Far from the attached particle the moving one flows parallel to the interface along the line connecting their centers. Similarly as in Fig. 11, if before the collision the distance between the moving particle and the wall is > 2.2 (scaled vs. particle radius) the trajectory is symmetrical. For closer initial distances the

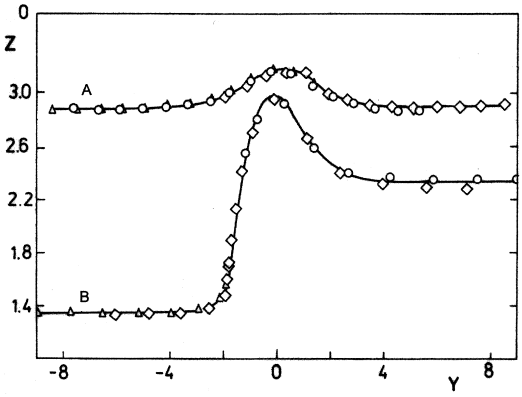


Fig. 11. The experimentally measured trajectory of the particle moving along the rigid wall and interacting with the second particle firmly attached to the wall at (0,0,1) [67]. Solid lines represent the trajectory obtained from the solution of the creeping flow equation using the method of Dąbroś and van de Ven [66] for two initial positions of moving particle relatively to the wall, curve A for $Z_i = 2.9$ and curve B for $Z_i = 1.36$. The distance scaled vs. particle radius.

trajectory becomes non-symmetrical, the final distance Z_f after the collision being common for any initial distance $Z_i = z_i/a < 2.2$. In the trajectories presented in Fig. 13 the initial distance from the interface is $Z_i = 1.6$ and the lateral coordinate $X_i = x_i/a$ of the starting position ranges from -2 to 2 and $Y_i = y_i/a = -8$. It can be seen that the semicircular scattering pattern is formed by the final positions ($Y_f = 8$) of moving particles. This characteristic pattern is shown in more detail in Fig. 14a. Here 200 initial positions (white circles) of the virtual particle moving in the simple shear flow were randomly chosen within the semicircle having radius $R = 2$ which origin was situated at $(0, -8, 1)$. The adsorbed particle, which contour is marked by the dotted line, was located at (0,0,1). The black circles denote final positions (at $Y_f = 8$) of the virtual particles. It can be seen that some of them form almost semicircular collision pattern which practically coincides with the contour of the attached particle. The initial position of particles starting outside this pattern exactly overlapped with the final position (symmetric trajectories) while the final position of particles starting inside are located at the collision pattern (non-symmetrical trajectories).

The Brownian motion of the moving particle causes the collision scattering pattern to become diffuse but for a sufficiently high Peclet number the probability of finding the final position of the moving particle just behind the sitting one, i.e. inside the collision pattern is substantially decreased. It is illustrated in Fig. 14b where are shown the initial and final positions of 200 virtual particle trajectories colliding with the adsorbed one in the simple shear flow. The BD simulation method, developed according to Eq. (19) and similar to one used in Ermak and McCammon [26] was applied to obtain the virtual particle trajectories.

The pronounced shift of the particle trajectory out of the interface after the

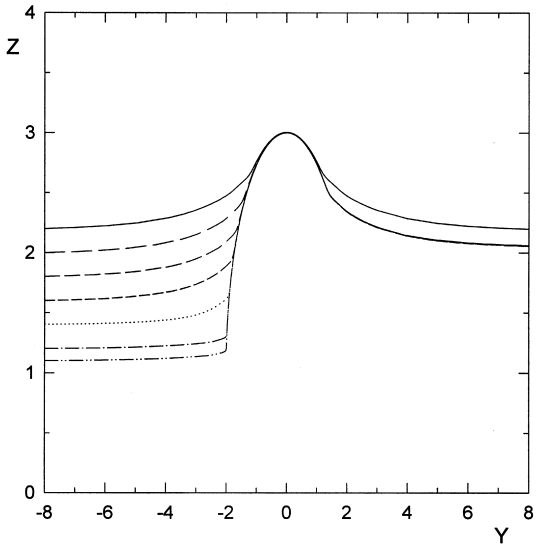


Fig. 12. The set of trajectories of a particle moving in the simple shear flow around the second one attached to the wall at (0,0,1). Motion along the axis connecting the particle centers. The initial distance of moving particle to the wall ranges from $Z_i = 2.2$ to $Z_i = 1.1$. Trajectories obtained from the solution of Eq. (15).

collision with the particle adsorbed at the interface occurs in any type of flow with the component parallel to the interface. Fig. 15 illustrates the trajectories in the slot impinging jet flow. The position of the adsorbed particle is (0,20,1), i.e. it is located 20 particle radii from the symmetry line. Since this type of flow possesses the component directed towards the interface [cf. Eq. (4)], the flow induced particle adsorption is possible. As it is shown in Fig. 15, particles following the trajectories close to the wall before the collision are considerably shifted out of interface after the collision. Thus, the presence of pre-adsorbed particle causes that the interface is no longer uniformly accessible. Particles cannot adsorb immediately behind the sitting one. It is better illustrated in Fig. 16. The final positions of the 2000 moving virtual particles, i.e. the (X,Y) coordinates at the interfaces where the particles where adsorbed are shown as a black dot. The location of the center of the immobile particle, shown as a solid circle, was at (0,10,1). The initial positions of flowing particles were chosen at random at the distance $Z_i = 10$ from the interface. Only the hydrodynamic and hard sphere interactions between particles were taken into account in the calculations of trajectories. The van der Waals attraction between flowing particles and the wall was calculated according to Eqs. (52) and (53) for $Ad = 0.4$. Further details of calculations are given in Warszyński [68]. As it can be seen in Fig. 16, due to the presence of the particle attached to the wall, part of it is not accessible for particle adsorption. In the absence of flow this excluded area is circular (dashed circle) and equals $4\pi a^2$[31]. In the presence of a flow, additionally the hydrodynamic shadow appears behind the attached particle. The

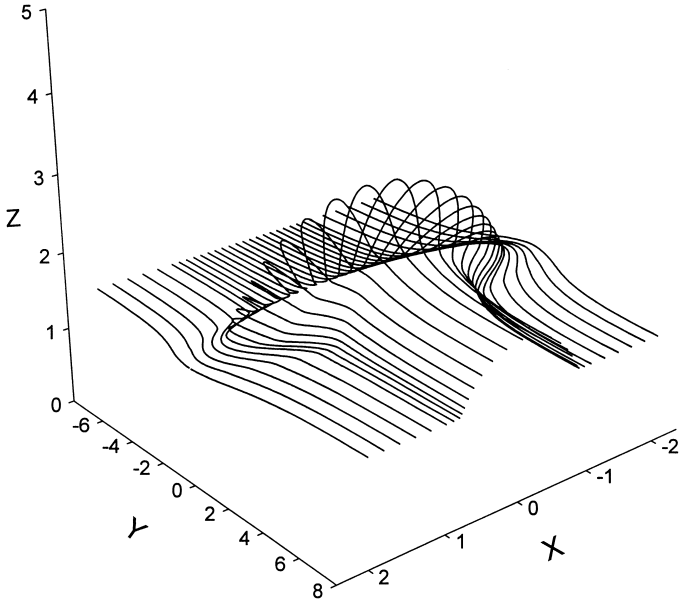


Fig. 13. The collision pattern formed by the trajectories of particles moving in the simple shear flow at the rigid wall and colliding with the particle firmly attached to it. The starting distance to the wall $Z_i = 1.6$ and the lateral position ranges from $X_i = -2$ to $X_i = 2$ and $Y_i = -8$.

area of this shadow is dependent on the local flow as well as on particle-wall interactions. In the case of stagnation point (line) flow the size of the hydrodynamic shadow depends on the local shear rate and the ratio of normal and tangential fluid flow components, i.e. on the distance from the stagnation point (line). It is illustrated in Fig. 17 which shows the hydrodynamic shadow for various flow intensities expressed in terms of the Peclet number, formed behind the adsorbed particle located at $Y_d = 10$. The calculations were performed for the stagnation line flow [Eq. (4)] and the stagnation line was situated at $Y = 0$. The length of the hydrodynamic shadow strongly increases with the flow intensity (Pe number). The increase is far more pronounced if the adsorbed particle is located far from the stagnation line since the local shear rate $G = 2\alpha_0 YdV_m$ increases linearly with the distance. The dependence of the area blocked by adsorbed particles in the stagnation line flow C_{1h}, scaled vs. the area blocked in the absence of flow C_1, on the Pe number is shown in Fig. 18. Symbols denote the results of the numerical calculations using the particle trajectory equation, Eq. (15), for various locations of the deposited particle Y_d. Solid lines represent the dependence $C_{1h}/C_1 = 1 + a_c Pe^n$ used for interpolation. The van der Waals attraction between flowing particles and the wall was calculated according to Eqs. (52) and (53) for $Ad = 0.4$ while only hydrodynamic and hard sphere interactions were considered between flowing and deposited particle. It appeared that for $Pe > 0.1$ for all locations of the deposited particle Y_d (provided $Y_d > 5$) the satisfactory fit was

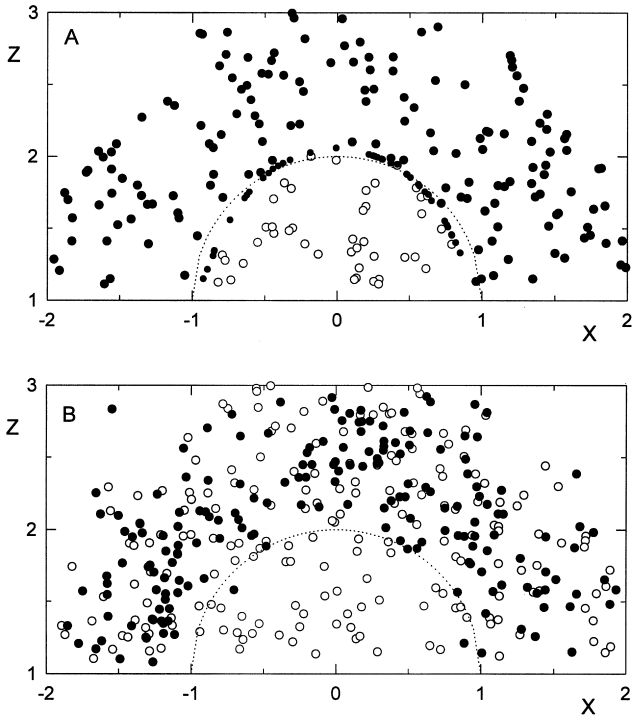


Fig. 14. The collision pattern formed by the final positions of 1000 virtual particles colliding in the simple shear flow with the particle attached to the interface at (0,0,1). Open circle represents the initial positions of virtual particles chosen at random, solid circles their final positions. The contour of the immobile particle is shown as a dotted line. Part a — collisions of non-Brownian particles, part b — collisions of Brownian particles with $Pe = Ga^2/D_\infty = 150$.

obtained for exponent $n \approx 0.67$ while the coefficient a_c was proportional to the distance of the adsorbed particle to the stagnation line.

The size of the hydrodynamic shadow also depends on the interaction between the solid interface and flowing particles. This is illustrated in Fig. 19 where the dependence of the area blocked by adsorbed particle in the stagnation line flow C_{1h}, scaled vs. C_1, on the Pe number is shown for various values of the dimensionless Hamaker constant Ad. Again, symbols denote the results of the numerical calculations using the particle trajectory equation and solid lines represent the dependence $C_{1h}/C_1 = 1 + a_c Pe^n$ used for interpolation. Strong dependence of the size of the hydrodynamic shadow on the magnitude of attractive forces between particles and interface can be observed. The weaker the attraction is, the more effective the surface is blocked by adsorbed particles. The coefficient a_c of the interpolation function is roughly inversely proportional to Ad while the exponent $n \approx 0.67$ is independent of Ad.

The Brownian motion of the flowing particles decrease the effect of the hydrody-

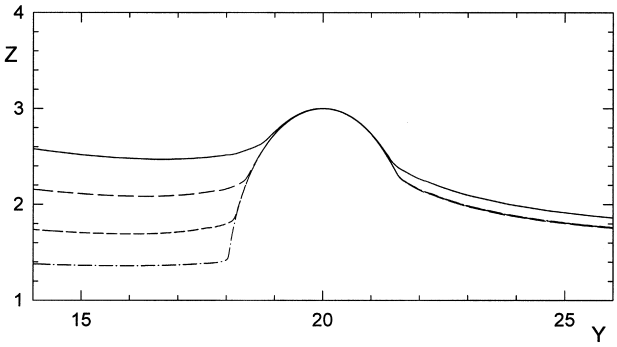


Fig. 15. The set of trajectories of a particle moving in the stagnation line flow around the second one attached to the wall obtained from the solution of Eq. (15). Motion along the axis connecting the particle centers. The initial distance at $Y_i = 10$ of moving particle to the wall ranges from $Z_i = 2.6$ to $Z_i = 1.4$. The immobile particle center is located at the distance $Y_d = 20$ from the stagnation line.

namic shadowing especially for weak flows and/or close to the stagnation point (line) where the local shear rate is low. This is demonstrated in Fig. 20 where 2000 final positions of the flowing particles, i.e. the coordinates of the spots where the virtual flowing particles are adsorbed at the interface, are marked as black dots. The hydrodynamic shadow in the absence of Brownian motion is marked by the dash-dot line while the contour of the deposited particle is shown as a solid circle. The calculations of virtual particle trajectories were performed in the stagnation line flow using the Brownian dynamics algorithm (for details see Warszyński [68]) for $Y_d = 10$ and $Y_d = 100$ and for two Pe numbers 0.2 and 1.0. One can see that in the presence of Brownian motion the outermost part of the hydrodynamic shadow is now accessible for the particle adsorption to the same extent as the interface outside of the shadow. On the remaining inner part of the hydrodynamic shadow

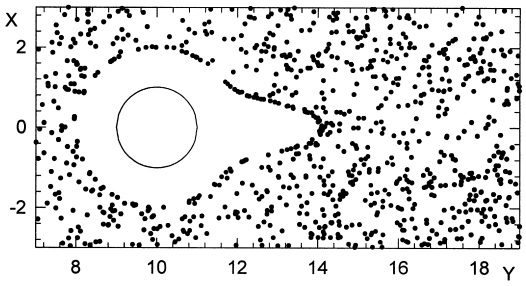


Fig. 16. The non-Brownian hydrodynamic shadow formed by the particle attached to the interface in the stagnation line flow, $Pe = 0.1$, at the distance $Y_d = 10$ from the stagnation line. Solid circle represent the contour of the attached particle, points are final positions of virtual particles after they reached the interface. The van der Waals attraction between the moving particle and the wall calculated according to Eqs. (52) and (53) for $Ad = 0.4$.

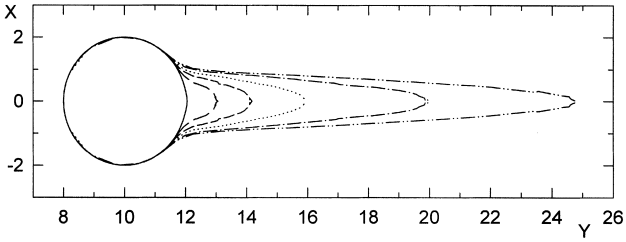


Fig. 17. The dependence of the size of the non-Brownian hydrodynamic shadow formed behind the particle attached to the interface in the stagnation line flow at the distance $Y_d = 10$ from the stagnation line on the flow intensity expressed in terms of the Peclet number. Solid line $Pe = 0.01$, long dash line $Pe = 0.05$, short dash $Pe = 0.1$, dotted line $Pe = 0.2$, dash–dot line $Pe = 0.5$ and dash double dot $Pe = 1$; $Ad = 0.4$.

adsorption is still possible but its probability is strongly decreased. The area of the reduced adsorption probability increases with the *Pe* number and with the distance of the deposited particle from the stagnation line. Thus, in real systems of adsorbing Brownian particles, in the region close to the stagnation point (line) and at low *Pe* number the effects of the hydrodynamic shadow play a minor role and adsorption kinetics for the partially covered interface can be described in terms of the geometrical blocking [31]. Due to the hydrodynamic shadow, particle adsorbed far from the stagnation point (line) or in the intense flow effectively blocks much

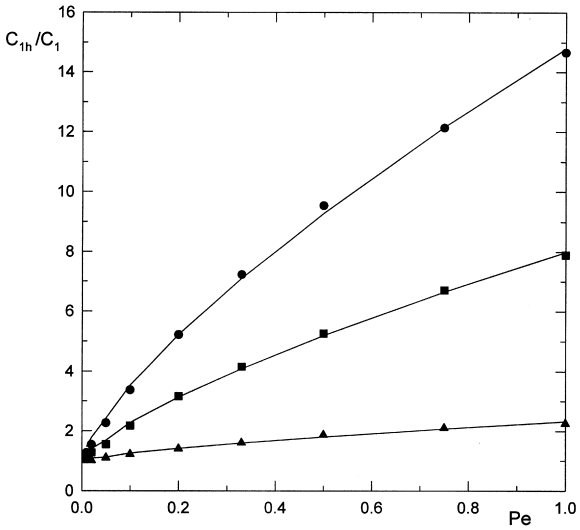


Fig. 18. The dependence of the size of the non-Brownian hydrodynamic shadow on *Pe* for various positions of the particle attached to the wall with respect to the stagnation line. Points represent the shadows obtained from the solution of trajectory equation, Eq. (15), lines are the interpolations. Circles $Y_d = 10$, triangles $Y_d = 50$ and squares $Y_d = 100$, $Ad = 0.4$.

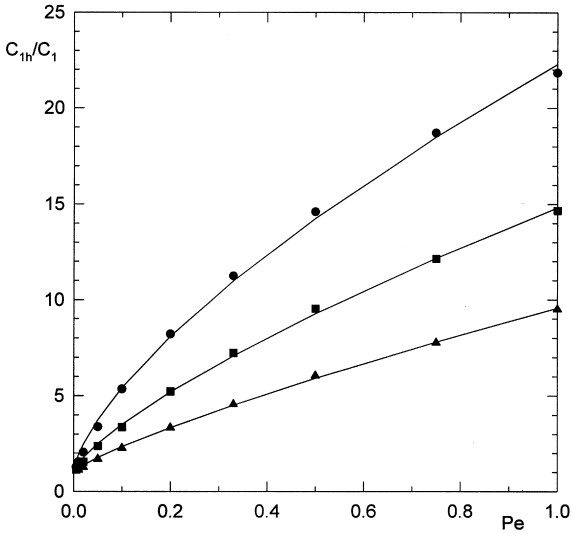


Fig. 19. The same as in Fig. 18 but for one position of the attached particle $Y_d = 100$ and various magnitude Ad. Circles $Ad = 0.2$, triangles $Ad = 0.4$ and squares $Ad = 0.8$.

larger area than in the absence of flow. This leads to a dramatic decrease of the adsorption rate. This phenomenon is illustrated in Fig. 21 where the normalized surface coverage Θ is given as a function of the dimensionless time τ. The deposition of 1-μm polystyrene latex particles ($a = 500$ nm) at the positively charged mica plates under various flow intensity was observed 'in situ' by optical microscope. The experiments were carried out in the slot impinging jet cell relatively far from the stagnation line for $Y = 316$. The experimental details are reported in Adamczyk et al. [23]. The results shown in Fig. 21 show that after the period of the initial growth of the surface coverage the apparent stationary state is attained. The apparent stationary coverage Θ_{st} depends strongly on the flow intensity, decreasing as the Re number of the flow increases. It equals $\Theta_{st} \approx 24\%$ for $Re = 2$ ($Pe = 0.01$), $\Theta_{st} \approx 10\%$ for $Re = 8$ ($Pe = 0.09$) and $\Theta_{st} \approx 7\%$ for $Re = 16$ ($Pe = 0.30$). A quantitative analysis of these non-linear adsorption phenomena under flow conditions can be performed only in terms of SBD simulations. The results of such simulations are shown as the solid lines in Fig. 21. The fair agreement between the experimental results and the SBD simulations is achieved in spite of very approximate treatment of many particle hydrodynamic interactions. The details of computations are given in Adamczyk et al. [23].

It should be emphasized that the stationary coverages observed for high Re at distances far from the slot impinging jet cell center were not the saturation or equilibrium value since particle adsorption was irreversible. This apparent saturation is purely kinetic and has its origin in the hydrodynamic shadowing effect. This is spectacularly illustrated in Fig. 22. First, the particle adsorption experiment was carried out in the slot impinging jet cell for the flow rate corresponding to $Re = 16$

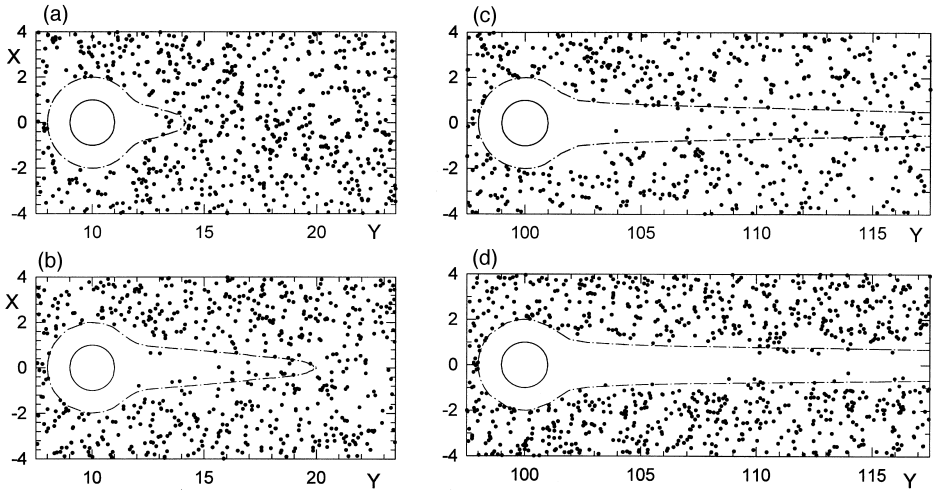


Fig. 20. The effect of Brownian motion on the size of the hydrodynamic shadow for various stagnation line flow intensity and distance of the attached particle to the stagnation line. Points represent the final positions of virtual particles colliding with the attached one after they were deposited at the interface. Part a for $Y_d = 10$ and $Pe = 0.2$, part b for $Y_d = 10$ and $Pe = 1.0$, part c for $Y_d = 100$ and $Pe = 0.2$ and part d for $Y_d = 100$ and $Pe = 1.0$, $Ad = 0.4$. Solid circle stands for the countour of the attached particle and dash–dot line represents the hydrodynamic shadow for non-Brownian particles.

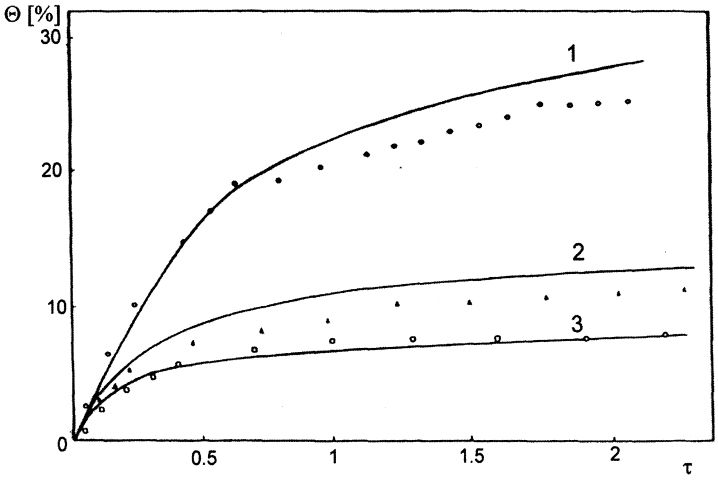


Fig. 21. The kinetics of adsorption of the monodisperse latex particles ($a = 0.5$ μm) on mica observed in the slot impinging jet cell [23] at the distance of 158 μm from the center. Points represent the experimental results for $Re = 2$ (1), $Re = 8$ (2) and $Re = 16$ (3). Lines are the result of SBD simulations.

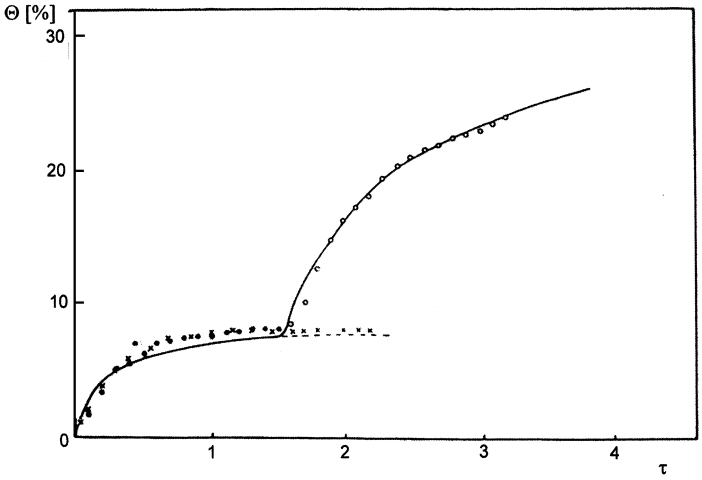


Fig. 22. The kinetics of adsorption of monodisperse latex particles (a = 0.5 μm) on mica observed in the slot impinging jet cell at the distance of 158 μm from the center. The adsorption carried out first at Re = 16 (solid circles) then at Re = 2 (hollow circles). Crosses represent the result of experiment for Re = 16 only. Lines are the result of SBD simulations.

until the apparent saturation at the level of $\Theta_{st} \approx 7\%$ was attained. Then, the flow rate was reduced by eightfold to $Re = 2$ and this resulted in an increase of particle surface coverage which approached the new quasi-stationary value three times higher than the previous one. The theoretical simulations performed according to the SBD algorithm (solid lines in Fig. 22) are in quantitative agreement with the experimental results.

Since the hydrodynamic shadowing is strongly dependent on the particle–wall interactions (c.f. Fig. 19) for the similarly charged particles and the interface, i.e. in the adsorption in the presence of the energetic barrier, it should have a pronounced effect on the adsorption rate. The apparent saturation can be attained already when only a few percent of the interface is covered by particles [69].

The effect of the hydrodynamic shadowing is enhanced when the particles are charged due to the long range electrostatic repulsion. This indication of the coupling between hydrodynamics and electric interactions will be considered later on.

4. Electric interactions

4.1. General considerations

When the suspension of colloidal particles is stagnant, i.e. $\boldsymbol{v} = 0$ everywhere, system of Eqs. (8)–(11) reduces to:

$$-\sum_i n_i z_i e \nabla\psi = \nabla p \tag{59}$$

$$\nabla^2\psi = -\frac{1}{\varepsilon}\sum_i n_i z_i e \tag{60}$$

$$\frac{\partial n_i}{\partial t} = \nabla\cdot\left(D_i\nabla n_i + n_i\frac{z_i e}{kT}D_i\nabla\psi\right) \tag{61}$$

Additionally, when we consider the steady-state, i.e. for $\frac{\partial n_i}{\partial t} = 0$, Eq. (61) describes the Boltzmann distribution of the ionic charges in the liquid:

$$n_i = n_{i\infty}\exp\left(\frac{z_i e\psi}{kT}\right) \tag{62}$$

where $n_{i\infty}$ is the bulk concentration of the i-type ions.

Accordingly, Eq. (60) assumes the form:

$$\nabla^2\psi = -\frac{e}{\varepsilon}\sum_i z_i n_{i\infty}\exp\left(\frac{z_i e\psi}{kT}\right) \tag{63}$$

Eq. (63) is usually referred to as the Poisson–Boltzmann equation and its solution describes the equilibrium electric potential distribution around particles in the colloidal system. On the other hand Eq. (59) describes the balance between the electrostatic and osmotic stress in the system in the mechanical equilibrium. The latter can be described by the osmotic pressure tensor [70]:

$$\underline{\underline{\boldsymbol{P}}} = kT\sum_i n_i \underline{\underline{\boldsymbol{I}}} \tag{64}$$

while, since the fluid is at rest, the hydrodynamic pressure is the same everywhere.

The force exerted on the particle suspended in the solution can be obtained by applying Eqs. (13) and (14):

$$\boldsymbol{F}_p = \int_{S_p}\left[\left(\underline{\underline{\boldsymbol{\Delta P}}} + \frac{\varepsilon}{2}E^2\underline{\underline{\boldsymbol{I}}}\right) - \boldsymbol{EE}\right]\cdot \mathrm{d}S \tag{65}$$

where $\underline{\underline{\boldsymbol{\Delta P}}} = kT\sum_i (n_i - n_{i\infty})\underline{\underline{\boldsymbol{I}}}$.

Knowing the dependence of the force on the particle position in the system one can calculate the energy of the electric interactions as work done by bringing a particle from the infinity to the given position along path l:

$$\Phi = -\int_l \boldsymbol{F}_p\cdot \mathrm{d}\boldsymbol{l} \tag{66}$$

For the symmetric electrolyte the Poisson–Boltzmann equation simplifies to:

$$\nabla^2\psi = -\frac{z_e e n_\infty}{\varepsilon}\left[\exp\left(-\frac{z_e e\psi}{kT}\right) - \exp\left(\frac{z_e e\psi}{kT}\right)\right] = \frac{kT}{z_e}\kappa^2\sinh\left(\frac{z_e e\psi}{kT}\right) \tag{67}$$

where $\kappa = \sqrt{\frac{2n_\infty z_e^2 e^2}{\varepsilon kT}}$ is the Debye–Hückel parameter and its inverse $1/\kappa$ is a measure of the EDL thickness.

The boundary condition for the Poisson–Boltzmann equation is determined by the electric charge distributed over the particle surface. The normal component of the electric displacement vector on the particle surface has to fulfill the condition:

$$(D_\perp^{\text{out}} - D_\perp^{\text{in}})_{|S_p} = \sigma_p\left(\psi_{|S_p}\right) \tag{68}$$

where $\boldsymbol{D}^{\text{in}}$, $\boldsymbol{D}^{\text{out}}$ are the electric displacement vectors inside and outside the particle respectively and σ_p is the electric surface charge density. The surface charge density is in general a function of the local electric potential at the particle interface and depends on the mechanism of the charge formation. Thus, in general the electric charge density is dependent on the configuration of particles in the colloidal system. As the simplifying assumption the limiting case is often considered that the charge density is not dependent on the local electrostatic potential and consequently on the distance between the interacting particles (the constant charge model — c.c.). The other limiting case often considered, the constant potential model (c.p.), assumes that the electrostatic potential is determined entirely by the concentration of the potential determining ions in the dispersion and is maintained constant during the interactions between colloidal particles.

The analytical solution of the Poisson–Boltzmann equation is possible only in the simple planar geometry. For other systems either numerical methods or analytical approximations are used to determine the electric potential distribution around particles and to calculate the electric interactions.

If the electric potential is small, i.e. $\frac{z_e e\psi}{kT} \ll 1$ ($\psi < 25$ mV for a simple 1–1 electrolyte in room temperature) Eq. (67) can be linearized and assumes the form:

$$\nabla^2\psi = \kappa^2\psi \tag{69}$$

A simple analytical solution of Eq. (69) can be derived for an isolated spherical particle immersed in arbitrary electrolyte, isolated spheroidal particle [71] or for a two plate system. An application of the linear Poisson–Boltzmann equation for more complicated systems, e.g. two spheres, leads to complicated iterative or series solutions.

Eq. (69) is also applicable for non-linear systems (high surface charges on particles) at distances large in comparison with $1/\kappa$ where the potential decreases to low values due to electrostatic screening. This observation is the basis of the linear superposition approximation which is often used to find the electric potential

distribution in the multiparticle system. Another method of finding approximate, closed form solution of the non-linear Poisson–Boltzmann equation, for the two body systems was pioneered by Derjaguin [72] who applied an integration procedure exploiting known solution for two flat plates. This method usually referred to as 'Derjaguin approximation' has been also successfully applied to calculate electric, van der Waals [3,4], adsorbed polymer [73], etc. In the subsequent section the application of the above mentioned approximations to the electric interactions in the sphere–plane geometry is discussed in some detail.

4.2. Exact solution of the non-linear Poisson–Boltzmann equation

The exact analytical solution of the non-linear Poisson–Boltzmann equation for the two-body problem is possible only in the case of the flat plate geometry. Integrating twice Eq. (67) one obtains:

$$\int \frac{d\bar{\psi}}{\sqrt{2\cosh\bar{\psi} + C_1}} = \bar{\xi} + C_2 \tag{70}$$

where $\bar{\xi} = \kappa\xi$, ξ being the coordinate normal to the plate surfaces, $\bar{\psi} = \frac{z_e e\psi}{kT}$ is the reduced electric potential and C_1, C_2 are the integration constants to be found from the boundary conditions at the surface of flat interfaces. For the boundary conditions according to constant charge, constant potential or mixed models Eq. (70) can be expressed in the closed form in terms of elliptic integrals and Jacobi elliptic functions [74]. Introducing the electric potential found from Eq. (70) to Eq. (65) one can express the force per unit area of plates as:

$$\Delta\Pi = 2n_\infty kT\left[\cosh\bar{\psi} - 1 - \frac{1}{2}\left(\frac{d\bar{\psi}}{d\bar{x}}\right)^2\right] = \text{const} \tag{71}$$

For the symmetrical plates, i.e. for the same boundary condition at both surfaces, Eq. (71) assumes particulary simple form:

$$\Delta\Pi = 2n_\infty kT\left[\cosh\bar{\psi}_{d/2} - 1\right] \tag{72}$$

where $\bar{\psi}_{d/2}$ is the potential in the midplane at a half distance between plates.

For more complicated geometries of interacting bodies, i.e. sphere–sphere or sphere–plane, only the numerical solution of the non-linear Poisson–Boltzmann equation is possible. Hoskin and Levine [75,76] were the first who solved it for two equal sized spheres in an exact way using the system of bispherical coordinates (see Fig. 4) and applying the finite difference method. The same method (with more mesh points) was subsequently used by Camie et al. [77] who performed calculations of the interaction force for two spherical particles in a 1–1 electrolyte. The authors also determined the range of validity of the Derjaguin approximation as a

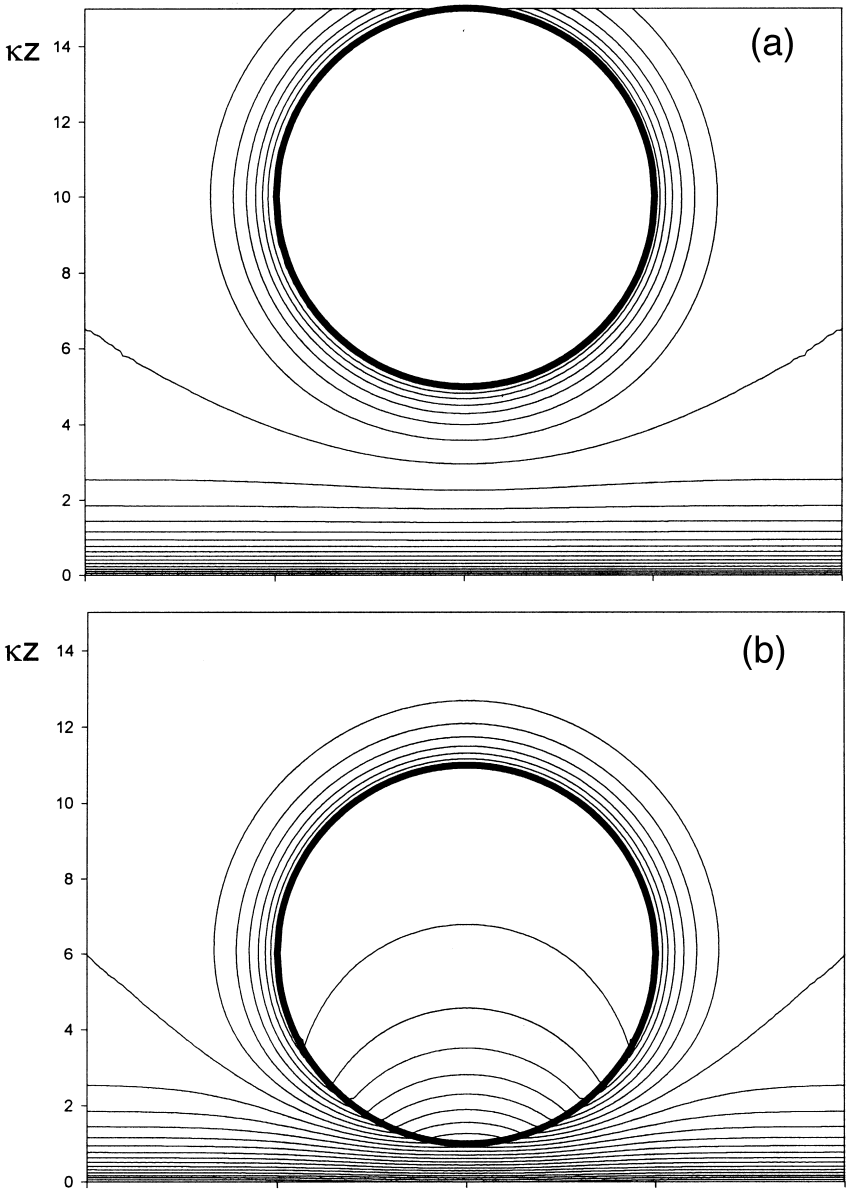
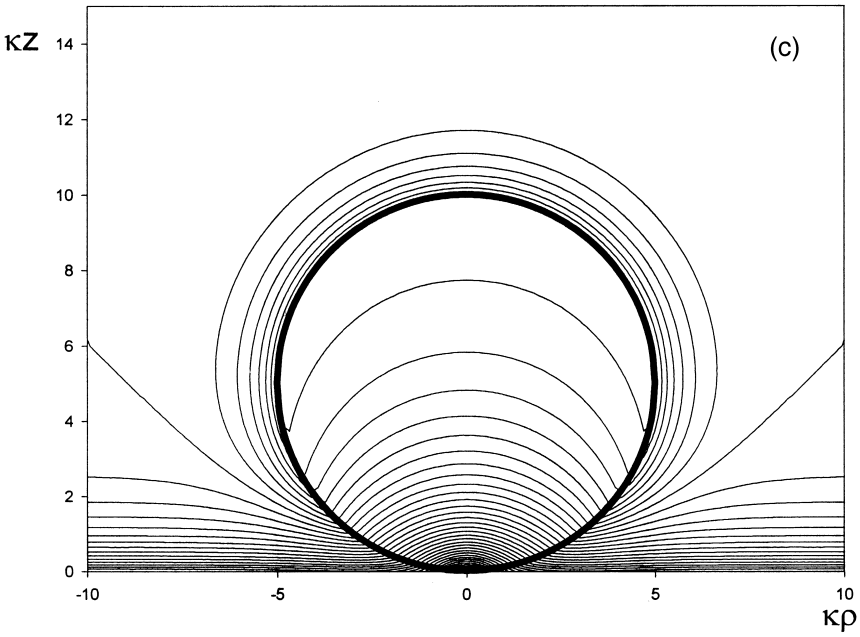


Fig. 23. Distribution of the electrostatic potential within and outside a spherical particle ($\varepsilon = 2.5$) approaching a planar interface in aqueous solution ($\varepsilon = 78$). Lines denote equipotential surfaces obtained from the solution of the non-linear PB equation in bispherical coordinates for $\bar{\psi}_p = -1.5$ and $\bar{\psi}_w = 3.0$, $\kappa a = 5$ assuming the c.c. boundary conditions on both particle and interface. Part a for $\kappa h = 5$, part b for $\kappa h = 1$ and part c for $\kappa h = 0.05$.



function of reduced particle radius κa and particle surface potential. Later on, they extended their approach for the dissimilar spheres system [78]. The electrostatic potential distribution and the interaction energy for the dissimilar spheres system, including sphere interaction with the plane, were calculated by Warszyński and Adamczyk [79] using also bispherical coordinates with a fine grid of 160×120 mesh points. The particles were assumed as dielectric and the electric potential distribution within the particles was also found. The calculations were performed for a 1–1 electrolyte, κa ranged from 0.25 to 10 and the scaled surface potential of particles at the infinite separation reached 4 (100 mV).

Figs. 23 and 24 present the distribution of the electric potential in the sphere–plane geometry as the sphere approaches oppositely charged solid surface for two values of the reduced sphere radius $\kappa a = 5$ and $\kappa a = 0.5$. The numerical calculations were performed in bispherical systems of coordinates assuming constant charge boundary conditions on surfaces of the sphere and of the plane. As it can be seen while for $\kappa a = 5$ at every sphere–plate separation the potential distribution around the major part of the sphere remains practically identical as for isolated particle, for $\kappa a = 0.5$ the particle gradually sinks in the electric field of the flat interface. The electric field to a large extent penetrates inside the particle. When the particle is stuck at the interface the distribution of the electric potential around it is strongly modified when compared to the distribution around the isolated particle. The degree of modification strongly depends on κa and on surface potential asymmetry [79]. In extreme cases the electric potential over the

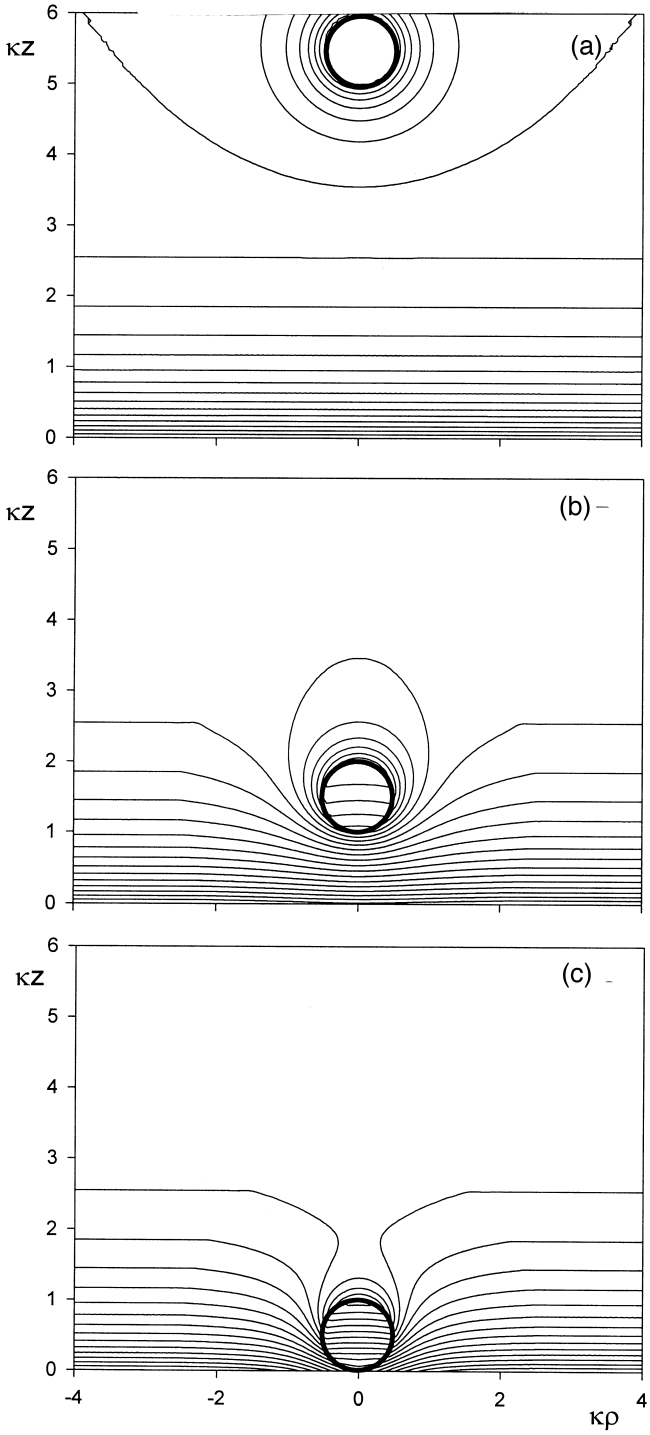


Fig. 24. The same as in Fig. 23 but for $\kappa a = 0.5$.

surface of the deposited particle can even inverse the sign facilitating adsorption of other particles on the ones already deposited.

The numerical results for the electric interaction force and/or energy are difficult to implement directly in the particle transport equations. Therefore, it is necessary to derive some approximate analytical formulae and to determine the range of their applicability. Two approximate analytical approaches will shortly be discussed in Section 4.3, the linear superposition method developed by Bell et al. [80] and generalized Derjaguin method [72,81,82]. For a more detailed discussion of the approximate methods also for non-spherical particles refer to Adamczyk and Warszyński [70].

4.3. Approximate methods for electric interactions

4.3.1. Linear superposition method

The basic assumption of the LSA method is that the solution of the PB equation for the two-particle system can be constructed as a linear superposition of the solutions for isolated particles in an infinite medium. This is justified by the fact that at large separations, due to screening, the absolute value of the electric potential decreases considerably and its distribution is governed by the linear PB equation. Thus, the LSA approach assumes:

$$\psi = \psi_1 + \psi_2 \tag{73}$$

where ψ_1 and ψ_2 are the electric potentials at a given point of space due to isolated spheres or to the sphere and the plane. Consequently the same applies to the electric field:

$$\boldsymbol{E} = \boldsymbol{E}_1 + \boldsymbol{E}_2 \tag{74}$$

The potentials ψ_1 and ψ_2 can be obtained either from an exact solution of the non-linear PB equation for a single sphere or from an approximate analytical solution reported in the literature. The latter approach seems especially attractive because a closed-form analytical solution for the force and energy of interaction can be formulated.

The starting point of the derivation of the analytical formulas is the cognizance that any solution of the non-linear PB equation for an isolated sphere immersed in an arbitrary electrolyte at distances $r \gg \kappa^{-1}$ should be in the form [80]:

$$\psi = \frac{kT}{e}\kappa a\,\overline{Y}_p^{\,0}\frac{\mathrm{e}^{-(\bar{r}-\kappa a)}}{\bar{r}} \tag{75}$$

where $\bar{r} = \kappa r$ is the dimensionless radial distance from the sphere center, a is the sphere radius, and $\overline{Y}_p^{\,0}$ is its 'effective' surface potential.

Loeb et al. [83] were the first to solve the non-linear PB equation numerically for spherical geometry and to calculate $\overline{Y}_p^{\,0}$ in a tabulated form for various electrolyte types. Stigter [84] presented interpolation equations for the tabulated solutions of

Loeb et al. White [85], Bentz [86] and Oshima et al. [87] developed useful solutions for both the potential distribution and the effective surface potential $\bar{Y}_p^0$ by replacing the original PB equation with a simplified version which they were able to solve analytically. For the 1–1 electrolyte, Oshima et al. [87] derived the relationship:

$$\bar{Y}_p^0 = 8\tanh\left(\frac{\bar{\psi}^0}{4}\right)\frac{1}{1+\left[1-\frac{(2\kappa a+1)}{(\kappa a+1)^2}\tanh\left(\frac{\bar{\psi}^0}{4}\right)\right]^{1/2}} \tag{76}$$

It can be deduced from Eq. (76) that the limiting value of $\bar{Y}_p^0$ in the case where $\kappa a \to \infty$ (planar interface) and for a symmetric electrolyte is given by:

$$\bar{Y}_w^0 = 4\tanh\left(\frac{\bar{\psi}^0}{4}\right) \tag{77}$$

It was shown in Oshima et al. [87] that the deviation of the results derived from Eq. (76) from the exact results of Loeb et al. [83] is confined to a few percent for $\kappa a > 0.1$ and $\bar{\psi}^0 < 5$.

Knowing the potential and field distributions and applying the LSA principle one can calculate the force between spheres using Eq. (65). The surface integration can be conveniently carried out over a plane located at an arbitrary position between spheres. In this way one obtains [80]:

$$\boldsymbol{F} = F_{pp}\frac{(1+\bar{R})}{\bar{R}}A_1A_2\mathrm{e}^{-\bar{h}}\boldsymbol{r}_{12} \tag{78}$$

where $\bar{h} = \kappa h$ is the dimensionless separation between the sphere surfaces, $\boldsymbol{r}_{12}$ is the vector connecting the sphere centers, $A_1 = \kappa a_1$ and $A_2 = \kappa a_2$ are the dimensionless radii, $\bar{R} = A_1 + A_2 + \bar{h}$, $F_{pp} = 4\pi\varepsilon\left(\frac{kT}{e}\right)^2\bar{Y}_{p_1}^0\bar{Y}_{p_2}^0$ and $\bar{Y}_{p1}^0$ and $\bar{Y}_{p2}^0$ are the effective surface potentials of particles 1 and 2, respectively.

The interaction energy can be calculated by integrating the force with respect to the dimensionless distance $\bar{h}$. In this way one obtains:

$$\Phi = \frac{F_{pp}}{\kappa}\frac{A_1A_2}{A_1+A_2+\bar{h}}\mathrm{e}^{-\bar{h}} \tag{79}$$

When one of the sphere's radii tends to infinity, Eqs. (78) and (79) reduce to the forms describing the plane/sphere interactions:

$$\boldsymbol{F} = F_{pw}\mathrm{e}^{-\bar{h}}\,\hat{\boldsymbol{n}}$$
$$\Phi = \Phi_{pw}\,\mathrm{e}^{-\bar{h}} \tag{80}$$

where $\Phi_{pw} = F_{pw}a = 4\pi\varepsilon a\left(\frac{kT}{e}\right)^2 \bar{Y}_p^0\bar{Y}_w^0$ and $\bar{Y}_w^0$ is the effective potential of the planar double layer given by Eq. (77).

It should be mentioned that the advantage of the above formulae is that they are applicable to arbitrary surface potentials and κa values provided $\bar{h} > \kappa a$.

4.3.2. Derjaguin method

According to the Derjaguin approximation the interaction of spheres is calculated as a sum (integral) of the corresponding interaction energy per unit area of plane parallel half spaces. The great advantage of the method is that it enables to find the interactions between bodies of complicated geometry, e.g. spheres, cylinders and spheroids for which exact solutions are often inaccessible, in terms of the well-studied cases of plate–plate interactions provided that the distance between the interacting bodies is smaller than the local radii of curvature. Using the Derjaguin method, the force F and interaction energy Φ between convex particles can be calculated as:

$$\boldsymbol{F} = 2\pi\, G_d\, \Phi_p(h)\hat{\boldsymbol{n}}$$

$$\Phi = 2\pi\, G_d \int_h^\infty \Phi_p(h_p)\mathrm{d}h_p \tag{81}$$

where h_p is the separation distance between plates, Φ_p is the interaction energy of plates (per unit area) and G_D denotes the geometric Derjaguin factor connected with the principal radii of curvature of interacting bodies, a'_1, a''_1, a'_2 and a''_2 through the dependence:

$$G_d = \sqrt{\frac{a'_1 a''_1 a'_2 a''_2}{(a'_1 + a'_2)(a''_1 + a''_2)}} \tag{82}$$

Eq. (81) indicates that the interaction force between particles can be simply obtained by multiplying the interaction energy of plates Φ_p by some correction geometrical factor G_D. For two unequal spheres $G_D = a_1a_2/(a_1 + a_2)$, for two equal spheres $G_D = a/2$ and for sphere/plane $G_D = a$.

The Derjaguin method can be applied to both linear and non-linear regimes of a plate–plate interaction. However only for linear regime, i.e. for small potential, it is possible to obtain simple analytical expression. Hogg et al. [88] derived an analytical expression for energy and force of interaction of two parallel plates. Then, using the Derjaguin approximation, they obtained the expressions for the interaction energy between dissimilar spheres:

$$\Phi = \varepsilon\left(\frac{kT}{ze}\right)^2 G_D\left[\mp\left(\bar{\psi}_1^{0^2} + \bar{\psi}_2^{0^2}\right)\ln(1 - \mathrm{e}^{-2\bar{h}}) + 2\bar{\psi}_1^0\bar{\psi}_2^0\ln\left(\frac{1 + \mathrm{e}^{-\bar{h}}}{1 - \mathrm{e}^{-\bar{h}}}\right)\right] \tag{83}$$

where the signs denote the constant charge and constant potential model, respectively. This equation is also valid for the sphere/plane case with $G_D = a$. For two equal spheres, Eq. (83) simplifies to the equation derived originally by Derjaguin [72]

$$\Phi = \mp \frac{\varepsilon}{2} a \bar{\psi}^{0^2} \left(\frac{kT}{ze} \right)^2 \ln(1 \mp \mathrm{e}^{-\bar{h}}) \tag{84}$$

For the mixed case, i.e. when the surface of particle 1 is maintained at a constant potential and the surface of particle 2 at a constant charge, the expression for the interaction energy Φ becomes [89]:

$$\Phi = \varepsilon \left(\frac{kT}{ze} \right)^2 G_D \left[-\left(\bar{\psi}_1^{0^2} - \bar{\psi}_2^{0^2} \right) \ln(1 + \mathrm{e}^{-2\bar{h}}) + 2\bar{\psi}_1^0 \bar{\psi}_2^0 \left(\frac{\pi}{2} - \tan^{-1}(\sinh(\bar{h})) \right) \right] \tag{85}$$

The Derjaguin approach can also be used in combination with the exact numerical solutions of the non-linear Poisson–Boltzmann equation for plates as it was done in Derjaguin [72] and Adamczyk et al. [82].

A quantitative comparison of the results of various analytical approximations for the EDL interaction energy with exact numerical computations is given in Figs. 25–28. There the normalized interaction energy is plotted as a function of dimensionless separation $\bar{h}$ both for the two-particle system (upper part of these figures) and for particle–plane system (lower part of the figures). The data originating from the exact numerical solutions of the non-linear Poisson–Boltzmann equation are compared with the results of LSA model and linear Derjaguin approximation. The constant charge (c.c.), constant potential (c.p.), and mixed (c.c. on particle, c.p. at the plane) boundary conditions are considered. As it can be seen in Fig. 25 the particle/particle interaction profile for $\kappa a = 5$ is well described by the LSA model, whereas linear Derjaguin approximation shows the tendency to overestimate the interactions in particular for c.c. boundary conditions. Also the particle/interface interaction energy profiles are well reflected by the LSA model, with slightly lower accuracy for the c.p. boundary conditions. The predictions of linear Derjaguin model differ significantly from the exact solution. The same conclusions can be drawn from the comparison of the exact numerical results with the approximate analytical formulae for smaller κa as shown in Figs. 26 and 27. The deviations of the linear Derjaguin model from the exact results increase with the decrease of κa while the LSA model describe surprisingly well the exact data for the interaction energy between two spheres or between sphere and planar interface at the c.c. boundary conditions. However, the accuracy of the LSA approximation depends strongly on the surface potential asymmetry. As illustrated in Fig. 28 the LSA satisfactorily reflects exact results when the absolute values of the surface potentials of the particle and the interface do not differ to much. For high asymmetry the LSA overestimates the interaction energy for distances $\bar{h} < 1$.

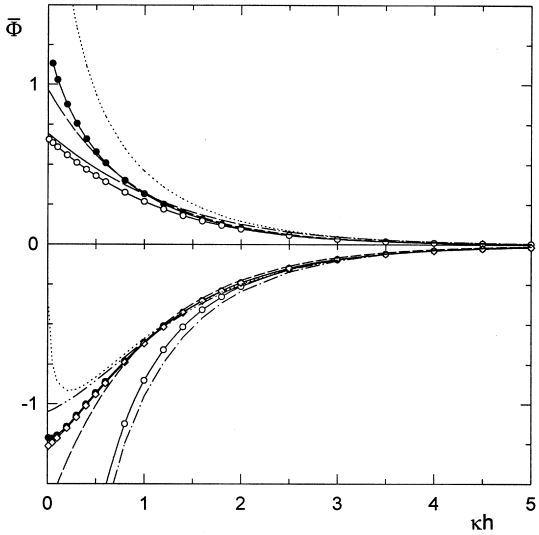


Fig. 25. The dimensionless electrostatic interaction energy $\overline{\Phi}$ between two particles (upper part) $\overline{\Phi} = \Phi/2\pi\varepsilon\psi_p^2$ and particle and interface (lower part) $\overline{\Phi} = \Phi/2\pi\varepsilon|\psi_w\psi_p|$ calculated for $\kappa a = 5$ from the exact solution of the non-linear PB equation,–●–●– for c.c. model, –○–○– for c.p. model, –◇–◇– for mixed model (c.c. at the interface) and from approximation analytical formulae; LSA — dashed line and linear Derjaguin approximation: dotted line c.c. model, dash–dot line c.p. model and dash double dot line mixed model. Other parameters as in Fig. 23.

Carnie et al. [77] performed systematic comparison of the results obtained from the approximate models of LSA and Derjaguin with exact numerical solution for the electric interaction force between spherical colloidal particles. They concluded that for the c.p. boundary conditions the non-linear Derjaguin approximation is accurate for large κa and small κh, especially for high surface potentials while the LSA holds in the complementary regime of small κa and large κh. The same conclusion can be drawn for the c.c. boundary conditions. However, since there is no closed form expressions available for the non-linear Derjaguin model its applicability for the prediction of the adsorption behavior of colloidal particles at interfaces is limited. The linear Derjaguin approximation is quite accurate in predicting the interaction force for the c.p. boundary conditions provided that the reduced surface potential of the particle does not exceed $\overline{\psi}_0 = 2$. For the c.c. boundary conditions the linear Derjaguin model fails to be a good approximation [77].

From the analysis performed above one can conclude that for applications associated with colloid particle adsorption the LSA has proved a useful model reflecting properly the interaction energy profiles in terms of simple analytical formulae. They are easy to apply in any description of the adsorption process either in terms of differential equations or Monte Carlo or Brownian dynamics simulations.

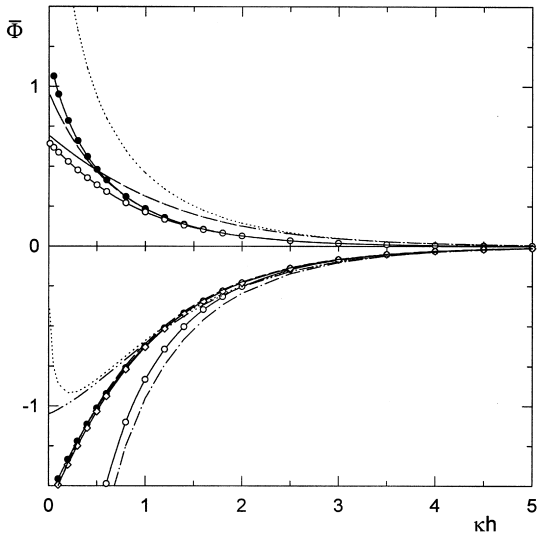


Fig. 26. The same as in Fig. 25 but for $\kappa a = 1$.

4.4. Implications for particle adsorption

The crucial role of the electrostatic interactions in the colloidal system is described in terms of the well established DLVO theory [1,2,72]. The stability of the colloidal dispersion can be understood as a result of the competition of van der

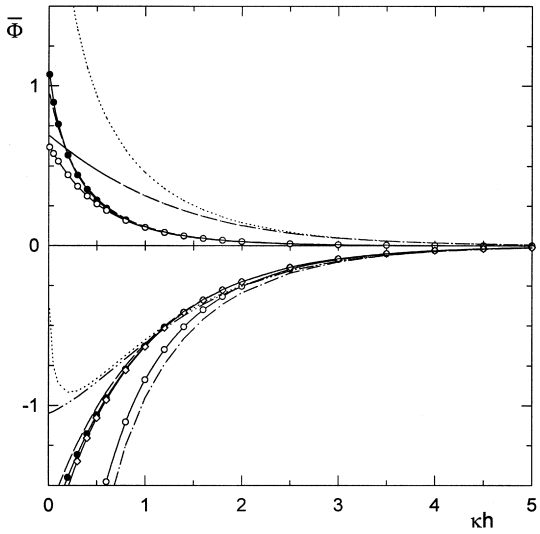


Fig. 27. The same as in Fig. 25 but for $\kappa a = 0.25$.

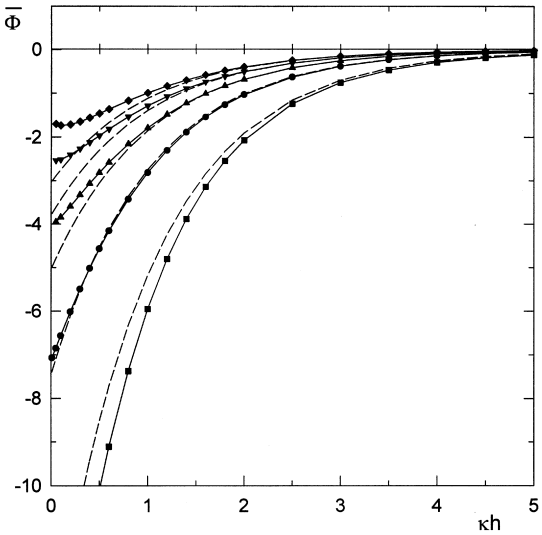


Fig. 28. The dimensionless electrostatic interaction energy $\overline{\Phi} = \Phi/2\pi\varepsilon|\psi_w\psi_p|$ between particle and interface calculated for $\kappa a = 1$ from the exact solution of the non-linear PB equation for various surface potentials at the interface and for c.c. model; $\overline{\psi}_p = 3$ and $\overline{\psi}_w = -3$ (squares), $\overline{\psi}_w = -1.5$ (circles), $\overline{\psi}_w = -1$ (triangles), $\overline{\psi}_w = -0.75$ (reversed triangles) and $\overline{\psi}_w = -0.6$ (diamonds). Dashed curves — analytical LSA approximation.

Waals attraction and the electrostatic repulsion of likely charged particles. The same description can be applied for the nano- and microparticle adsorption phenomena. Consider for example a colloid particle of a radius $a = 100$ nm in 1–1 electrolyte of 10^{-3} M concentration ($\kappa a \approx 10$) having the surface potential 50 mV at the solid interface bearing the same potential of 50 mV. The interaction energy as calculated from Eq. (58) is equal to thermal energy of 1.5 kT when the particle is at a distance of 50 nm from the flat interface (see Fig. 29). When the same particle is immersed in 10^{-5} M electrolyte the electric interaction energy exceeds thermal energy for the distances around 500 nm, i.e. much bigger than particle size. In an analogous way one can estimate the range of electrostatic attraction between oppositely charged particle and interface.

Based on these estimations one can expect that the electric interaction should considerably affect particle adsorption process. For low surface coverages when the particle–interface interactions dominate, the adsorption rate is strongly decreased by the electric forces between likely charged particles and interface. For the oppositely charged interfaces the EDL attraction facilitates the adsorption.

On the other hand the repulsive lateral electric interactions between adsorbed and adsorbing particles decrease the rate at the later stages of adsorption when the interface is partially covered. Simultaneously, they determine the structure of the adsorbate. Stronger is the electric repulsion between particles, the less packed is the adsorbate and consequently lower is the final coverage for the adsorption time

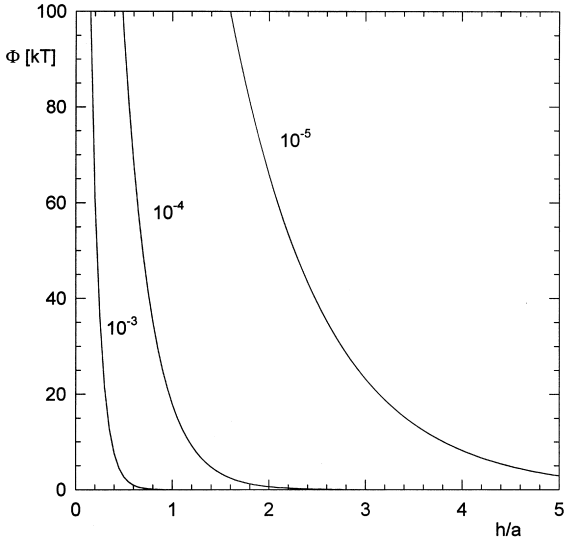


Fig. 29. The EDL interaction energy of 100-nm particle with the planar interface as a function of the distance between them (normalized by particle radius) for three concentrations of 1–1 electrolyte (marked in the figure) and $\psi_w = \psi_p = 50$ mV.

at $t_{ads} \to \infty$. This is experimentally demonstrated in Fig. 30 where the kinetics of adsorption of $a = 500$ nm, negatively charged latex particles on the positively charged silanized mica plates is shown. The experiments were carried out in the slot impinging jet cell at $Re = 2$ and the adsorptions were observed close to the cell symmetry line in order to minimize the hydrodynamic blocking effects. As it can be seen in Fig. 30 initially, up to $\Theta \approx 0.15$, the adsorption rate is the highest for the lowest ionic strength of the suspension ($I = 2 \times 10^{-5}$ M) due to the longest range of the attractive EDL forces. However, as the surface concentration of adsorbed particles increases the reverse phenomenon is observed. The rate of adsorption is the highest for the highest ionic strength ($I = 10^{-3}$ M) of the suspension as the EDL repulsive forces between adsorbing and already adsorbed particles are of the shortest range. As it is illustrated by the solid lines in Fig. 30 the experimental adsorption kinetics in the conditions of absence of the hydrodynamic blocking effects can be satisfactorily described in terms of the RSA simulations when the lateral EDL interactions between adsorbing particles are taken into account. The RSA simulation method was discussed extensively in Adamczyk and Warszyński [70].

A very similar trend as that shown in Fig. 30 was also observed in the measurements of Johnson and Elimelech [90] concerning particle adsorption in packed bed columns. The positively charged polystyrene latex of average diameter $2a = 480$ nm was adsorbed on 0.023 cm soda glass beads. Similarly like in the results presented in Fig. 30 the particle adsorption rate was initially the largest for the

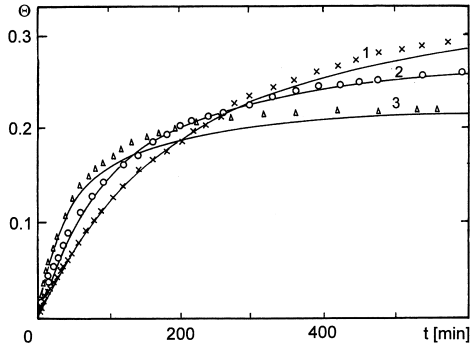


Fig. 30. Adsorption kinetics of 1-μm latex particles on silanized mica measured in the center of the slot impinging jet cell for $Re = 2$, $n_b = 4 \times 10^8$ cm^3 in: (1) $I = 10^{-3}$ M, (2) $I = 10^{-4}$ M and (3) 2×10^{-5} M KCl. Solid lines represent the RSA simulations using LSA approximation.

lowest ionic strength of the suspension whereas for longer times the trend was reversed.

As was shown in the previous chapter, for larger surface coverages under flow conditions, the presence of uncharged particles already adsorbed at the interface significantly decreases the adsorption rate. This phenomenon is due to the effect of the hydrodynamic shadowing and strongly depends on the local flow conditions. For charged particles the long ranged electric double layer interactions coupled with hydrodynamic interactions are expected to further decrease the adsorption rates.

All these effects of coupling between the electric and hydrodynamic interactions and their influence on adsorption of colloidal particles are discussed in Section 5.

5. Electrokinetic effects

5.1. General considerations

Electrokinetic effects are the result of the interplay between the charge at interfaces and the flow field around and between them. The theoretical and experimental research on the electrokinetic effects like: electrophoresis, electroosmosis, streaming potential, primary and secondary electroviscous effects has been extensively carried out for many years. The thorough summary of the results can be found in the monographs of Hunter [4,91] and Dukhin and Derjaguin [92]. However, the influence of the electrokinetic effects on the motion of a single particle at the vicinity of interface has started to be considered only recently [9,93]. As it was mentioned in Section 2.1 the general framework of the electrokinetic phenomena in colloidal systems is provided by the Stokes equation containing the electric force term [Eq. (8)] together with the continuity equation [Eq. (2)] for incompressible fluids. The distribution of the electric potential is governed by the

Poisson equation [Eq. (9)] while the distribution of ionic charges is determined by the ionic convective–diffusion equation, Eq. (11), one for each type of ion present in the system. The solution of this set of equations is a formidable task even in the simplest geometry (e.g. isolated particle) due to the presence of non-linear coupling terms $n_i \boldsymbol{v}$ in Eq. (11) and $\rho_e \nabla \psi$ in Eq. (8). Therefore, only either analytical approximations for small potentials $\psi \ll 1$ and/or large particles $a \gg 1/\kappa$ or in general case complicated numerical algorithms are used to solve the problem. However, the most crucial assumption, the meaning of which is often overseen [94], concerns the ratio of the convective and diffusive contribution to the ionic transport mechanism described by the ionic Peclet number Pe_i.

For the sake of simplicity we consider binary symmetric electrolyte of which the ionic charge is equal to $q = z_e e$. We can write Eq. (11) in the dimensionless form as:

$$d_+ - \frac{\partial \bar{n}_+}{\partial \tau} = \bar{\nabla} \cdot \left(\bar{\nabla} \bar{n}_+ + \bar{n}_+ \bar{\nabla} \bar{\psi} - d_+ Pe_i \bar{n}_+ \bar{\boldsymbol{v}} \right) \tag{86}$$

and

$$d_- \frac{\partial \bar{n}_-}{\partial \tau} = \bar{\nabla} \cdot \left(\bar{\nabla} \bar{n}_- - \bar{n}_- \bar{\nabla} \bar{\psi} - d_- Pe_i \bar{n}_- \bar{\boldsymbol{v}} \right) \tag{87}$$

for the positive and negative ions, respectively. The ionic Peclet number is defined as $Pe_i = V_{ch} l_{ch} / D_s$, where V_{ch} is the characteristic fluid flow velocity (e.g. particle velocity), l_{ch} is the characteristic length (e.g. particle radius a), $D_s = 2D_+ D_- / (D_+ + D_-)$ is the harmonic average of ionic diffusion coefficients. Other symbols in Eqs. (86) and (87) are defined as follows: $\tau = D_s t / l_{ch}^2$ is the dimensionless time, $\bar{\nabla} = l_{ch} \nabla$ is the dimensionless gradient operator, $d_+ = D_s / D_+$, $d_- = D_s / D_-$, $\bar{n}_+ = n_+ / n_\infty$, $\bar{n}_- = n_- / n_\infty$ and $\bar{\boldsymbol{v}} = \boldsymbol{v} / V_{ch}$. If we consider a 1-μm particle moving with the velocity of several μm/s in an aqueous solution of inorganic electrolyte with $D_s = 2 \times 10^{-5}$ cm^2/s the resulting $Pe_i \approx 10^{-3} \div 10^{-2}$. For faster particle motion (more intense flows) or in more viscous media (lower ionic diffusion coefficient) Pe_i can be considerably greater.

We can formally express the ionic concentrations and the electric potential in terms of expansions:

$$\bar{n}_+ = \bar{n}_{+,eq} + Pe_i \bar{n}_{+,1} + Pe_i^2 \bar{n}_{+,2} + \ldots \tag{88}$$

$$\bar{n}_- = \bar{n}_{-,eq} + Pe_i \bar{n}_{-,1} + Pe_i^2 \bar{n}_{-,2} + \ldots \tag{89}$$

$$\bar{\psi} = \bar{\psi}_{eq} + Pe_i \bar{\psi}_1 + Pe_i^2 \bar{\psi}_2 + \ldots \tag{90}$$

and for low Pe_i we can retain only the first two terms. The first term for $Pe_i = 0$ corresponds to the case when the fluid is at rest, i.e. to the equilibrium. For larger ionic Peclet numbers higher terms in the expansions have to be also considered.

The solution of the electrokinetic equations by the expansion in terms of Pe_i up

to the linear terms was successfully used by O'Brien and White [95]. Their numerical algorithm allowed calculation of the dependence of the electrophoretic mobility on zeta potential of particles at arbitrary κa and ψ and was then effectively used for other electrokinetic problems [96,97].

Another simplifying assumption frequently applied for the description of the electrokinetic effects in colloidal systems was introduced by Dukhin [92]. He proposed the 'thin double layer' approximation valid if the size of particles is much bigger than the equilibrium double layer thickness. Recently Cox [94] developed rigorous mathematical background of the thin double layer approach. Its advantage is that almost entire volume of the system can be treated as uncharged fluid and therefore its motion can be described in terms of simple Stokes and continuity equations which are much easier to solve. The full electrokinetic problem has to be solved only in the very thin layer adjacent to a particle surface which can be considered as locally flat. Thus, the solution of the complex electrokinetic problem is not additionally complicated by the surface curvature.

The electric potential in the vicinity of a moving particle can be expressed as a sum of three terms:

$$\psi = \psi_{eq} + \psi_d + \psi_s \tag{91}$$

where ψ_{eq} is the potential in the equilibrium double layer, ψ_d the distortion of the double layer from the equilibrium due to particle motion, and ψ_s is the long range 'streaming' potential (outside the double layer).

Since the relaxation of the thin electric double layer is very fast (10^{-7}–10^{-6} s), the ionic fluxes normal to the surfaces are compensated and the local equilibrium is always established [92,94]. Thus, the ion concentration can be written as:

$$n^{\pm} = n_{\infty}(1 + \bar{n}_s)\exp\left(\mp\left(\bar{\psi}_{eq} + \bar{\psi}_d\right)\right) \tag{92}$$

where $\bar{\psi}_d = (q / kT)\psi_d$, $n_{\infty}\bar{n}_s$, is the long range concentration polarization, which along with the streaming potential ψ_s has to be determined from the solution of the convective diffusion equation [Eqs. (86) and (87)].

The origin of the double layer distortion and appearance of the long range ionic concentration $\bar{n}_s$ and electric 'streaming' potential $\bar{\psi}_s$ is illustrated in Fig. 31. Motion of the particle induces the fluid flow through the electric double layer surrounding the particles. As a result, the convective flux of the oppositely charged ions (with respect to particle charge) in the EDL is enhanced in comparison to the flux of coions. To fulfill the continuity condition there must be the excessive flux of counterions to the EDL in the upstream part of the particle surface and out of the EDL in the downstream part. In the steady state this flux is short circuited outside the EDL by the diffusive and conductive flux due to the gradient of long range ionic concentration $\bar{n}_s$ and electric potential $\bar{\psi}_s$ fields which are formed. The range of these fields is much larger than the extent of the equilibrium EDL and is of the order of particle size.

The potential distribution in the distorted double layer can be determined combining Eqs. (67) and (92):

$$\nabla^2\left(\bar{\psi}_{eq} + \bar{\psi}_d\right) = \kappa^2(1 + \bar{n}_s)\sinh\left(\bar{\psi}_{eq} + \bar{\psi}_d\right) \tag{93}$$

Since $\bar{n}_s$ is practically constant in the thin double layer, Eq. (93) can be readily solved obtaining [94]:

$$\bar{\psi}_d = 4\left[\tanh^{-1}\left(\tanh\left(\frac{\bar{\psi}_p}{4}\right)\exp\left(-\bar{\xi}\sqrt{1 + \bar{n}_s}\right)\right) - \tanh^{-1}\left(\tanh\left(\frac{\bar{\psi}_p^{eq}}{4}\right)\exp(-\bar{\xi})\right)\right] \tag{94}$$

where $\bar{\psi}_p^{eq}$ is the electric potential at the particle surface in the equilibrium and $\bar{\psi}_p$ is the non-equilibrium particle surface potential which has to be established from the boundary conditions determined by the particle charge (see the considerations of the boundary conditions for the Poisson–Boltzmann equation in Section 4.1).

For the low Pe_i, Eq. (93) can be linearized (the lowest order term of $\bar{n}_s$ and $\bar{\psi}_d$ are linear in Pe_i) obtaining [98,99]:

$$\nabla^2\bar{\psi}_{eq} + Pe_i\nabla^2\bar{\psi}_{d,1} = \kappa^2\left[\sinh\left(\bar{\psi}_{eq}\right) + Pe_i\left(\bar{n}_{s,1}\sinh(\phi_0) + + \bar{\psi}_{d,1}\cosh(\psi_{eq})\right)\right] \tag{95}$$

Equating terms of the same order in Pe_i we find that:

$$\nabla^2\bar{\psi}_{eq} = \kappa^2\sinh\left(\bar{\psi}_{eq}\right) \tag{96}$$

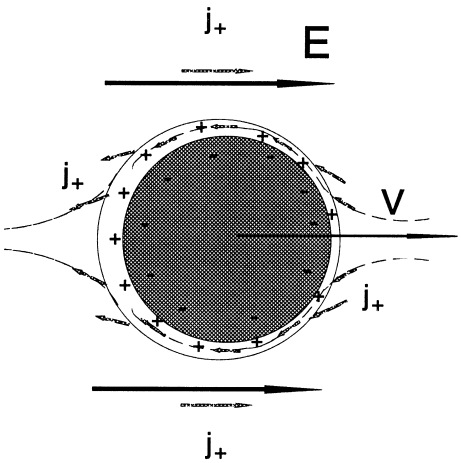


Fig. 31. Schematic representation of the distorted EDL and ionic flux around charged particles moving in the polar liquid.

determines the equilibrium potential distribution in the EDL [cf. Eq. (67)], whereas:

$$\nabla^2\bar{\psi}_{d,1} = \kappa^2\left(n_{s,1}\sinh\left(\bar{\psi}_{eq}\right) + \bar{\psi}_{d,1}\cosh\left(\bar{\psi}_{eq}\right)\right) \tag{97}$$

describes double layer polarization potential due to the particle motion.

Using the ion distribution defined by Eq. (92), we can find that outside the double layer, where $\bar{\psi}_{eq} = \bar{\psi}_d = 0$, the convective diffusion equation for positive and negative ions assumes form:

$$\nabla^2\bar{n}_s + \nabla\bar{n}_s\cdot\nabla\bar{\psi}_s + (1+\bar{n}_s)\nabla^2\bar{\psi}_s - \frac{1}{D_+}\nabla\bar{n}_s\cdot\boldsymbol{v} = \frac{1}{D_+}\frac{\partial\bar{n}_s}{\partial t} \tag{98}$$

$$\nabla^2\bar{n}_s - \nabla\bar{n}_s\cdot\nabla\bar{\psi}_s - (1+\bar{n}_s)\nabla^2\bar{\psi}_s - \frac{1}{D_-}\nabla\bar{n}_s\cdot\boldsymbol{v} = \frac{1}{D_-}\frac{\partial\bar{n}_s}{\partial t} \tag{99}$$

Adding and subtracting these two equations we obtain:

$$\nabla^2\bar{n}_s = \frac{1}{D_s}\left(\nabla\bar{n}_s\cdot v + \frac{\partial\bar{n}_s}{\partial t}\right) \tag{100}$$

$$(1+\bar{n}_s)\nabla^2\bar{\psi}_s + \nabla\bar{n}_s\cdot\nabla\bar{\psi}_s = \frac{1}{D_s}\left(\frac{D_- - D_+}{D_- + D_+}\right)\left(\nabla\bar{n}_s\cdot v + \frac{\partial\bar{n}_s}{\partial t}\right) \tag{101}$$

When the stationary state is established during the motion of the particle with the constant speed the partial derivative in Eqs. (100) and (101) can be further transformed. Since:

$$\frac{\partial\bar{n}_s}{\partial t} = -\frac{\partial\boldsymbol{r}_p}{\partial t}\cdot\nabla\bar{n}_s = -\boldsymbol{v}_p\cdot\nabla\bar{n}_s \tag{102}$$

where $\boldsymbol{r}_p$ is the position vector of the particle and $\boldsymbol{v}_p$ its velocity, Eqs. (100) and (101) assume the form:

$$\nabla^2\bar{n}_s = \frac{1}{D_s}\nabla\bar{n}_s\cdot(\boldsymbol{v}-\boldsymbol{v}_p) \tag{103}$$

$$(1+\bar{n}_s)\nabla^2\bar{\psi}_s + \nabla\bar{n}_s\cdot\nabla\bar{\psi}_s = \frac{1}{D_s}\left(\frac{D_- - D_+}{D_- + D_+}\right)\nabla\bar{n}_s\cdot(\boldsymbol{v}-\boldsymbol{v}_p) \tag{104}$$

The boundary conditions for Eqs. (103) and (104) have to be formulated at the outer-edge of the double layer (ODL). If we consider the local frame of reference in which a surface is at rest, the continuity of the ionic flux in the double layer requires that at any surface:

$$-j_{n\pm}|_{\mathrm{ODL}} = \nabla_s\cdot\int_{DL}\boldsymbol{j}_{s\pm}\mathrm{d}\xi \tag{105}$$

where $j_{n\pm}$ and $\boldsymbol{j}_{s\pm}$ are normal and tangential components of ionic flux, respec-

tively, $\nabla_s \cdot$ is surface divergence. Here we additionally assume that the surface is impenetrable for ions. If we express Eq. (105) in terms of ionic concentrations and potential we obtain the required boundary conditions for Eqs. (103) and (104):

$$\left(D_{\pm}\left(\nabla\bar{n}_s \pm (1+\bar{n}_s)\nabla\bar{\psi}_s\right) - \bar{n}_s\boldsymbol{v}\right)\cdot\boldsymbol{n}|_{\mathrm{ODL}}$$
$$= \nabla_s \cdot \int_{DL}\left(-D_{\pm}\left(\nabla_s\bar{n}_s \pm (1\pm\bar{n}_s)\nabla_s\bar{\psi}_s\right)\mathrm{e}^{\mp\bar{\psi}_{01}} + (1\pm\bar{n}_s)\boldsymbol{v}_s\mathrm{e}^{\mp\bar{\psi}_{01}} - \boldsymbol{v}_s\right)\mathrm{d}\xi \tag{106}$$

where n is the unit vector normal to the surface, ∇_s is the surface gradient operator, $\boldsymbol{v}_s$ is the component of the fluid flow velocity parallel to the surface and $\bar{\psi}_{01} = \bar{\psi}_{eq} + \bar{\psi}_d$.

Introducing dimensionless variables we can transform Eqs. (103) and (104) to:

$$\bar{\nabla}^2\bar{n}_s = Pe_i\bar{\nabla}\bar{n}_s\cdot\left(\bar{\boldsymbol{v}} - \bar{\boldsymbol{v}}_p\right) \tag{107}$$

$$(1+\bar{n}_s)\bar{\nabla}^2\bar{\psi}_s + \bar{\nabla}\bar{n}_s\cdot\bar{\nabla}\bar{\psi}_s = Pe_i\left(\frac{D_- - D_+}{D_- + D_+}\right)\bar{\nabla}\bar{n}_s\cdot\left(\bar{\boldsymbol{v}} - \bar{\boldsymbol{v}}_p\right) \tag{108}$$

Using the expansion defined by Eqs. (88)–(90) and retaining only linear terms one obtains [98,99]:

$$\bar{\nabla}^2\bar{n}_{s,1} = 0 \tag{109}$$

$$\bar{\nabla}^2\bar{\psi}_{s,1} = 0 \tag{110}$$

i.e. the linear terms in Pe_i fulfill the Laplace equation, which is in agreement with Dukhin theory of the thin polarized double layer [92,94]. The boundary conditions to Eqs. (109) and (110) can be obtained from the general ones [Eq. (106)] by applying expansion in Pe_i.

It was pointed out in Warszyński et al. [98,99] that the higher terms of the Pe_i expansion of ionic concentration and streaming potential cannot be described in terms of the Laplace equation, i.e. for faster particle motion there exists a distribution of the electric charge outside the EDL sometimes called the secondary double layer [100,101]. The range of this layer profoundly exceeds the range of the equilibrium double layer.

The linearity of the hydrodynamic equations allows to decompose the fluid flow into 'hydrodynamic' and electroosmotic flows $\boldsymbol{v} = \boldsymbol{v}_h + \boldsymbol{v}_e$. The electroosmotic flow $\boldsymbol{v}_e$ vanishes for uncharged particles. The Stokes equation is fulfilled for each flow separately, i.e.:

$$\mu\nabla^2 v_h = \nabla Ph \tag{111}$$

and

$$\mu\nabla^2 v_e = \nabla p_e + \rho\nabla\psi \tag{112}$$

where p_h and p_e are pressure due to the 'hydrodynamic' and electroosmotic flow respectively. Taking into account that Eq. (93) implies that the charge density in the distorted double layer is given by:

$$\rho = -2n_\infty(1 + \bar{n}_s)q\sinh\left(\bar{\psi}_{01}\right) \tag{113}$$

and for low Pe_i, $\rho = 0$ outside the double layer one can conclude that the electroosmotic flow outside the distorted EDL is governed by the same equation as hydrodynamic flow for uncharged systems.

If the particle surface can be assumed as locally flat Eq. (112) can be expressed in terms of local Cartesian coordinates as:

$$\mu\frac{\partial^2 \boldsymbol{v}_{e,s}}{\partial\xi^2} = \nabla_s p_e + \rho\nabla_s\psi$$
$$\frac{\partial p_e}{\partial\xi} + \rho\frac{\partial\psi}{\partial\xi} = 0 \tag{114}$$

where $\boldsymbol{v}_{e,s}$ is the component of the electroosmotic flow velocity parallel to the particle surface. Since $\partial\psi_s/\partial\xi \approx 0$ and $\partial\bar{n}_s/\partial\xi \approx 0$ the second equation can readily be integrated to receive:

$$p_e = 2n_\infty(1 + \bar{n}_s)kT\left(\cosh\left(\bar{\psi}_{01}\right) - 1\right) + p_{es} \tag{115}$$

where p_{es} is a function of the coordinates lateral to the surface. Introducing Eq. (115) into Eq. (114) one obtains:

$$\mu\frac{\partial^2 \boldsymbol{v}_{e,s}}{\partial\xi^2} = \nabla_s p_{es} + 2n_\infty kT\left(\cosh\left(\bar{\psi}_{01}\right) - 1\right)\nabla_s\bar{n}_s - 2n_\infty kT(1 + \bar{n}_s)\sinh\left(\bar{\psi}_{01}\right)\nabla_s\bar{\psi}_s \tag{116}$$

Neglecting small terms of the order of $1/(\kappa a)^2$ the electroosmotic flow velocity at the outer edge of the EDL can be obtained from Eq. (116) by double integration [94,95]. Eq. (116) simplifies for the small Pe_i when the polarization potential ψ_d can be neglected. The electroosmotic flow at the outer edge of the EDL is usually referred to as an electroosmotic slip [92] and its velocity can be used as the boundary condition for Eq. (112) outside the double layer. Thus, the motion of charged particle creates additional electroosmotic flow field. This flow field exerts some force on the particle which can be determined from Eqs. (13) and (14). For low ionic Peclet numbers this force is a linear function of the particle velocity, therefore, one can write that for $Pe_i \ll 1$ the resistance force is given by:

$$\boldsymbol{F}_p = \left(\underline{\underline{\boldsymbol{R}}}_t^h + \underline{\underline{\boldsymbol{R}}}_t^{ev}\right)\cdot\boldsymbol{U}_p \tag{117}$$

where $\underline{\underline{\boldsymbol{R}}}_t^h$ is the resistance matrix for uncharged particle, due to the hydrodynamic flow field v_h and $\underline{\underline{\boldsymbol{R}}}_t^{ev}$ is the electroviscous resistance matrix originated from the electroosmotic flow field. For $Pe_i \approx 1$ and higher also the non-linear terms appear and the force exerted on the particle induced by its motion depends also on higher powers of the particle velocity.

In the Section 5.2 the electroviscous drag force acting on the isolated particle and on the particle in the vicinity of the charged planar interface is considered. The drag force depends linearly on the particle velocity according to Eq. (117). Then the electrokinetic lift force acting on the charged particle moving parallel to the charged wall, being the example of the non-linear electrokinetic phenomena, is discussed.

5.2. Electroviscous drag

5.2.1. Isolated particle

When the uncharged particle moves in the liquid far from any external boundaries the relationship between the particle velocity and the drag force which in the steady-state is equal to the force causing particle motion is given by Eq. (17). For the charged particle this equation has to be modified to include electroviscous drag force, i.e.

$$\boldsymbol{U}_p = f_{\mathrm{Stev}} \boldsymbol{F}_p \tag{118}$$

where f_{Stev}, is the Stokes mobility coefficient corrected for the electroviscous drag which for uncharged particle reduces to f_{St}. Oshima et al. [97] used the low Pe_i linearization method of O'Brien [95] and calculated the sedimentation velocity of a charged particle for the arbitrary κa and particle surface potential ψ_p. For small electric potentials $q\psi_p / kT < 1$ they obtained analytical approximation which for the symmetric electrolytes can be expressed as [97]:

$$\begin{aligned}\frac{f_{\mathrm{Stev}}}{f_{\mathrm{St}}} = 1 - \frac{m_+ + m_-}{8}\bar{\psi}_p^2\Big(&\exp(2\kappa a)[3E_4(\kappa a) - 5E_6(\kappa a)]^2 \\ &+ 8\exp(\kappa a)[E_3(\kappa a) - E_5(\kappa a)] - \exp(2\kappa a) \\ &\times[4E_3(2\kappa a) + 3E_4(2\kappa a) - 7E_8(2\kappa a)]\Big)\end{aligned} \tag{119}$$

where

$$E_n(x) = x^{n-1}\int_x^\infty \frac{\exp(-\mathrm{t})}{\mathrm{t}^n}\mathrm{d}t \tag{120}$$

is the exponential integral of the order n [102] and the scaled drag coefficient of the positive and negative ions is defined as:

$$m_{\pm} = \frac{2\varepsilon kT}{3\mu q^2}\lambda_{\pm} \tag{121}$$

where the ionic drag coefficient $\lambda_{\pm}$ can be found from the limiting conductance $\Lambda^0_{\pm}$ according to Koryta et al. [103]:

$$\lambda_{\pm} = \frac{N_A q^2}{|z_{\pm}|\Lambda^0_{\pm}} = \frac{kT}{D_{\pm}} \tag{122}$$

For the arbitrary particle potential but for $\kappa a \gg 1$ another analytical approximation can be obtained [94,97]:

$$\frac{f_{\mathrm{Stev}}}{f_{\mathrm{St}}} = 1 - \frac{8\varepsilon}{\mu(\kappa a)^2}\left(\frac{kT}{q}\right)^2 \frac{D_- I^+_{1p} + D_+ I^+_{1p}}{D_+ D_-} \tag{123}$$

where

$$I^{\pm}_{1p} = \int_0^{\infty} \bar{\xi}(\mathrm{e}^{\pm\bar{\psi}_{eq}} - 1)\mathrm{d}\bar{\xi} = \mp\bar{\psi}^p_{eq} + 4\ln\left[\cosh\left(\frac{\bar{\psi}^p_{eq}}{4}\right)\right] \tag{124}$$

is often referred to as the first moment of the double layer [92].

Fig. 32 shows the results of the numerical calculation of Oshima et al. [97] of the ratio of the Stokes mobility coefficient for charged particle to the one for uncharged particle as a function of the reduced surface potential, calculated for several κa. The results of the analytical approximation valid for low particle surface potentials are also presented as the dashed lines. The general conclusion can be drawn that the correction to the mobility coefficient due to the particle charge is quite small not exceeding 10% for $\kappa a = 1$ and for the extremely high surface potentials. The correction is at its maximum for $\kappa a = 1$ for any value of the reduced surface potential. For κa approaching zero, i.e. for small particles in the diluted electrolytes as well for the opposite limit $\kappa a \to \infty$ (big particles in the concentrated electrolyte solutions) the mobility of charged particles approaches one for the uncharged particles.

The theoretical results presented in Fig. 32 were confirmed experimentally by Schumacher and van de Ven [104]. They determined the dependence of the diffusion coefficient of colloidal particles on the ionic concentration of the suspension using the dynamic light scattering technique. The measured values of the diffusion coefficient agreed well with one obtained from the theoretical mobilities shown in Fig. 32 by the Stokes–Einstein relationship [Eq. (21)].

5.2.2. Charged particle approaching the charged wall

The more complicated problem of a charged particle approaching the charge wall has been solved so far only in the 'thin double layer' approximation. Bike and Prieve [105] considered the electroviscous drag exerted by fluid squeezing out from the gap between identical spheres when the distance between them was much

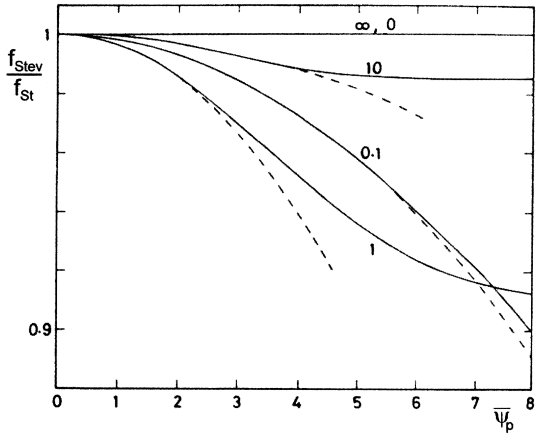


Fig. 32. Ratio of the Stokes mobility coefficient for charged particles to the one for uncharged particles as a function of the reduced particle surface potential. Solid lines are the results of numerical calculations, dashed lines — Eq. (119) for small potentials. Reprinted with permission from Oshima et al. [97].

greater than the EDL thickness, $\kappa h \gg 1$. In their approach they used the lubrication approximation. As it was demonstrated in Section 3.2 this approximation can be applied if the distance between interacting bodies does not exceed 20%. They obtained the analytical formula for the correction of the resistance coefficient to account for the electroviscous drag, in the form:

$$\frac{r_s^{ev}(H)}{r_s(H)} = 1 + \frac{4\varepsilon}{\mu(\kappa a)^2}\left(\frac{kT}{q}\right)^2 \frac{\bar{\psi}_p^2}{D_+ + D_-} \tag{125}$$

Under the same conditions $\kappa h \gg 1$ and $\kappa a \gg 1$ similar formula can be derived for the charged sphere approaching the charged wall:

$$\frac{\lambda_{\parallel}^{ev}(H)}{\lambda_{\parallel}(H)} = 1 + \frac{e}{4\mu(\kappa a)^2}\left(\frac{kT}{q}\right)^2 \left[\frac{I_{1w}^{+}(I_{1w}^{+} + I_{1c}^{+})}{D_+} + \frac{I_{1w}^{-}(I_{1w}^{-} + I_{1c}^{-})}{D_-}\right] \tag{126}$$

where

$$I_{1w}^{\pm} = = \mp\bar{\psi}_{eq}^{w} + 4\ln\left[\cosh\left(\frac{\bar{\psi}_{eq}^{w}}{4}\right)\right] \tag{127}$$

is the first moment of the EDL at the wall and $\bar{\psi}_{eq}^{w}$ is its equilibrium surface potential.

From Eqs. (125) and (126) one can conclude that when the distance between charged particle and charged wall exceeds considerably the EDL thickness and also the size of the particle is large in comparison with EDL, the resistance coefficient

can be well approximated by the one for the uncharged system. For instance the correction for the electroviscous drag for the particle with the surface potential $\psi_p = 100$ mV moving towards the charge wall with the same potential is at the distance $\kappa h = 10$ only 8%. For the oppositely charged bodies the correction is even smaller.

When the distance between particles or between particle and the wall is of the order of the EDL thickness the analytical solution for the drag force is not possible. Muller [106] and Warszyński and van de Ven [107,108] used the squeezing flow approximation to consider the flow between two charged disks and then they used the Derjaguin transformation to obtain the drag force between two spherical particles and/or between particle and the wall. They solved the set of the electrokinetic equation numerically under the condition $\kappa a \gg 1$ for the case of overlapping double layers. They used the c.p. and c.c. boundary conditions at surfaces of interacting bodies, however, to simplify the computations, for the distribution of the electric potential between disks they used one obtained by Hogg et al. [88] from the solution of linearized Poisson–Boltzmann equation. Thus, the results for high surface potentials in particular for the c.c. boundary conditions are largely overestimated.

Fig. 33 illustrates the ratio of the drag coefficient for approaching charged particles to one for uncharged particles as a function of the scaled distance between them calculated for various particle surface potentials at $\kappa a = 50$. For high surface potentials at close approach of the particles the correction for the electroviscous drag is around 100% having the maximum at $\kappa h = 1$. That means that charged particle on the approach to other charged interface are retarded two times stronger than uncharged one. For smaller separations between particles the correction due to the electroosmotic flow decreases as for the c.p. boundary condition the electric field in the gap between particles decreases. Fig. 34 shows the same dependence as in Fig. 33 but calculated for the same value of surface potentials (100 mV) assuming different ionic composition of the suspension. It shows that the electroviscous drag increases in the presence of less mobile ions like Li^+. The difference of the drag in the presence of Li^+ $\{D_+ = 1.03 \times 10^{-5} cm^2/s)$ and K_+ $\{D_+ = 1.96 \times 10^{-5}$ $cm^2/s)$ [100] is almost twofold. It is confirmed by a dependence of the drag on the ionic diffusion coefficient also at large distances as it is indicated in Eqs. (126) and (127). In Section 3 it was presented that the drag force exerted on the particle determines the rate of particle coagulation or adsorption process at macroscopic interface. Since the drag for charged particles depends on the ionic composition of the suspension, the similar dependence for the aggregation rate should be observed. Fig. 35a illustrates the dependence of the coagulation rate represented by the inverse of the stability ratio W on the salt concentration in solution expressed as κa calculated for a various dimensionless ionic mobility $b_i = (\varepsilon / \mu D_i)(kT/q)^2 = (3/2)m_i$. The stability ratio is defined as the ratio of the rapid to slow coagulation rates. All curves shown on a log–log scale exhibit a typical behavior predicted by the DLVO theory. The stability coefficient calculated taking into account the electroviscous drag is always higher than one obtained when the electroviscous drag in neglected, i.e. for $b_i = 0$. The coagulation

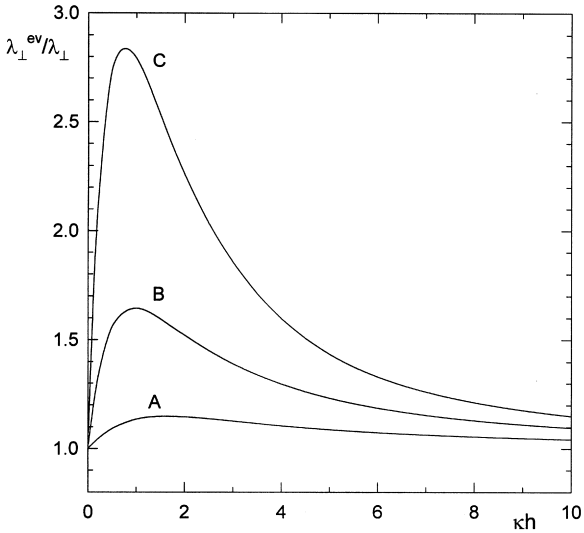


Fig. 33. Ratio of the resistance coefficient for the charged particle moving in the aqueous solution towards the surface of identical charged particles to the resistance coefficient for uncharged bodies as a function of interparticle distance. Curve A for $\bar{\psi}_p = -2$, curve B for $\bar{\psi}_p = -4$ and curve C for $\bar{\psi}_p = -6$, c.p. boundary conditions, $b_+ = 0.5$ and $\kappa a = 50$.

rate for a given κa is now dependent on the ionic mobility i.e. on their effective size [108]. In the slow coagulation regime for bigger ions (larger values b) a slower aggregation is predicted. The value of the critical coagulation concentration (CCC), defined here as the intercept of the extrapolation of the dependence of the aggregation rate on κa in the slow coagulation regime with axis $W = 1$ (cf. Fig. 35a) also depends on the ionic size.

Fig. 35b compares experimental data for the CCC obtained for As_2S_3 (closed circles) and AgI (open circles) [109] with the CCC predicted by the DLVO theory with the electroviscous drag taken into account [108]. It can be seen that the dependence of coagulation rate on the ionic composition for some ions — K^+, Na^+ and Li^+ — can be explained in terms of the electroviscous drag force. On the other hand for ions like H^+ and Rb^+ another mechanism, presumably preferential adsorption of this ion, must be involved.

The similar dependence of the deposition rate on the ionic size was also recorded for the adsorption of colloidal particles on macroscopic surfaces [110]. However, the observed effect was much stronger than it could be theoretically predicted [108]. This discrepancy still has to be resolved.

The electroviscous drag force can also be partly responsible for the experimentally observed independence of the slope of the colloid stability curve like one presented in Fig. 35a on particle size [111]. Since the relative importance of the drag force increases with decreasing κa it can partly compensate the decrease of the EDL repulsive forces with particle size making steeper the slope of the stability

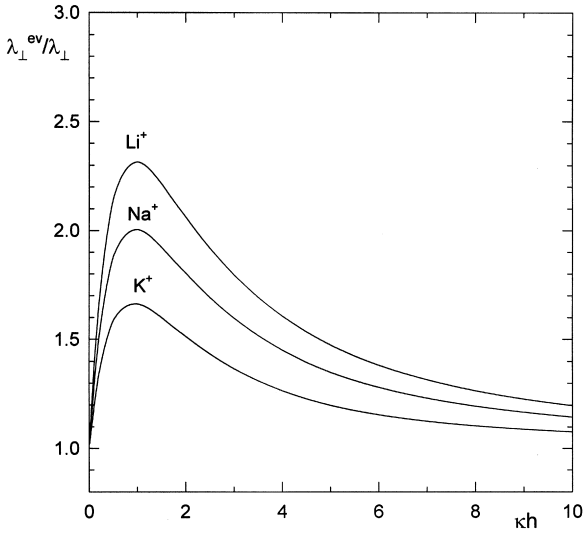


Fig. 34. The same relationship as in Fig. 33 compared for various counterions, $b_{K+} = 0.263$, $b_{Na+} = 0.386$, $b_{Li+} = 0.500$, $\bar{\psi}_p = -4$ and $\kappa a = 50$.

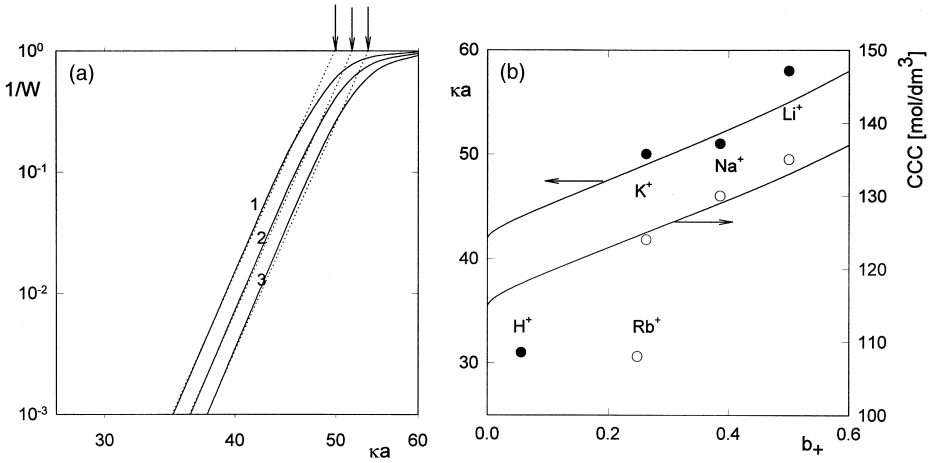


Fig. 35. Part a — the dependence of the coagulation rate expressed as the inverse of the stability ratio $(1/W)$ on the electrolyte concentration, κa. Curves 1–3 are for $b_+ = 0$, 0.25 and 0.50 respectively, $\bar{\psi}_p = -6$, c.p. boundary conditions, the Hamaker constant $A = 2 \times 10^{-20}$ J. The critical coagulation concentrations are marked by arrows. Part b — the comparison between the CCCs calculated taking into account the electroviscous drag force and the experimental data reported in Kruyt [109], closed circles — As_2S_3, open circles for AgI. Model parameters: $A = 2 \times 10^{-20}$ J, $\psi_p = -150$ mV, $a_{As2S3} = 35$ nm, $a_{Agi} = 50$nm.

curve. This phenomenon was extensively discussed in Warszyński and van de Ven [108].

The electroviscous drag forces play a much smaller role in the case where the particle approaches the oppositely charged interface. The dependence of the particle adsorption rate on the electrolyte composition exceeds the experimental error only at high *Pe* number and low electrolyte concentration when the coupling between the hydrodynamic interception and attractive double layer forces is the strongest [112].

The theory for the electroviscous drag for a case when the particle size is comparable with the EDL thickness is not yet available.

5.2.3. Charged particle moving along the charged wall

The problem of the charged particle moving along the charged wall was solved analytically using both the 'thin double layer' and lubrication approximation when the distance between the particle and the wall was much greater than double layer thickness by Warszyński and van de Ven [99]. They calculated the drag force on charged cylinders moving at the charged wall. Applying the Derjaguin transformation it is possible to transform their result and apply it to the motion of spherical particle. The resulting approximate formula for the drag coefficient is as follows:

$$\lambda_{\parallel}^{ev} = \lambda_{\parallel} + \frac{8\pi a\varepsilon}{(\kappa h)^2}\left(\frac{kT}{q}\right)^2 f_{ev}(\psi_w,\psi_p,D_+/D_-) \tag{128}$$

where:

$$fe_x(\psi_c,\psi_w,D_+/D_-) = 0.59\left(\frac{I_{1p}^{-\,2}}{D_-} + \frac{I_{1p}^{+\,2}}{D_+} - 0.472\left[\frac{I_{1p}^{-}I_{1w}^{-}}{D_-} + \frac{I_{1p}^{+}I_{1w}^{+}}{D_+}\right] + 0.106\left[\frac{I_{1w}^{-\,2}}{D_-} + \frac{I_{1w}^{+\,2}}{D_+}\right]\right) \tag{129}$$

and $\lambda_{\parallel} = (16/5)\pi\mu a^* \ln(a/h)$ is the drag coefficient for the uncharged spherical particle moving at the uncharged wall as calculated by Goldman et al. [50] in the lubrication approximation. Fig. 36 illustrates the dependence of the relative drag coefficient for the charged particle moving along the charged wall on the dimensionless particle wall separation for $\kappa a = 50$ and $\kappa a = 100$. As it can be seen in the figure for large particles, in comparison with the EDL thickness, moving far from the interface, i.e. at $\kappa h \gg 1$, the correction for the electroviscous effect is below 10%. The same conclusion was obtained by Bike and Prieve [102], however, they did not give any closed form formula. For the separations between the particle and the wall of the order of the EDL thickness neither analytical expression nor numerical calculation has not been yet published.

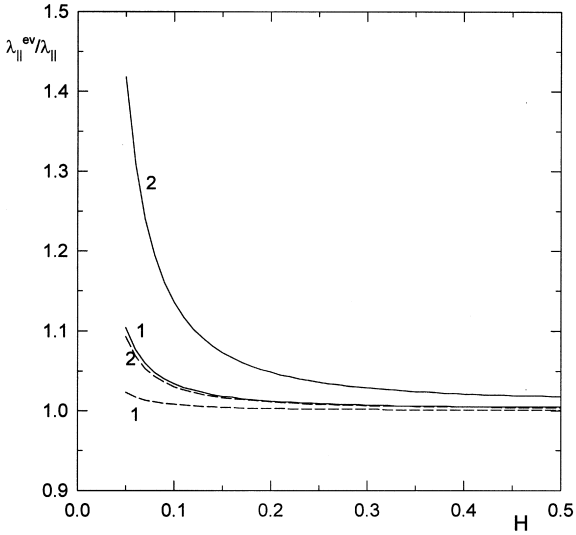


Fig. 36. The ratio of the drag coefficient for the charged particle moving in the aqueous solution along the charged interface to one for uncharged bodies as a function of the particle–interface separation calculated for $\kappa a = 50$ (solid lines) and $\kappa a = 100$ (dashed lines). (1) $\bar{\psi}_p = \bar{\psi}_w = 2$, (2) $\bar{\psi}_w = 4$.

5.3. Electrokinetic lift force

The electrokinetic lift force appears when the charged particle moves along the charged wall. Unlike the drag force which is directed oppositely to the particle motion the lift force acts in the direction normal to the wall. It was first observed by Prieve and co-workers [55] in the system of latex particles moving in the shear flow near the glass surface in the glycerol–water solution. They investigated the behavior of latex particles by means of direct microscopical observations. In the absence of flow the particles were trapped in the energy minimum determined by the balance of buoyancy and EDL repulsive forces. They observed that in concentrated 98% glycerol solutions after applying the shear flow the particle moved out from the surface to the new steady state position determined by the rate of shear. The new experimental results confirming the existence of the lift force in the same viscous, low conductivity medium were recently published [57,113]. Here the systematic studies of the effect of shear, medium conductivity, ionic strength and zeta potential were performed. Hollingsworth and Silebi [114] reported the dependence of the separation factor of the colloidal particles of various dimensions on the ionic strength of the eluent in the capillary hydrodynamic fractionation. They attributed that phenomenon to the presence of the electrokinetic lift.

For uncharged spherical particle moving along the smooth solid wall in the creeping flow the theory predicts no force in the direction perpendicular to the wall. Since the observed lift depends strongly on the medium conductivity it has to have electric origin.

The theory of the electrokinetic lift was initially developed for two limiting cases. For the case when the distance between the particle and the wall is much larger then the double layer thickness and much smaller than the particle radius Bike and Prieve [105] applied lubrication approximation and obtained the analytical expression for the lift force. The same authors [115] used the method of reflections to consider the case when the distance between particle and the surface is much greater than the particle radius. Similar case was also considered by van de Ven et al. [116]. Later on Bike and Prieve [117] extended their theory to the case of the spherical particle moving in a shear flow at an arbitrary distance to the wall. They used the 'thin double layer' approximation and solved set of the electrokinetic equation in the bispherical coordinates. However, in all this works the assumptions about low ionic *Pe* number was implicitly used, i.e. the set of the electrokinetic equations were linearized by expansion in terms of Pe_i retaining only linear terms. The lift force depending on the particle velocity squared originated from the Maxwell stress contribution of the long range streaming potential ψ_s. They predicted, in this way, electrokinetic lift force many orders of magnitude weaker than experimentally observed [57,105,113]. Whereas, in any experiment when the lift force was observed the conditions of the shear flow, particles velocity, medium viscosity and electrolyte concentration were such that the ionic Peclet number Pe_i were of the order of unity and higher. In their papers Warszyński et al. [57,98,99] recognized the electrokinetic lift as a non-linear elektrokinetic phenomenon and modified theory to account for the high Pe_i. They considered the motion of the charged cylinder at the charged wall in the lubrication approximation under the assumption of the 'thin double layer'. Then, they used Derjaguin transformation to find the lift force in the sphere–wall geometry. They solved the set of the electrokinetic equation by the expansion in terms of Pe_i retaining terms up to quadratic one [98]. They attributed the electrokinetic lift force to the viscous stress exerted on a charged particle by the electroosmotic flow calculated with the accuracy to the second order in Pe_i. Three types of hydrodynamic flow field were considered, connected with: the cylinder/sphere sliding along the wall without rotation, the cylinder/sphere rotating at the wall without translation and the stationary cylinder/sphere in the wall shear flow. Combining the contributions of these three types of the flow the lift force for the spherical particle moving in the wall shear flow can be obtained [49,51,98]. The resulting expression for the lift force exerted on the charged particle moving along the charged wall has the form [98]:

$$F_{\text{lift}}^{s-w} = \frac{\varepsilon\pi}{D_s^2\kappa^2}\left(\frac{kT}{q}\right)^2 \frac{G^2a^2}{H^2}\frac{(D_+I_{1w}^- + D_-I_{1w}^+)}{16(D_+ + D_-)^2}\left[H + \frac{8}{3}(u(H) + \omega(H))\right]^* \times[g_1(H) + u_1(H) + \omega_1(H)] \tag{130}$$

where

$$g_1(H) = 8H(D_-(I_{1w}^+ + I_{1s}^+) + D_+(I_{1w}^- + I_{1s}^-))$$

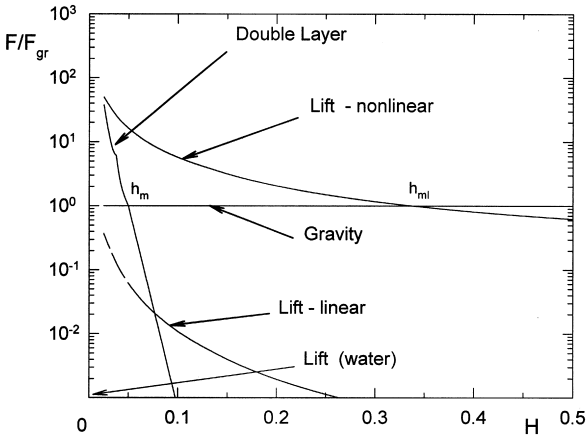


Fig. 37. The comparison between the gravity, the double layer forces and the lift forces, calculated according to non-linear and linear, for $Pe \ll 1$, theories acting on a charged particle moving in the simple shear flow in the vicinity of a charged wall. All forces scaled vs. gravity force are drawn as a function of scaled particle–wall separation. The parameters used for the calculations: $G = 6.6\ \mathrm{s}^{-1}$, $a = 5\ \mu\mathrm{m}$, $\kappa a = 145$, $\varepsilon = 42.5\varepsilon_0$, ($\varepsilon_0$ is the vacuum permittivity), $\bar{\psi}_p = \bar{\psi}_w = 2$ (50 mV), $D_s = D_+ = D_- = 2.5 \times 10^{-12}\ \mathrm{m}^2/\mathrm{s}$, which correspond to the conductivity $K = 0.7\ \mu\mathrm{S/m}$ of almost pure glycerol. The particle velocity in the shear flow calculated according to Eq. (48). h_m and h_{ml} denote the most probable separation between the particle and the wall in the absence and presence of shear respectively. The electrokinetic lift force in water ($K = 0.5$ mS/m) is given for a comparison.

$$\omega_1(H) = \frac{0.8436}{0.6376 - 0.2\ln H}(D_-(9I_{1w}^+ + I_{1s}^+) + D_+(9I_{1w}^- + I_{1s}^-)) \tag{131}$$

$$u_1(H) = \frac{0.7431}{0.6376 - 0.2\ln H}(D_-(I_{1w}^+ + 9I_{1s}^+) + D_+(I_{1w}^- + 9I_{1s}^-))$$

and the functions $u(H) = \dfrac{0.7431}{0.6376 - 0.2\ln H}$ and $\omega(H) = \dfrac{0.8436}{0.6376 - 0.2\ln H}$ describing the tangential and rotational motion of the sphere in the wall shear were calculated in Goren and O'Neill [49] and Goldman et al. [51] for uncharged bodies. In the case of thin double layers and for $\kappa h \gg 1$ they can be used for the charged sphere at the charged wall without considerable error.

Fig. 37 presents the comparison of the elektrokinetic lift force calculated according to linear and non-linear theory with other forces exerted on the 5-μm particle moving in the shear flow ($G = 6.6\ \mathrm{s}^{-1}$) in almost pure glycerol. The zeta potentials of the latex particle and glass surface were assumed equal to 50 mV and the ionic strength of the solution corresponded to $\kappa^{-1} = 35$ nm. The magnitude of the forces is scaled by the gravity force. As it can be seen in Fig. 37 the non-linear theory predicts several orders of magnitude stronger force than linear one in accordance with experimental findings [57,105,113]. For the sake of comparison the lift force in water under the same flow conditions is shown. It is evident that in water the lift force is negligible, unless the shear flow is very strong.

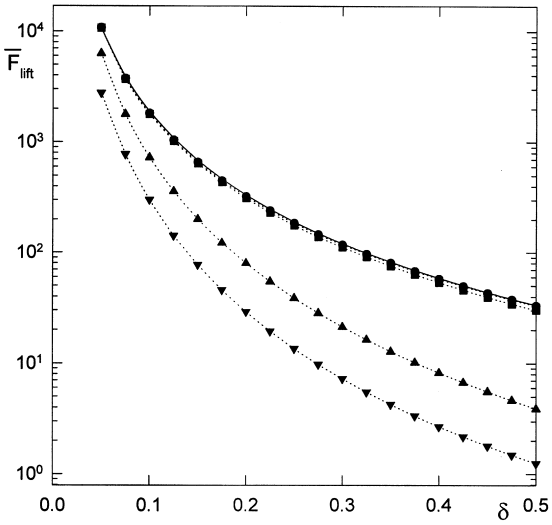


Fig. 38. The electrokinetic lift force scaled vs. $\frac{\varepsilon}{2D_s^2\kappa^2}(\frac{kT}{q})^2\frac{V^2}{a}$ for $\kappa a = 100$ and $\bar{\psi}_p = \bar{\psi}_w = 3$ as a function of the dimensionless cylinder–wall separation calculated according to: solid line analytical approach [99] and numerical solution of the ionic balance and electroosmotic flow equations for: circles — $Pe = 0.1$, squares — $Pe = 1$, upside triangles — $Pe = 10$ and downside triangles — $Pe = 20$.

For a sliding motion of the cylinder, Warszyński and van de Ven [99] checked validity of the analytical expression obtained by the expansion in terms of Pe_i. They solved the full set of electrokinetic equation in the 'thin double layer' and lubrication approximation but without resorting to expansion according to Eqs. (88)–(90). The results are presented in Fig. 38. Here the normalized lift force exerted on a charged cylinder moving at a charged wall is demonstrated as the function of the cylinder–wall separation. It is shown that for the Pe_i up to unity the lift force is well approximated by the analytical formula, whereas for higher Pe_i the analytical approximation seriously overestimates the elektrokinetic lift force.

As it is illustrated in Fig. 37 in the absence of shear flow the particle remains at the distance from the wall h_m determined by the gravity and the repulsive double layer forces. Since the particle is subjected to Brownian motion h_m is the average distance around which the particle position fluctuates. When the particle is subjected to flow along the surface, the new quasi-equilibrium distance has to be reached. That distance is determined by the balance of the gravity and lift force. The gradual motion of the particle out of the wall upon applying the shear flow allows detection of the lift force. Fig. 39 illustrates the principle of the experiment carried out in the 'colloidal particle collider' [56]. The wall shear flow was generated by moving the bottom plate of the measuring chamber with the constant speed. It was placed parallel to the glass surface, where the motion of particle was observed by the optical microscope, at the distance, 200 μm. From the measured velocity of the particle in the wall shear flow, its distance to the surface could be

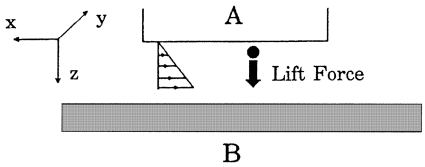


Fig. 39. The principle of the electrokinetic lift force measurement in the 'colloid particle collider' [57].

readily calculated by inverting relationship given by Eq. (48). As it was explained in the previous section when the double layers of the particle and the wall do not overlap and if $\kappa a \gg 1$, which was the case in the experimental conditions, there was no need to correct Eq. (48) for the electroviscous drag. Other details of the experiment can be found in Wu et al. [57].

The motion of the charged particle perpendicular to the solid charged wall during shearing can be described by:

$$\frac{\mathrm{d}h}{dt} = \frac{F_1(h/a)}{6\pi\eta a}\left[F_{\mathrm{lift}}(h)\tau(t) + F_{\mathrm{EDL}}(h) - \frac{4}{3}\pi a^3(\rho_f - \rho_p)g\right] \tag{132}$$

where ρ_f is the fluid density, ρ_p is the particle density, g is the gravitational acceleration. Function $\tau(t)$ expresses the dependence of the lift force on the effective shearing time as described in Wu et al. [57]. For the repulsive double layer forces, Eq. (80) describing the electric force in the sphere–plane geometry was used. Since in the experimental conditions the electric double layers around particle and at the wall did not overlap, to account for the hydrodynamic interactions we used the hydrodynamic correction function F_1 [see Eqs. (39) and (40)] for uncharged bodies, not adjusted for the electroviscous drag.

Applying Eq. (130) for the lift force in Eq. (132) we can compare our theoretical predictions with the results of experiments performed in the 'colloidal particle collider'. That is illustrated in Figs. 40 and 41 where the dependence of the distance of the latex particle to the wall on the time of applying shear flow is presented. The symbols represent the experimental results for the particle distance measured according to the procedure described above, lines correspond to the solution of Eq. (132). The results presented in Fig. 40 reflect the experimental conditions when the conductivity, i.e. the ionic strength of the solution, was kept constant for the variable shear rate whereas the results illustrated in Fig. 41 correspond to the variable conductivity and constant shear rate. All theoretical curves presented in figures were calculated using one adjustable parameter, zeta potential considered the same for sphere and the glass wall. Its best fit value -28 mV is lower than one predicted from extrapolation of electrophoresis data (-45 mV) [57], but taking into account rather severe approximations the agreement is quite good. Additionally it has to be taken into account that the approximate analytical approach based on the expansion in terms of Peclet number overestimates the lift force for $Pe > 1$ (see Fig. 38), which is the case in our experimental

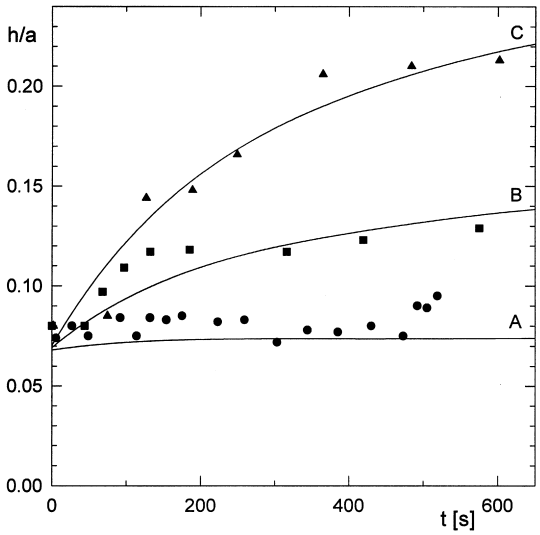


Fig. 40. Changes in the particle wall separation with the duration of shear flow as the result of the electrokinetic lift force observed for various shear rates. Curves: A for $G = 6.1\ s^{-1}$, curve B for $G = 14.0\ s^{-1}$ and curve C for $G = 19.1\ s^{-1}$ represent the results of the solution of Eq. (132) with $\psi_p = \psi_w = -28$ mV. Other parameters $a = 2.55\ \mu m$, $\varepsilon = 43.5\varepsilon_0$, $K = 5\ \mu S/m$. Symbols denote the experimental data obtained from the 'colloid particle collider' [57].

conditions. Thus, the obtained best fit value for the surface potential underestimates the real one.

As it can be seen in Figs. 40 and 41, in the low conductivity media and under strong shear due to the electrokinetic lift a charged particle is pushed away from the interface. When the average distance between the particle and interface increases the probability of adsorption due to diffusion decreases. It is worth to note that the lift force is always positive, even for the oppositely charged interfaces. Thus, application of strong lateral flow can prevent the adsorption even if in the absence of flow the adsorption in barrierless conditions occurs. The analysis presented here indicates that the electrokinetic lift force should be relevant for adsorption only in viscous, low conducting media. However, the results of Hollingsworth and Silebi [114] indicate that in conditions of intense flow the lift force may be important also in aqueous solutions.

6. Electro-hydrodynamic coupling in transport of particles to interfaces

6.1. The inverse salt effect

At the initial stages of the colloidal particle adsorption process the adsorption rate is determined by the transport mechanism and by the interactions between

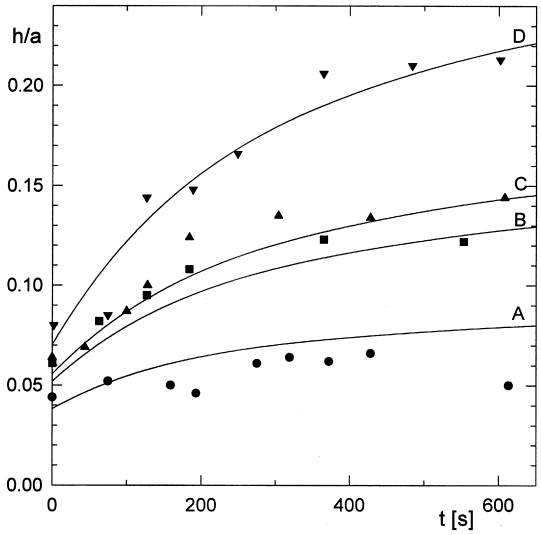


Fig. 41. Changes in the particle wall separation with the duration of shear flow as the result of the electrokinetic lift force for various suspension conductivities. Curve A for $K = 20$ μS/m, curve B for 10 μS/m, curve C for 8.4 μS/m and curve D for 5 μS/m, $G = 19.1\ s^{-1}$. Meaning of symbols and other parameters as in Fig. 40.

adsorbing particles and the interface. Strong attractive EDL interactions of oppositely charged surfaces facilitate the adsorption. This is the case in particular, when the attractive double layer forces act together with strong hydrodynamic flow field having a component directed perpendicular to the adsorbing surface. The increase of the particle flux to the interface in this condition is usually referred to as the 'inverse salt effect' [17]. It can be conveniently observed in the impinging jet or impinging slot experimental cells described in Section 2.1. Selected results are presented in Figs. 42 and 43.

In Fig. 42 the results for the deposition of negatively charged $a = 0.47$-μm latex particles on the positively charged mica surface are presented, obtained in the cylindrical impinging jet cell [17]. The dependence of the normalized particle flux on the ionic strength of the suspension are plotted for three Reynolds numbers: $Re = 150$ ($Pe = 0.61$), $Re = 100$ ($Pe = 0.34$) and $Re = 30$ ($Pe = 0.05$). The experimental points are in quantitative agreement with the theoretical predictions (solid lines) calculated from the numerical solution of the convective diffusion equation, Eq. (50). The attractive EDL forces were taken into account by applying Eq. (80) for the interaction between spherical particle and semi-infinite plane with $\psi_p = -50$ mV and $\psi_w = 100$ mV in accordance with elektrokinetic measurements of zeta potential of latex particles and mica sheets. The hydrodynamic correction functions in Eq. (50) were taken as for uncharged particles as for the oppositely charged surfaces the correction for the electroviscous drag force can be neglected.

Similar experiments were performed in the slot impinging jet cell, stagnation line

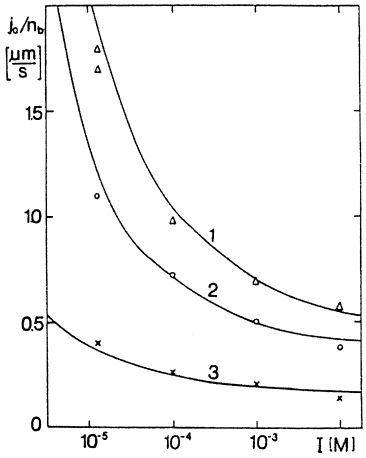


Fig. 42. The normalized particle flux j_0/n_b vs. the ionic strength I determined experimentally for the monodisperse latex suspension ($a = 0.47$ μm) in the impinging jet cell [17] for various Reynolds number. (1) $Re = 150$, (2) $Re = 100$ and (3) $Re = 30$. The solid lines denote the theoretical results obtained from numerical solution of transport equation.

flow [23]. Fig. 43 illustrates the dependence of the normalized flux of negatively charged $a = 0.5$ μm latex particles to the positively charged mica sheets on the Reynolds number of the flow for three selected ionic concentrations of the suspension. Symbols are experimental points, solid lines are the result of the solution of Eq. (50) and dashed line represent the approximate model of Smoluchowski–Levich, Eq. (54). As it can be seen in Fig. 43 for low Reynolds numbers all curves tend to the common limit which can be well described by the Smoluchowski–Levich approximation neglecting both hydrodynamic and EDL interactions. It means that for a weak flow the role of any attractive interactions either van der Waals or EDL is only secondary and the adsorption rate is determined by the transport mechanism.

The results presented in Figs. 42 and 43 suggest that for intense flow particle, adsorption is significantly enhanced by the attractive double layer forces. This phenomenon is of an electrophoretic origin caused by the non-uniform electric field at the interface accelerating particle transport through the diffusion layer. Decreasing the electrolyte concentration in the suspension the range of attractive EDL forces increases. On the other hand the effect can be interpreted as an enhanced interception effect. Due to the attractive EDL forces the effective particle size increases, enhancing the probability of particle capture by the interface. The effective particle size can be conveniently approximated by:

$$a_{\text{eff}} = a(1 + H^*)^2 \tag{133}$$

where H^* can be treated as the effective interaction range. In the case of the

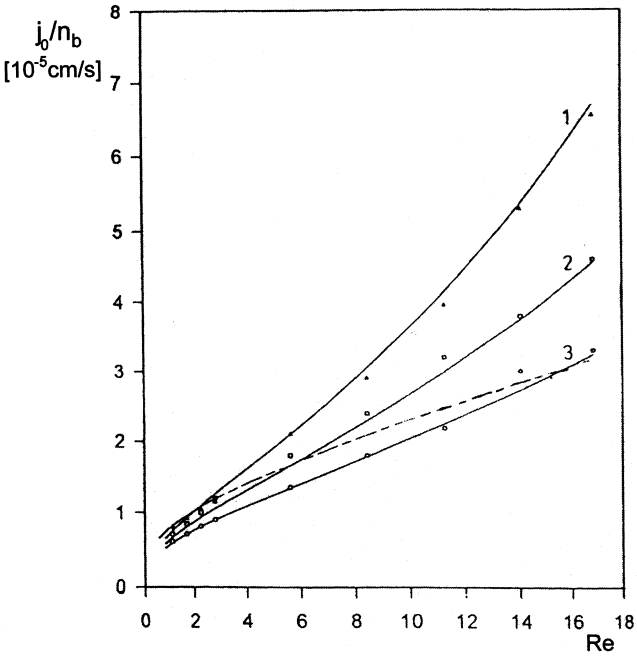


Fig. 43. The normalized particle flux j_0/n_b vs. the Reynolds number *Re* determined experimentally for the monodisperse latex suspension ($a = 0.5$ μm) in the slot impinging jet cell [23] for various ionic strength. (1) $I = 2 \times 10^{-5}$ M, (2) $I = 10^{-4}$ M and (3) $I = 10^{-3}$ M. The solid lines denote the theoretical results obtained from numerical solution of transport equation, the dashed lines are the results of the Smoluchowski–Levich approximation.

attractive EDL interactions between a spherical particle and a wall it can be found from the relationship [17]:

$$H^* = \frac{1}{\kappa a}\left[Z - 2\ln\left(1 + \frac{Z}{\kappa a}\right)\right]$$
$$Z = \ln\left(\frac{2|\Phi_{pw}|\kappa a}{kTPe}\right) \tag{134}$$

where the interaction energy Φ_{pw} was defined in Eq. (80).

Analogous observations were done by Elimelech [118] who used the packed bed column of glass beads of average diameter 0.046 cm to perform adsorption of positively charged latex particles of various size ranging from 0.08 to 2.51 μm. The obtained adsorption rates were in quantitative agreement with the theoretical predictions [17].

For the case of particle adsorption in the presence of repulsive double layer interactions, i.e. in the barrier-controlled adsorption regime, the mechanism of

particle transport to the interface plays only a secondary role [8,61]. The other factors like fluctuations in the surface charge distribution and/or surface heterogeneities to larger extent influence particle adsorption rate [119]. The theory developed mostly for the smooth, uniform surfaces predicts usually much lower adsorption rates than it is observed in experiments [120,121].

6.2. Electro-hydrodynamic blocking

As it was mentioned before, with the progress of adsorption as the surface concentration of particles increases, the interaction of the adsorbing particles with ones already adsorbed at the interface becomes increasingly important. In Section 3.3 the hydrodynamic interactions between uncharged hard spheres in the vicinity of interface were discussed and their effect on particle adsorption was examined. In this section the effect of the lateral repulsive EDL interactions between adsorbing and adsorbed particles in absence and presence of the suspension flow is considered.

As it was illustrated in Figs. 13 and 14, when the particle moves along the interface in the simple shear flow and encounters the second particle which is firmly attached to the interface the hydrodynamic scattering occurs. The trajectory of the moving particle after the encounter is shifted out of the interface. If we study the set of non-Brownian, virtual particles of which the positions before the collision with the attached one is chosen at random, the coordinates of their trajectories after the collision form the characteristic, almost semicircular collision scattering pattern as shown in Fig. 14a. When the particle is charged the long range EDL repulsion changes the size of the collision pattern. It is illustrated in Fig. 44 where the collision patterns obtained by the solution of the trajectory equation for 200 virtual particles are shown for three values of κa, equal to 20 — part a, 60 — part b and 180 — part c. The repulsive EDL interactions were calculated according to Eq. (78) for the sphere–sphere geometry with $F_{pp}a/kT = 1000$. The interface was assumed uncharged and only hydrodynamic interactions between moving particle and interface were taken into account. Other details of the simulations were discussed in Section 3.3.

According to the expectations the radius of the scattering pattern raises when κa decreases as the range of the EDL interaction changes. This behavior can easily be interpreted in terms of the effective particle size. The geometrical size of particles is increased by the range of the EDL forces, since the particles cannot approach each other closer than thickness of the electrostatic barrier.

In the simulations illustrated in Fig. 44 the electroviscous drag forces were neglected since there is no available expression to account for these forces at arbitrary particle separation and the EDL thickness. One can expect that the electroviscous drag should result in broader collision pattern for small κa. For large κa the correction should become negligible as it was discussed in the previous section.

The size of the collision pattern is a measure of interactions between particles and therefore this phenomenon can be applied for studying interparticle forces.

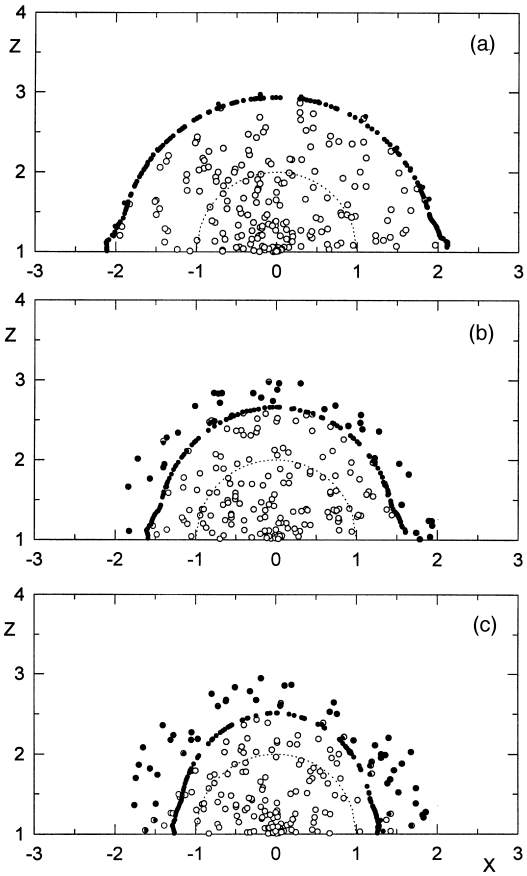


Fig. 44. The collision pattern formed by the final positions of 200 non-Brownian virtual particles colliding in the simple shear flow with the particle attached to the interface at (0,0,1). Open circles are the initial positions of virtual particles chosen at random, solid circles are the final positions. The contour of the immobile particle is shown as a dotted line. Part a for $\kappa a = 20$, part b for $\kappa a = 60$ and part c for $\kappa a = 180$.

That was accomplished in the 'colloidal particle collider' developed by van de Ven et al. [56]. Its principle of operation is shown in Fig. 45. The collision of two colloidal particles, one attached to the transparent surface and the other moving in the simple shear flow, is observed by means of the optical microscope. The shear flow is produced between two parallel plates, positioned at the typical distance of 200 μm, by moving the bottom plate mounted at the platform of the computer driven manipulator. The collision of particles is recorded and then analyzed using the image analyzing techniques. The details of the experiment and processing the experimental results can be found in van de Ven et al. [56] and Wu et al. [58].

Figs. 46 and 47 present the experimental scattering patterns obtained using

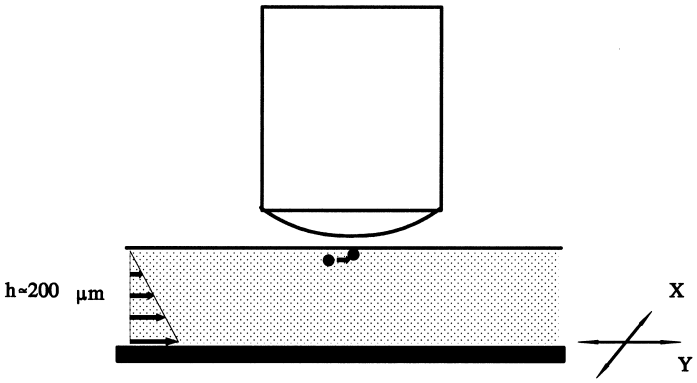


Fig. 45. Principle of experimental observations of particle collisions in the 'colloidal particle collider' [56].

'colloidal particle collider'. In Fig. 46 the results of surface collisions of $a = 1.7\ \mu m$ latex particles in 28% glycerol–water solutions with 10^{-4} M KCl are shown. The open circles represent the initial positions of the moving particles and the filled circles their final positions after the collision. As it can be seen the collision pattern is dispersed by the Brownian motion of particles during the collisions. Applying the statistical analysis (for details see van de Ven [56] and Wu et al. [58]) it is possible to calculate from the experimental collision pattern, the relationship between the interparticle interaction force and the distance between them. That kind of relationship, obtained basing on the collision results shown in part a of Fig. 46, is presented in part b of the same figure. The obtained force distance curve corresponds to a typical DLVO with the van der Waals forces calculated according to Eq. (37)) and EDL forces according to Eq. (78). The fitted Hamaker constant $A = 2.1 \times 10^{-20}$ J and surface potential $\psi_p = -54$ mV are in good agreement with the values obtained from spectroscopical data and measured by microelectrophoresis [56]. The solid triangles in Fig. 46a stand for the hypothetical final positions of the particle calculated from the solution of the trajectory equation using the interaction force profile shown in the part b of this figure without the Brownian motion taken into account.

Fig. 47 shows the similar results as in Fig. 46 but obtained in a 60% D_2O–water mixture with 10^{-3} M KCl. Since the concentration of the electrolyte is higher the size of the collision pattern is smaller than one presented in Fig. 46a. It is reflected in the force–distance relationship shown in Fig. 47b. The range of the repulsion is shorter and the particles can approach each other at much smaller distance. Also, the secondary minimum can be observed, however, its depth is within the experimental error.

The magnitude of the interaction force as shown at the right vertical axis of the graph presented in Fig. 47a proves that using the 'colloidal particle collider' one can record forces between the particles of the order of several fN (10^{-15} N).

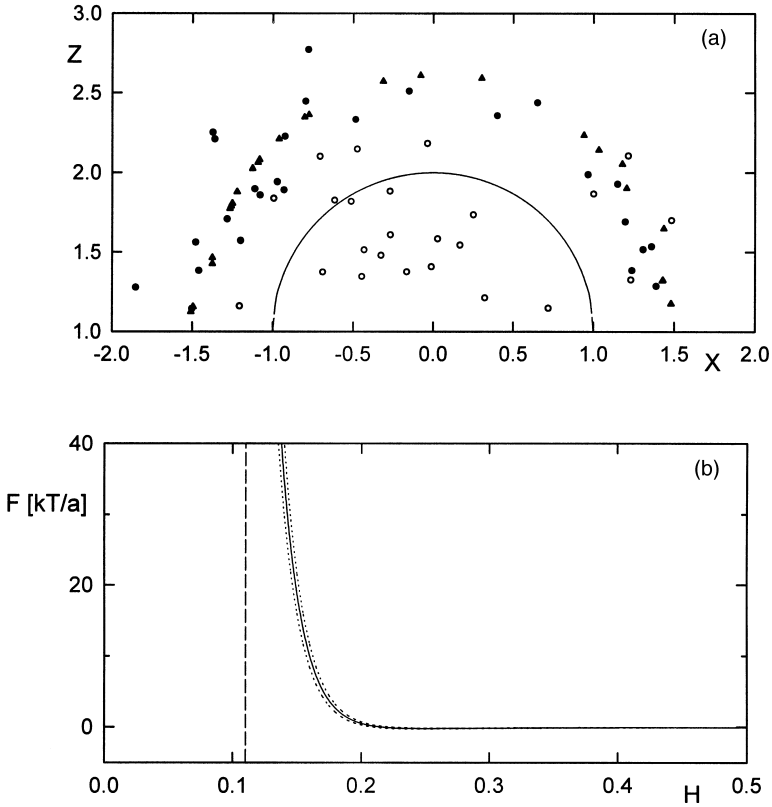


Fig. 46. Part a — experimental collision pattern of $a = 1.7$-μm latex particle in a 28% glycerol–water solution with 10^{-4} KCl [56]. The coordinates are scaled by the particle radius. The open circles represent the initial positions of moving particles, the filled circles represent the final positions of the particles. The filled triangle stand for the theoretically calculated final positions of the particle assuming the interaction force shown in part b and neglecting the Brownian motion. The semicircle in the middle is the projection of the stationary particle. Part b — fitted interaction force vs. interparticle distant relationship obtained by inversion of trajectory equation from the experimental data shown in part a. The solid line is the force–distant curve, the dotted lines are the confidence intervals (90%) of the force profile and the dashed vertical line marks the minimum separation distance between two particles observed in experiment.

From the results presented in Figs. 45–47 one can conclude that the long range of repulsive EDL interactions increase the effect of hydrodynamic scattering. The shift of particle trajectory out of the interface strongly depends on the electrolyte concentration of the suspension. As it was shown in Section 3.3, the hydrodynamic scattering for the uncharged particles is responsible for the decrease of the adsorption rate when the interface is partially covered by already adsorbed particles. If the adsorbing particles are charged, this effect of hydrodynamic shadowing is enhanced by the repulsive EDL interactions. Thus, behind the particle attached

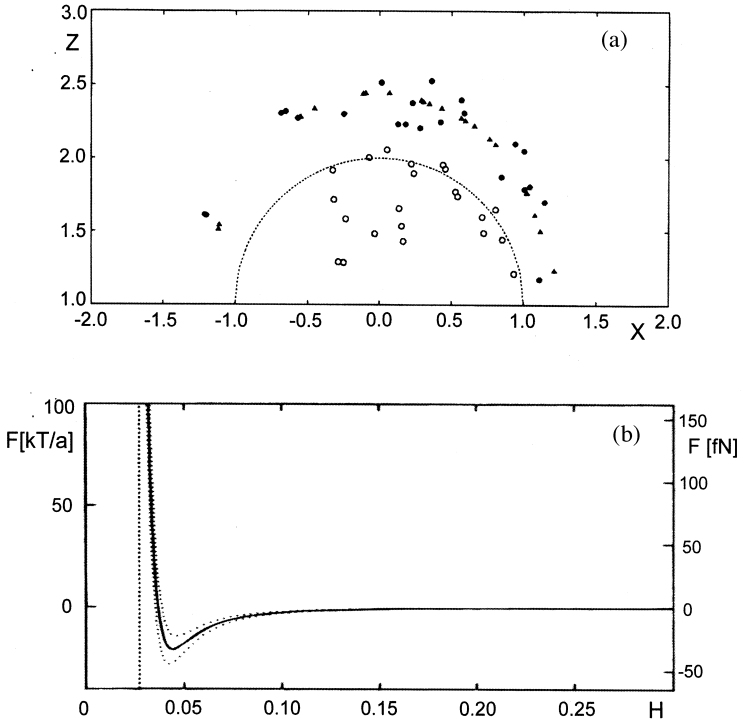


Fig. 47. The same as in Fig. 46 but for collisions of a = 2.5-μm latex particle in a 60% D_2O–water mixture with 10^{-3} M KCl.

to the interface, the electro-hydrodynamic shadow is formed which is inaccessible for further adsorption of particles. That is illustrated in Fig. 48 where the area blocked by a single particle at the interface in the stagnation line flow (scaled vs. C_1 — the area blocked by the uncharged particle in the absence of flow) is shown as a function of the Pe number. The adsorbed particle was placed far from the stagnation line, $Y_d = 100$, and the particle trajectories were calculated without the Brownian motion taken into account, by solving Eq. (15) with the mobility matrices calculated using the multisubunit method of Dąbroś and van de Ven [66]. The electroviscous drag forces were neglected in the trajectory calculations. The EDL interactions were accounted for in terms of LSA approximation with $F_{pp}a/kT = 2000$. As it can be seen in Fig. 48 the electrostatic repulsion strongly increases the area formed behind the attached particle inaccessible for further adsorption. For instance at $Pe = 0.5$ and $\kappa a = 20$ the size of the electro-hydrodynamic shadow is six times bigger then the respective one for uncharged particle (compare curve 1 and dotted line for uncharged particle). For low concentrations of electrolyte in suspension, if particles and interface are oppositely charged, the attractive interaction between them tends to reduce the size of the excluded area. As it is illustrated in Fig. 48 for $\kappa a = 20$ and high Pe the size of the shadow is strongly reduced by

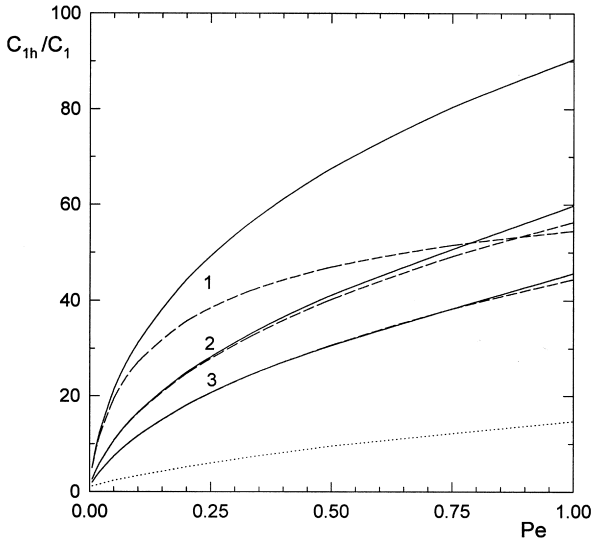


Fig. 48. The dependence of the scaled size of the non-Brownian electro-hydrodynamic shadow formed behind the particle attached to the interface in the stagnation line flow at the distance $Y_d = 100$ from the stagnation line on the flow intensity expressed in terms of the Peclet number. The particle–particle EDL interactions were calculated for $F_{pp} = 2000\ kT/a$ and particle–wall interactions for $F_{pw} = 0$ (solid lines) and $F_{pw} = -2000\ kT/a$ (dashed lines); curves (1) for $\kappa a = 20$, (2) for $\kappa a = 60$ and (3) for $\kappa a = 180$. $Ad = 0.4$ for both particle–particle and particle–wall van der Waals interactions.

the EDL attraction to the oppositely charged interface and becomes comparable with one for the higher electrolyte concentration (higher κa) for uncharged surface. For high electrolyte concentration the effect of interface attraction on the size of the shadow becomes negligible as the range of particle–wall EDL interactions is strongly reduced.

The electroviscous drag forces should further increase the size of the electro-hydrodynamic shadow especially for low κa, but since the general theory for arbitrary interparticle distance is not available, it is not considered here.

That assumption seems to be justified since the size of the electro-hydrodynamic shadow is mainly determined by the Brownian motion of adsorbing particles. That is illustrated in Figs. 49 and 50 where the results of Brownian dynamics simulations of particle adsorption in the stagnation line flow are shown. The stationary particle, which contour is marked by the solid line, were placed at the distance $Y_d = 316$ from the stagnation line in the flow with $Pe = 0.09$ (Fig. 49) and $Pe = 0.30$ (Fig. 50) for $\kappa a = 7$, 16 and 50. For the particles with $a = 500$ nm this corresponds to the monovalent electrolyte concentration in the suspension of 2×10^{-5} M, 10^{-4} M and 10^{-3} M, respectively. The EDL interactions were calculated for $F_{pp} = 2000\ kT/a$ ($\psi_p = -50$ mV) and particle–wall interactions for $F_{pw} = -4000\ kT/a$ ($\psi_w = 100$ mV). Points represent the final positions of virtual particles at the

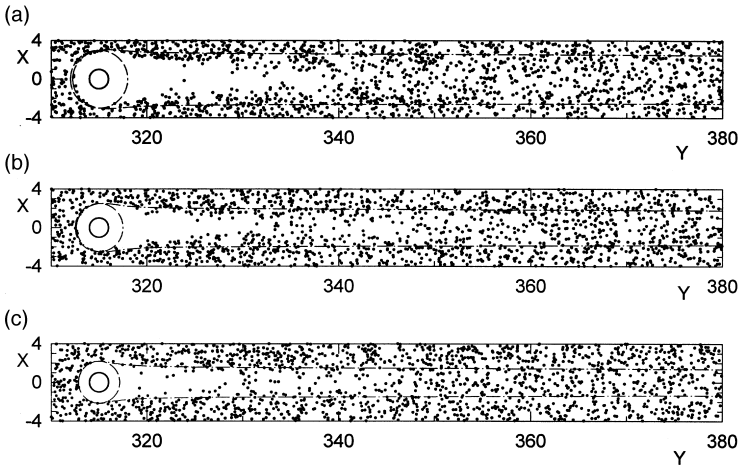


Fig. 49. The simulated Brownian electro-hydrodynamic shadow behind the particle attached to the interface at $Y_d = 316$ from the stagnation line for $Pe = 0.09$. Part a for $\kappa a = 7$, b for $\kappa a = 16$ and c for $\kappa a = 50$. The particle–particle EDL interactions were calculated for $F_{pp} = 2000\ kT/a$ and particle–wall interactions for $F_{pw} = -4000\ kT/a$, $Ad = 0.4$ as previously. Solid circle represents the contour of the attached particle, dashed circle is the area blocked by the charged particle in the absence of flow and dash–dot line is the non-Brownian electro-hydrodynamic shadow.

interface which trajectories were found solving Eq. (19). The dash–dot lines illustrate the electro-hydrodynamic shadow calculated for non-Brownian particles, whereas the dash circles symbolize the area blocked by the charged particle in the absence of flow. As it was discussed in Section 2.3 this area (scaled by the particle geometrical cross section) is equal to the first coefficient in the virial expansion of the available surface function $B(\Theta)$ [see Eq. (27)]. The latter can be evaluated from Adamczyk and co-workers [32,70]:

$$C_1 = 4(1 + H^*)^2 \tag{135}$$

where $H^* = h^*/a$, is the dimensionless effective interaction range which can be approximately calculated from the dependence [32,70]:

$$H^* = \frac{1}{2}\kappa a \ln\frac{\Phi_{pp}}{\Phi_{ch}} - \kappa a \ln\left(1 + \frac{1}{2}\kappa a \ln\frac{\Phi_{pp}}{\Phi_{ch}}\right) \tag{136}$$

where Φ_{ch} the characteristic energy close to 10 kT units.

Next virial coefficients can be analogously determined from:

$$C_2 = \frac{6\sqrt{3}}{\pi}(1 + H^*)^4$$

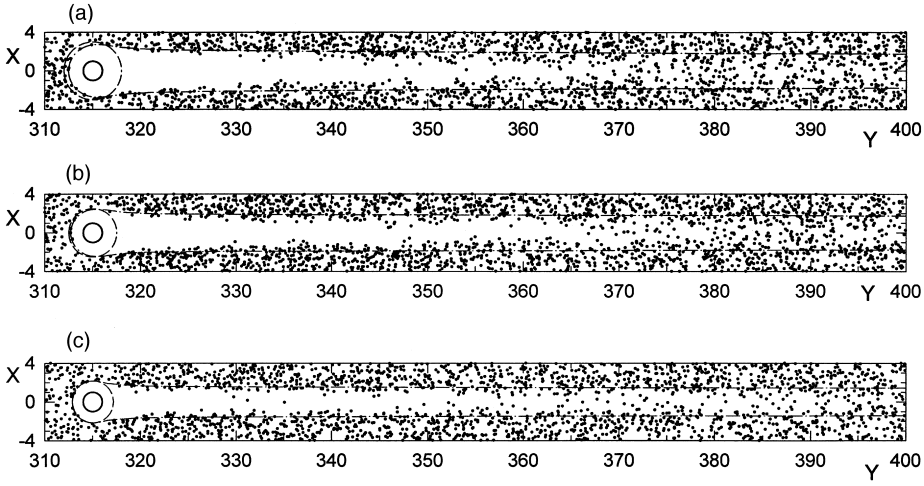


Fig. 50. The same as in Fig. 49 but calculated for $Pe = 0.30$.

$$C_3 = 1.407\,(1 + H^*)^6 \tag{137}$$

It can be seen that the effect of blocking for charged particles in the absence of flow can be adequately described using the concept of the effective hard spheres having the radius increased over their geometrical dimensions by h^*.

In the presence of flow the electro-hydrodynamic shadow is formed behind the deposited particle. As it is shown in Figs. 49 and 50 the size of this shadow is much smaller than for non-Brownian particles and its boundaries are diffused due to the thermal motion. The results of simulations presented in Figs. 20, 49 and 50 suggest that the length of the shadow depends mainly on the ratio of the local flow intensity to the diffusion, i.e. on the Pe number and Y_d the location of the attached particle with respect to the stagnation line, while the width of the shadow is determined by the EDL interactions, i.e. by the effective particle size $a^* = a\,(1 + H^*)$.

Analogously to Eq. (27) in the presence of flow one can determine the available surface function as a virial expansion:

$$B(\Theta) = 1 - C_{1h}\Theta + C_{2h}\Theta^2 + C_{3h}\Theta^3 + \cdots \tag{138}$$

where now the virial coefficients C_{ih} depend on the local flow conditions and the first coefficient C_{1h} has a meaning of the effective area (scaled by particle geometrical cross section) of the electro-hydrodynamic shadow. Taking into account the simulation results presented in Figs. 49 and 50, and ones presented in Section 3.3, one can conclude that C_{1h} is a complicated function of many parameters as: flow geometry, the Pe number, the location of attached particle, particle and wall surface potential, ionic strength of the suspension and its Hamaker constant. Thus, in general it can be written that:

$$C_{1h} = 4(1 + H^*)^2 + F_C(Pe, Y_d, \kappa a, F_{pp0}, F_{pw0}, Ad) \tag{139}$$

where F_C expresses the correction for the electro-hydrodynamic shadow for Brownian particles.

In the absence of flow, $Pe = 0$, it reduces to C_1 for charged particles as described by Eq. (135) while for $Pe\ Y_d \to \infty$ it describes the electro-hydrodynamic shadow for non-Brownian particles as shown in Fig. 48. The detailed form of the function F_C can be determined only by the time consuming SBD techniques which has not yet been accomplished. The same concerns higher virial coefficients. Adamczyk et al. [18,36] proposed the simple expression for C_{1h} and C_{2h} assuming that the size of the hydrodynamic shadow is proportional to the local shear rate:

$$C_{1h} = 4(1 + H^*)^2 + C_h\overline{G} = 4(1 + H^*)^2 + C_h PeY_d$$
$$C_{2h} = \frac{6\sqrt{3}}{\pi}(1 + H^*)^4 + \frac{f_g}{\pi}\left[\frac{C_h}{f_g}PeY_d - 2\right]^2 \tag{140}$$

where C_h and f_g are the dimensionless parameters dependent on the electrolyte concentration. These expressions are valid when $C_h PeY_d/f_g > 2$.

For the low surface coverage, up to $\Theta = 0.2$, one can neglect in the virial expansion all terms higher than Θ^2 and use Eq. (138) to describe the adsorption kinetics, i.e. the dependence of surface coverage on time according to Eq. (25):

$$\frac{d\Theta}{d\tau} = 1 - C_{1h}\Theta + C_{2h}\Theta^2 - \overline{K}_d\Theta \tag{141}$$

Eq. (141) can readily be solved to obtain:

$$\Theta(\tau) = \Theta_{st}\frac{1 - \exp(-C_{2h}\sqrt{p^2 - 4q}\,\tau}{1 - \dfrac{\Theta_{st}}{\Theta_2}\exp\left(-C_{2h}\sqrt{p^2 - 4q}\,\tau\right)} \tag{142}$$

where

$$p = \frac{C_{1h} + \overline{K}_d}{C_{2h}}$$
$$q = \frac{1}{C_{2h}}$$
$$\Theta_{st} = \frac{P}{2}\left(1 + \sqrt{\frac{4q}{p^2}}\right) \tag{143}$$

$$\Theta_2 = \frac{p}{2}\left(1 - \sqrt{\frac{4q}{p^2}}\right)$$

$$\frac{4q}{p^2} < 1$$

Fig. 51 presents the experimentally determined dependence of the adsorption kinetics of $a = 500$ nm, negative polystyrene latex particles on the positively charged silanized mica sheets [122]. The experiments were performed in the slot impinging jet experimental cell at the distance $y = 158$ μm ($Y = 316$) from the stagnation line for $Re = 2$ ($Pe = 0.01$ — white symbols) and $Re = 16$ ($Pe = 0.30$ — black symbols) for three electrolyte concentrations of the suspension: $2 \cdot 10^{-5}$ M (triangles), 10^{-4} M (squares) and 10^{-3} M KCl (circles). Lines represent fits of the experimental data to Eq. (142) with $\bar{K}_d = 0$ since no desorption was observed even at the highest flow intensity. One can see that after the period of initial growth the surface coverage attains the quasi-saturation value Θ_{st} which is strongly depended on the flow intensity and the electrolyte concentration of the suspension as this saturation is a result of the electro-hydrodynamic blocking effect. At $Re = 16$ and $I = 2 \times 10^{-5}$ M the quasi-saturation is attained already when only 5% of the interface is occupied by adsorbed particles. It is only one-fifth of the saturation which can be reached in the absence of flow at this ionic strength (24%) and less than one-tenth of the jamming limit for uncharged particles without the flow (54.7%). The saturation is only apparent since, as it was demonstrated in Fig. 22, it can be easily exceeded either by reducing the flow or by increasing the ionic strength of the suspension.

Since the electro-hydrodynamic scattering depends on the position relative to the stagnation line, so does the adsorption rate even though the interface in the stagnation line flow is uniformly accessible. The surface concentration profile of particles is established during the adsorption process. The illustrative example of such a profile is shown in Fig. 52 where the microscopic picture of $a = 0.5$ μm latex particles deposited on mica for $Re = 16$ and $I = 10^{-4}$ KCl is presented [18]. The surface coverage at the symmetry line $\Theta = 20\%$ is close to the quasi-saturation limit Θ_{st}.

The concentration profiles of the negatively charged polystyrene latex particles ($a = 725$ nm) adsorbed on the positively charged mica plates at the quasi-saturation Θ_{st} observed in the slot impinging jet cell [122] at $Re = 2$ (white circles) and $Re = 16$ (black circles) in 2×10^{-5} KCl are illustrated in Fig. 53. For $Re = 16$ the apparent saturation concentration drops by 50% already at the distance of 50 particle radii from the stagnation line. For $Re = 2$ the concentration drop is less dramatic but noticeable.

The surface coverage at the quasi-saturation Θ_{st} is directly determined by the virial coefficients C_{1h} and C_{2h}. Thus, by fitting Eq. (142) to the experimental adsorption kinetics one can obtain the information concerning the parameters of the electro-hydrodynamic shadow. In Fig. 53 the dependence of C_{1h} on the distance (scaled vs. particle radius) from the stagnation line calculated from the

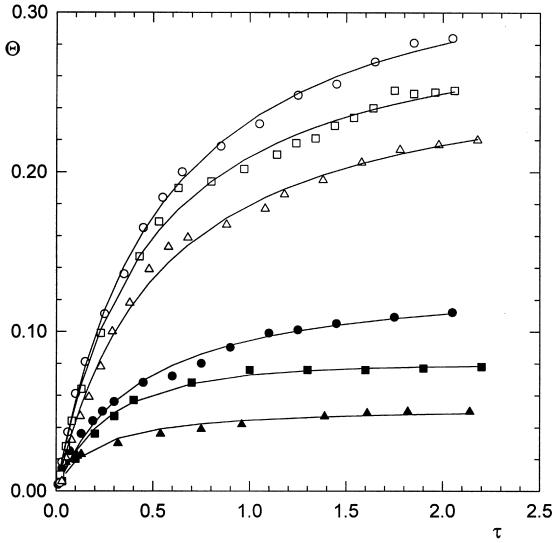


Fig. 51. The kinetics of adsorption of the monodisperse latex particles ($a = 0.5$ μm) on silanized mica observed in the slot impinging jet cell [122] at the distance of 158 μm from the center for $Re = 2$ (white symbols) and $Re = 16$ (black symbols). Triangles — $I = 2 \times 10^{-5}$ M, squares — $I = 10^{-4}$ M and circles — $I = 10^{-3}$ M KCl. Lines represent fits of the experimental data to Eq. (142).

quasi-saturation coverages Θ_{st} for $Re = 2$ (white triangles) and $Re = 16$ (black triangles) is presented. The results confirm that the size of the electro-hydrodynamic shadow depends linearly on the position with respect to the stagnation line as it was assumed by Adamczyk et al. [18,36].

Fig. 54 shows the dependence of the C_{1h} on the Peclet number Pe for the constant distance from the stagnation line $y = 158$ μm. In this figure the points represent data obtained by fitting Eq. (142) to the adsorption kinetics of polystyrene latex particles on mica plates under various Reynolds number of the stagnation line flow. The electrolyte concentration were: 2×10^{-5} M (triangles), 10^{-4} M (squares) and 10^{-3} M KCl (circles). The solid lines in Fig. 54 represent fits of the relationship:

$$C_{1h} = 4(1 + H^*)^2 + A_h(1 + H^*)Pe^n Y \tag{144}$$

to the points. A_h and n are the fitting parameters. The second virial coefficient C_{2h} for the whole range of Pe and I can be well approximated by:

$$C_{2h} = \alpha_h C_{1h}^2 \tag{145}$$

where α_h is the numerical coefficient ranging from 0.20 to 0.249. The best fit to the experimental points were obtained with exponent n around 0.5, thus, the assumption done by Adamczyk et al. [18,36] concerning the proportionality of the hydrodynamic shadow is not fully justified. The exponential dependence with

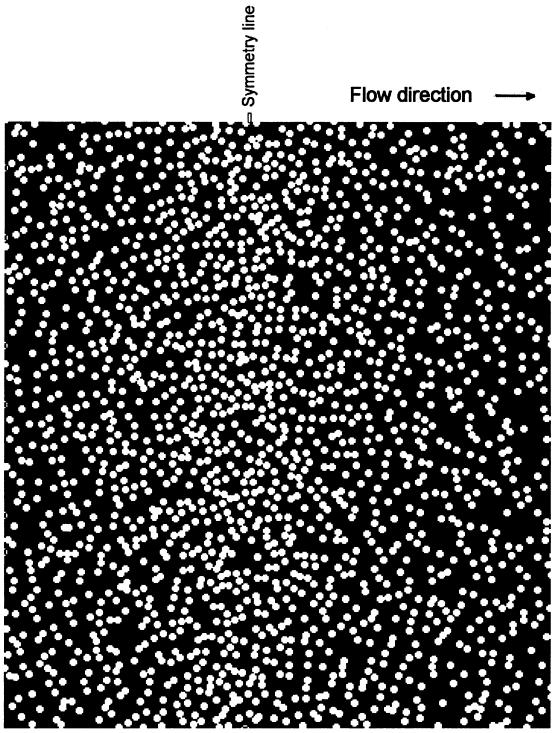


Fig. 52. The distribution of $a = 0.5$-μm latex particles adsorbed on mica plate experimentally observed in the slot impinging jet cell for $Re = 16$ and $I = 10^{-4}$ M KCl.

$n \approx 0.5$ like for the non-Brownian particles (cf. Figs. 18 and 48) much better describes the experimental data. The proportionality of the electro-hydrodynamic shadow size to the effective particle radius a^* is confirmed in Fig. 55 where C_{1h} is shown as a function of a^*/a for various Pe. Points represent the data obtained from the adsorption kinetics of polystyrene latex particles of radii $a = 0.5$ μm and $a = 0.725$ μm experimentally measured in the slot impinging jet cell at the distance $y = 158$ μm from the stagnation line [122]. Lines represent the linear regression with the slope increasing with Pe in agreement with Eq. (144). This is consistent with the results of BD simulation illustrated in Figs. 49 and 50 that the width of the hydrodynamic shadow is determined by the particle radius increased by the effective range of the repulsive EDL interactions between adsorbed and adsorbing particle, while length of the shadow depends mainly on the Pe, i.e. on the local flow conditions and Brownian motion of particles.

The detailed form of the dependence of the size of the electro-hydrodynamic shadow C_{1h} approximated here by Eq. (144) as well as higher terms of the virial expansion of the available surface function $B(\Theta)$ in the presence of flow can be only predicted by the SBD simulations which has not yet been accomplished.

The electro-hydrodynamic scattering influences not only the adsorption rate but

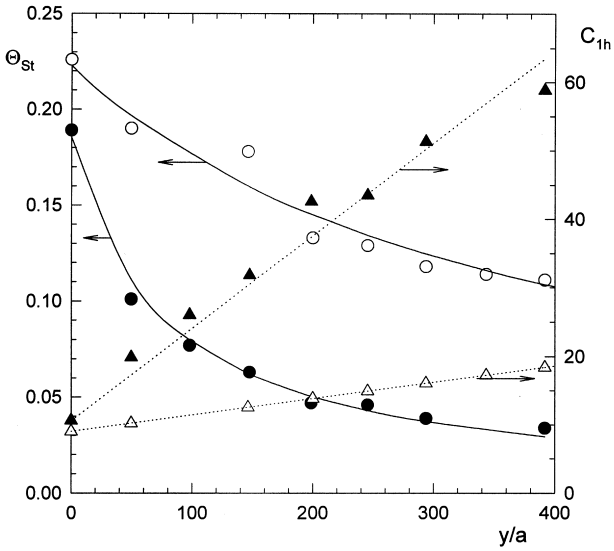


Fig. 53. The quasi-saturation coverage Θ_{st} of the monodisperse latex particles (a = 0.725 μm) on silanized mica (left axis, circles) and the size of the electro-hydrodynamic shadow C_{1h} (right axis, triangles) as a function of distance from the stagnation line. White symbols Re = 2 (Pe = 0.05), black symbols Re = 16 (Pe = 1.1).

also the local structure of adsorbed layer. This structure can be conveniently analyzed in terms of radial distribution function which is a measure of the probability of finding second particle at a given distance from the first one deposited at the interface. The radial distribution function can be experimentally determined by analyzing micrographs like the one shown in Fig. 52. The experimental procedure is widely discussed [32,121,123–125].

Fig. 56 presents the comparison between the radial distribution function obtained in the center of the slot impinging jet cell (black symbols) and at the distance y = 157 μm (white symbols) and for Re = 8; part a — κa = 16 and part b — κa = 10 [126]. The points are the experimental results for polystyrene latex particles adsorbed on mica and lines denote the Boltzmann distribution:

$$g_{r12} = \exp\left(-\frac{\Phi(r)}{kT}\right) \tag{146}$$

where $\Phi(r)$ is the EDL interaction energy calculated according to Eq. (79) with ψ_p = 50 mV. In the center of the cell where the effects of the electro-hydrodynamic scattering are small the radial distribution function is well described by the Boltzmann distribution. As it was shown by Adamczyk and co-workers [32,124,125] this good agreement between the experimentally measured radial distribution function and theoretical Boltzmann distribution is observed only for low surface coverage. Far from the cell center the effects of scattering are very pronounced. As

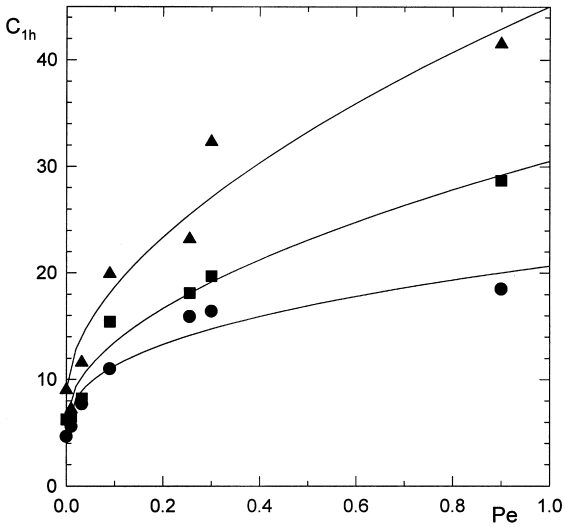


Fig. 54. The size of the electro-hydrodynamic shadow C_{1h} in the stagnation line flow as a function of *Pe*. Points represent the results obtained from the experimentally measured adsorption of polystyrene latex on mica in the slot impinging jet cell [122] for $I = 2 \times 10^{-5}$ M — triangles, $I = 10^{-4}$ M — squares and $I = 10^{-3}$ M KCl — circles. Solid lines represent fit of Eq. (144) to the points.

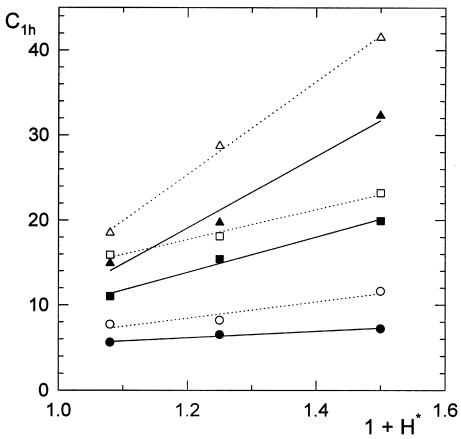


Fig. 55. The size of the electro-hydrodynamic shadow C_{1h} in the stagnation line flow as a function of the effective particle radius (scaled by *a*). Points represent the results obtained from the experimentally measured adsorption of polystyrene latex on mica in the slot impinging jet cell [122]; $a = 0.5$ μm (black symbols), $Pe = 0.01$ (circles), 0.09 (squares), 0.3 (triangles); $a = 0.725$ μm (white symbols), $Pe = 0.04$ (circles), 0.26 (squares) and 0.9 (triangles). Lines represent linear regression.

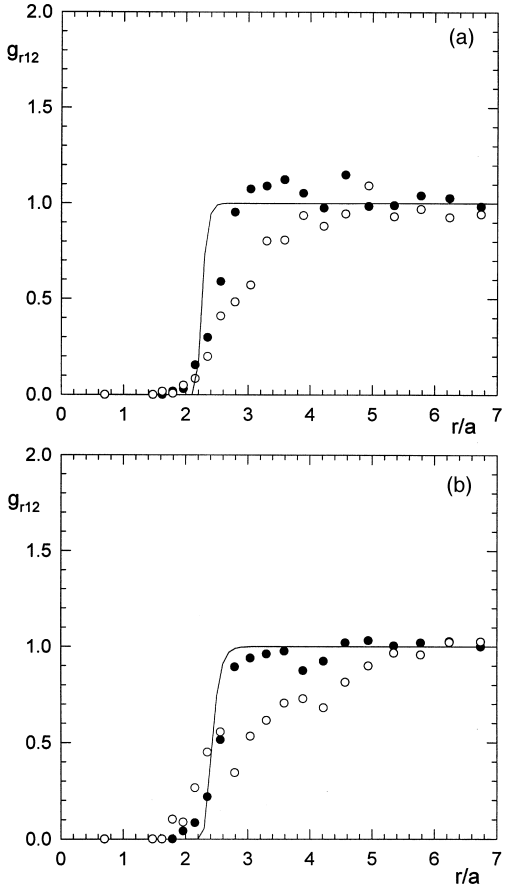


Fig. 56. The radial distribution function g_{r12} as a function of distance: a for $\kappa a = 16$, $\Theta = 0.09$ and b for $\kappa a = 10$, $\Theta = 0.07$. Symbols denote the experimental g_{r12} obtained in the slot impinging jet cell at $Re = 8$ in the center of the cell (black circles) and at the distance $y = 158$ μm from the center (white circles). Lines denote the Boltzmann distribution.

it can be seen in Fig. 56 the radial distribution function is strongly reduced up to $r/a = 4$. This is a consequence of the electro-hydrodynamic shadow created behind the particle attached to the interface. The probability of adsorption on this part of the interface is very much decreased and thus, the distribution function is depleted.

Fig. 57 presents the joint effect of the flow and the ionic strength on the radial distribution function obtained for the same system as in Fig. 56. It can be seen that the strongest depletion of the radial distribution function is observed for the lowest ionic strength of the suspension which is in agreement with the BD simulations presented in Figs. 49 and 50 supporting the hypothesis of the electro-hydrodynamic scattering.

The complete theoretical background for the observed phenomena can be

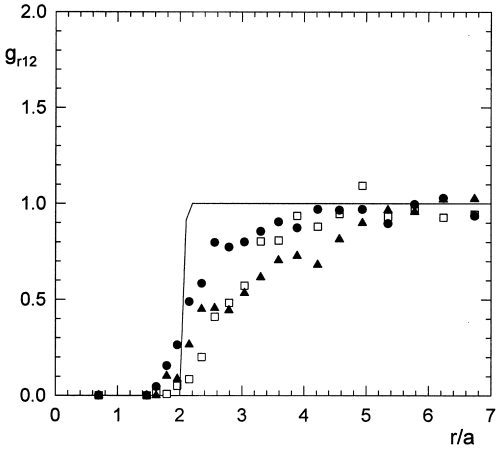


Fig. 57. The radial distribution function g_{r12} as a function of distance. Symbols denote the experimental g_{r12} obtained in the slot impinging jet cell at $Re = 8$ at the distance $y = 158$ μm from the center for $\kappa a = 50$, $\Theta = 0.11$ (circles), $\kappa a = 16$, $\Theta = 0.09$ (squares) and $\kappa a = 10$, $\Theta = 0.07$ (triangles). Line stands for the Boltzmann distribution.

established only using SBD simulation techniques. It involves, however, many-body hydrodynamic and electric interactions and that problem has not yet been solved theoretically.

7. Conclusions

It was demonstrated that the hydrodynamic and electric interactions between particles and between particles and macroscopic interfaces play a crucial role in the colloidal particle adsorption process. The influence of both interactions cannot be considered separately since in most cases they are strongly coupled. Despite that, in most theories of the particle adsorption so far developed, the hydrodynamic and electric interactions were treated independently. It was shown that this kind of independent treatment can be considered only in two limiting cases either when particles and interfaces in the system are uncharged or when the charged system is at rest. In any other case both hydrodynamic and electric interactions have to be considered simultaneously.

It was shown that the coupling of hydrodynamic and electric interactions appear already at the stage of formulation of fluid motion around moving particles. That leads to the electrokinetic effects like electroviscous drag or electrokinetic lift force. It was demonstrated that the electrokinetic drag plays the important role in the case of repulsive electrostatic forces between adsorbing particles, providing additional retardation for the particle aggregation and/or adsorption. In particular the dependence of the critical coagulation and critical deposition rates of certain colloids on the electrolyte composition can be explained in terms of the electrovis-

cous drag force. This force can also be partly responsible for the experimentally observed independence of the slope of the colloid stability curve on particle size. For the oppositely charged interfaces the electroviscous drag force plays a minor role.

The electrokinetic lift force which repels particles moving along charged interface can influence particle adsorption only in the high viscosity, low conductivity media. However, recent experimental observations show that it also has to be considered in aqueous suspensions under the condition of intense flow and low ionic strength of the suspension.

It was demonstrated that the coupling between the hydrodynamic and electric interaction directly determines the adsorption kinetics. At the early stages of the adsorption process, when the interface is not covered by deposited particle, the adsorption rate is determined only by the mechanism of transfer of particles from the bulk to the interface and by the particle–interface interactions. Strong fluid flow directed towards the interface enhances the effect of the attractive electric interactions and increases the adsorption rate. This phenomenon, sometimes referred to as 'the inverse salt effect' can be conveniently observed in the impinging jet or slot impinging jet experimental cells where the increase of the initial adsorption rate in the intense suspension flows under the condition of low ionic strength of the suspension was noticed. The 'inverse salt effect' can be quantitatively described by the convective–diffusion equation with the hydrodynamic and electric interaction taken into account.

When a single colloidal particle is attached to the interface it plays a role of an obstacle to motion of other particles. Due to the collision with the attached particle the trajectory of the moving one is modified which can be regarded as particle scattering. It strongly depends on the motion of particle in the external flow and on the interparticle interactions. The effect of the particle scattering can be used for the experimental determination of interparticle forces. Forces as low as 10^{-14} N can be effectively measured in recently developed 'colloid particle collider'.

When the interface is partially covered by particles the repulsive electric double layer interactions between adsorbing and adsorbed particles enhance the effect of hydrodynamic shadowing observed for uncharged particles. Due to the particle scattering the electro-hydrodynamic shadow is formed behind the particle deposited at the interface. The probability of adsoption of other particles within the area of the shadow is strongly decreased, thus, the substantial part of the interface becomes practically inaccessible for further adsorption. That leads to strong decrease of the adsorption rate. After a certain time of adsorption a quasi-equilibrium (saturation) coverage Θ_{st} is reached even if the adsorption is totally irreversible. The quasi-saturation level depends on the local flow conditions, Brownian motion of particles and also on the electric interactions between particles and particles and the interface. Even though in the stagnation point/line flow the surface close to the cell center is initially uniformly accessible for particle adsorption, the gradient of the surface concentration of adsorbed particles is established. The concentration decreases with the distance from the center of the cell due to the effect of electro-hydrodynamic scattering.

By the Brownian dynamics simulation it was established that in the case of stagnation point/line flow the length of the hydrodynamic shadow depends mainly on the *Pe* number which expressed the ratio between convection and diffusion and on the distance of the deposited particle relative to the stagnation point/line. On the other hand, the width of the electro-hydrodynamic shadow is mainly determined by the effective range of repulsive EDL interactions between particles. For intense flows and low ionic strength of the suspension, the quasi-saturation coverage is only a fraction of the coverage obtained in the absence of flow. The Θ_{st} can be easily exceeded, either by increasing the electrolyte concentration in the suspension or by reducing the flow intensity.

It was demonstrated that the Brownian dynamics simulation results concerning the electro-hydrodynamic shadow effect are in a good agreement with the experimental adsorption kinetics performed in the slot impinging jet cell.

The electro-hydrodynamic scattering influence not only the adsorption rate, but also determines the local distribution of adsorbed particles as it can be easily observed by examination of the radial distribution function in the absence and presence of flow.

Despite the progress in the field of electro-hydrodynamic coupling in the colloidal systems still many problems have to be addressed. The determination of hydrodynamic and electric many body interactions of particles at interfaces is of primary interest. Without the considerable progress in this field the detailed SBD simulations of particle adsorption occurring in the external flow at the surface coverages exceeding several percent cannot be done. More experimental work performed in various flow conditions and suspension compositions are required. Also the theory of electrokinetic phenomena for arbitrary particle size, surface potential and distance to the interface still waits to be formulated. There is a necessity of more experimental data concerning the influence of the electrokinetic effects on the particle adsorption in aqueous and non-aqueous media where they seem to be even more important. In particular, further experiments on the effect of the electrolyte type on particle adsorption rate are needed.

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